AI6126 Advanced Computer Vision

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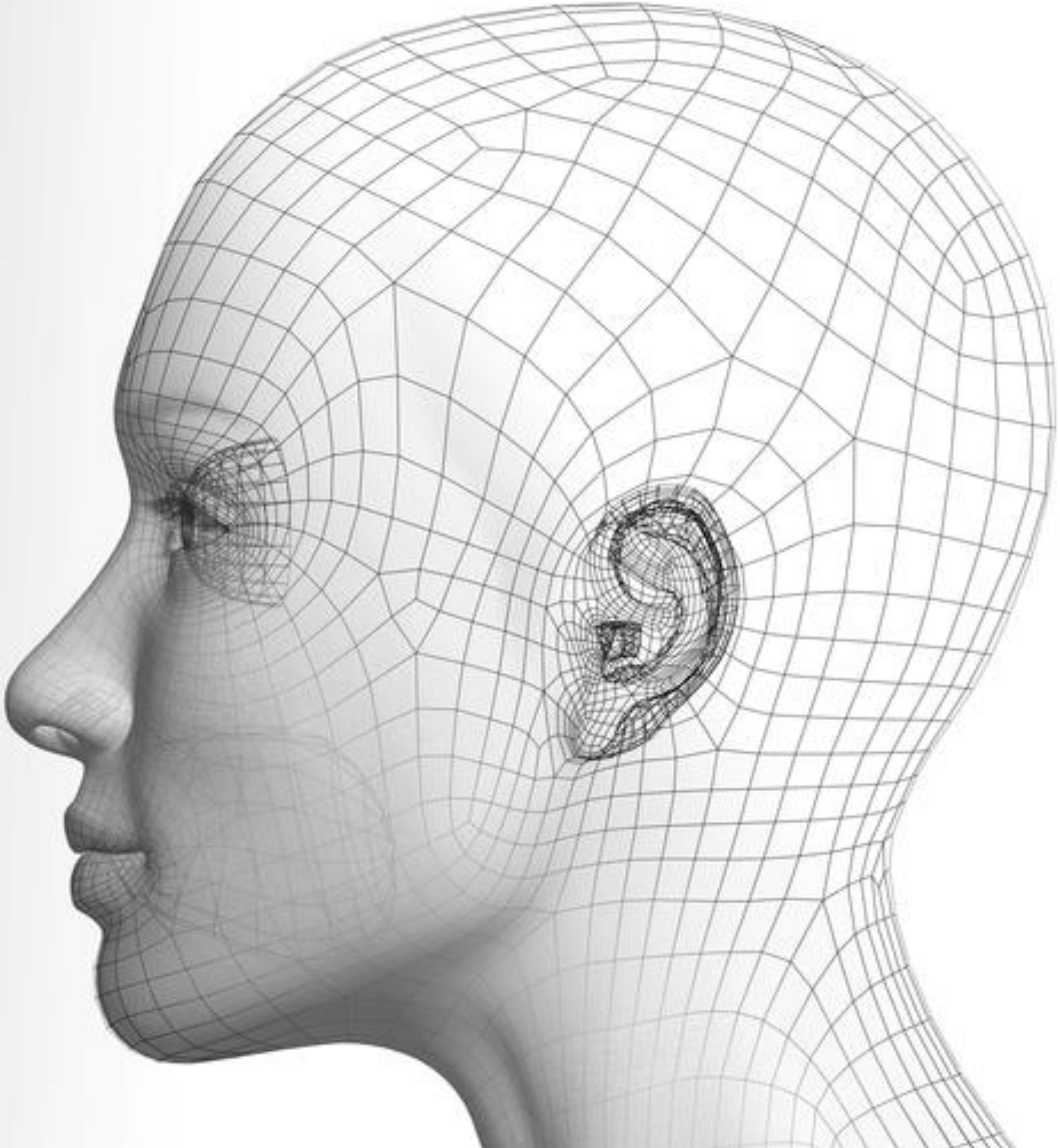
Generative Adversarial Nets

Ziwei Liu

刘子纬

https://liuziwei7.github.io/



Slide Credits

- Justin Johnson, EECS 498/598
- NIPS 2016 Tutorial, Generative Adversarial Networks



We are in the Metaverse

By 2027, creators won’t have to be technical, just creative, thanks to automation tools.

Outline

Generative Adversarial Networks (GANs):

- Generator & Discriminator
- Minimax Game – Optimality
- Zoo of GANs: DCGAN, Progressive GAN, StyleGAN

Advanced Topics in GANs:

- Conditional GAN
- DeepFake Detection
- GANs: Not Just for Images

Open Problems

Last Time: Supervised vs Unsupervised Learning

Supervised Learning

Data: (x, y)

x is data, y is label

Goal: Learn a *function* to map x -> y

Examples: Classification, regression, object detection, semantic segmentation, image captioning, etc.

Unsupervised Learning

Data: x

Just data, no labels!

Goal: Learn some underlying hidden *structure* of the data

Examples: Clustering, dimensionality reduction, feature learning, density estimation, etc.

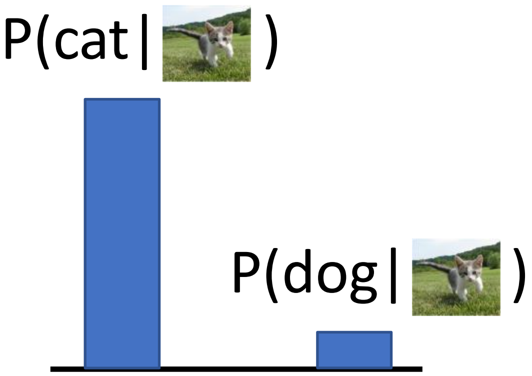
Last Time: Discriminative vs Generative Models

Discriminative Model: Learn a probability distribution p(y|x)

Generative Model: Learn a probability distribution p(x)

Conditional Generative Model: Learn p(x|y)

Data: x



Density Function
p(x) assigns a positive number to each possible x; higher numbers mean x is more likely

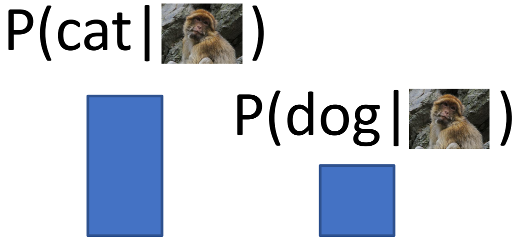
Density functions are **normalized**:

$$\int p(\mathrm{x})dx = 1$$

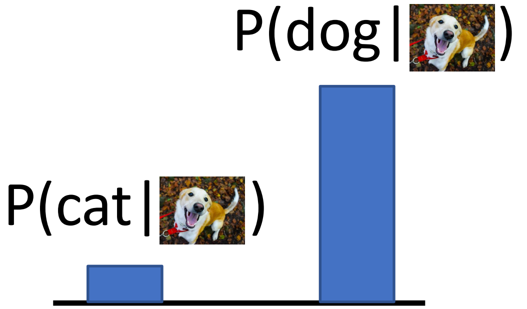
Different values of x **compete** for density

Last Time: Discriminative vs Generative Models

Discriminative Model:
Learn a probability distribution p(y|x)



Generative Model:
Learn a probability distribution p(x)



Conditional Generative Model: Learn p(x|y)

Discriminative model: No way for the model to handle unreasonable inputs; it must give label distributions for all images

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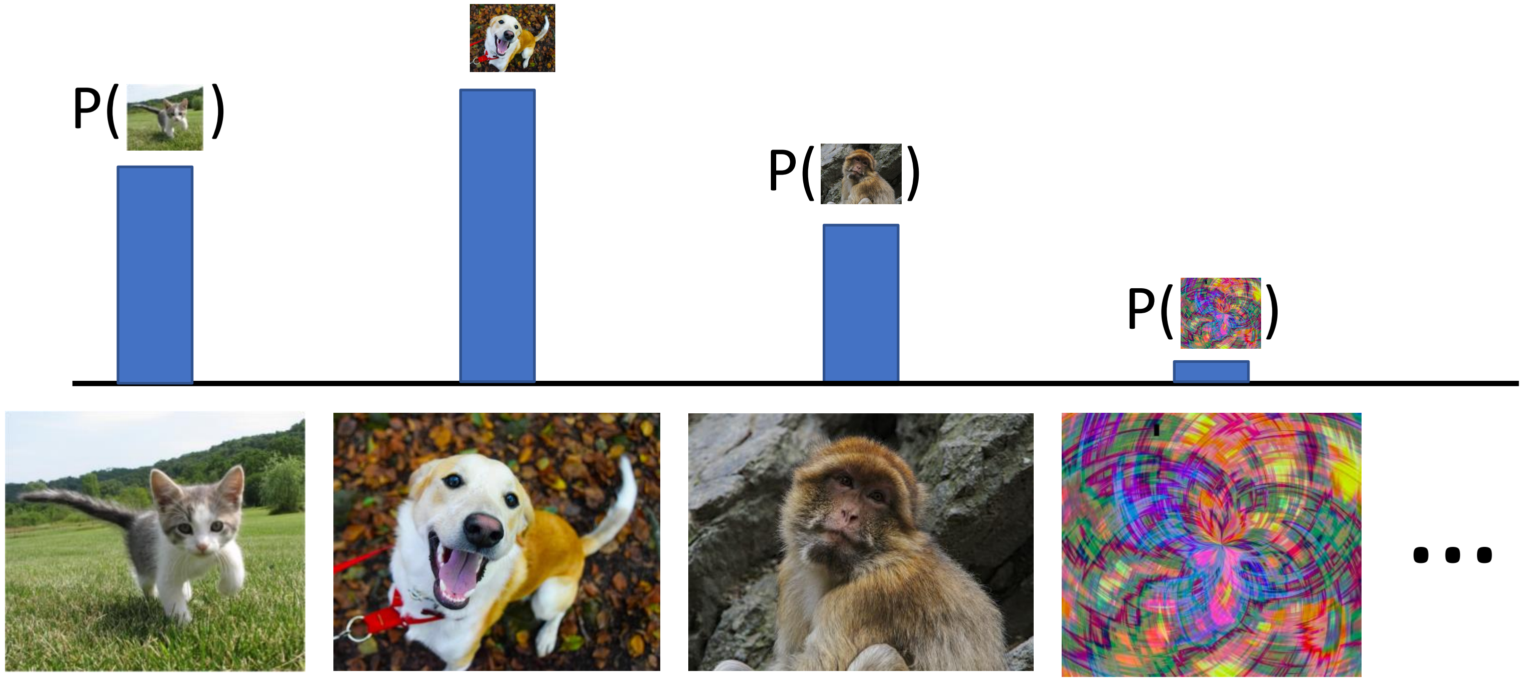

Generative model: All possible images compete with each other for probability mass

Requires deep image understanding! Is a dog more likely to sit or stand? How about 3-legged dog vs 3-armed monkey?

Last Time: Discriminative vs Generative Models

Discriminative Model:
Learn a probability distribution p(y|x)

Generative Model:
Learn a probability distribution p(x)

Conditional Generative Model: Learn p(x|y)

Recall **Bayes' Rule:**

$$\boxed{P(x \mid y)} = \frac{\boxed{P(y \mid x)}}{\boxed{P(y)}} \boxed{P(x)}$$

Discriminative Model

(Unconditional) Generative Model

Conditional Generative Model

Prior over labels

We can build a conditional generative model from other components!

Last Time: Taxonomy of Generative Models

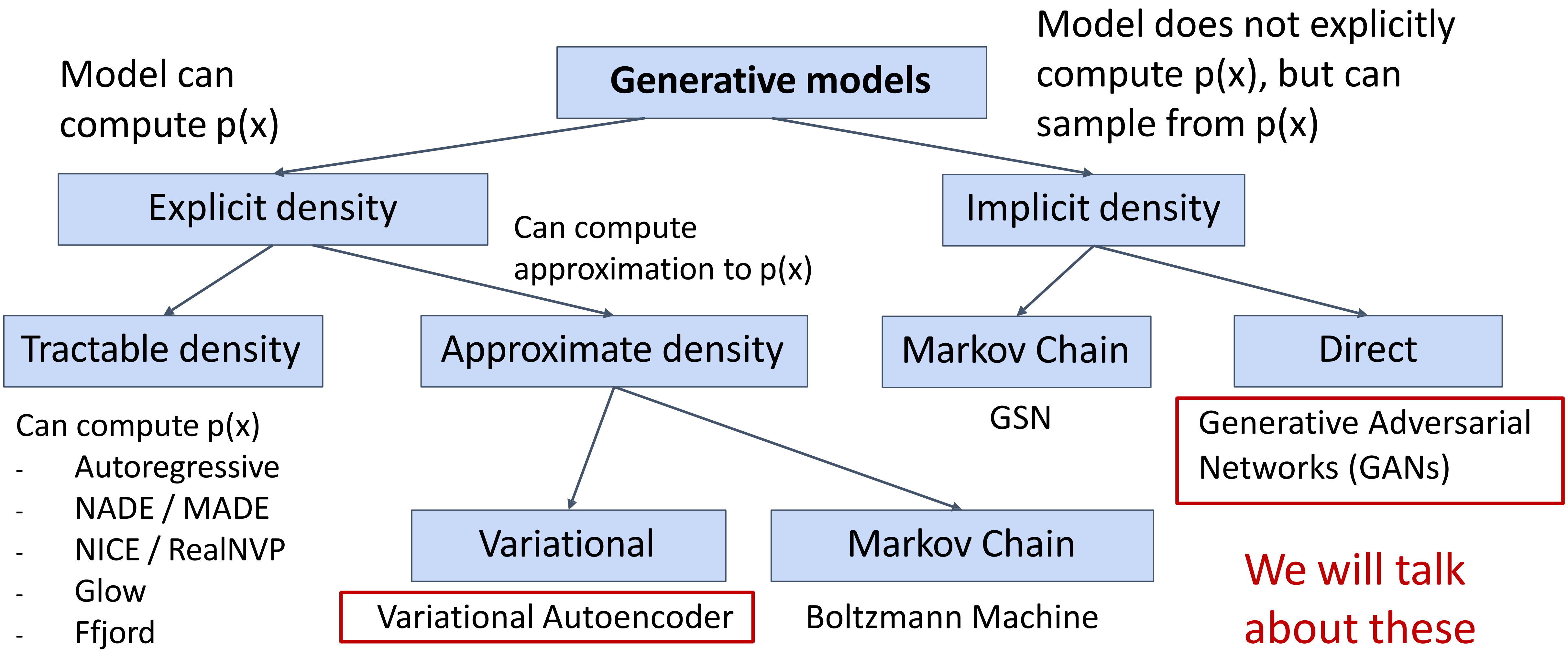

Figure adapted from Ian Goodfellow, Tutorial on Generative Adversarial Networks, 2017.

Generative Adversarial Networks

Generative Models So Far:

Variational Autoencoders introduce a latent z, and maximize a lower bound:

$$p_\theta(x) = \int_Z p_\theta(x|z)p(z)dz \geq E_{Z \sim q_\phi(Z|x)}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x), p(z)\right)$$

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Generative Adversarial Networks give up on modeling p(x), but allow us to draw samples from p(x)

Generative Adversarial Networks

Setup: Assume we have data x_i drawn from distribution $p_{data}(x)$. Want to sample from p_{data}.

Goodfellow et al, "Generative Adversarial Nets", NeurIPS 2014

Generative Adversarial Networks

Setup: Assume we have data x_i drawn from distribution $p_{data}(x)$. Want to sample from p_{data}.

Idea: Introduce a latent variable z with simple prior p(z).
Sample $z \sim p(z)$ and pass to a **Generator Network** x = G(z)
Then x is a sample from the **Generator distribution** p_G. Want $p_G = p_{data}$!

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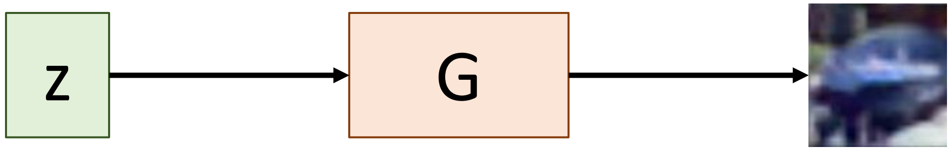
Generator Network

Generated Sample

Sample z from p_z



Train **Generator Network** G to convert z into fake data x sampled from p_G

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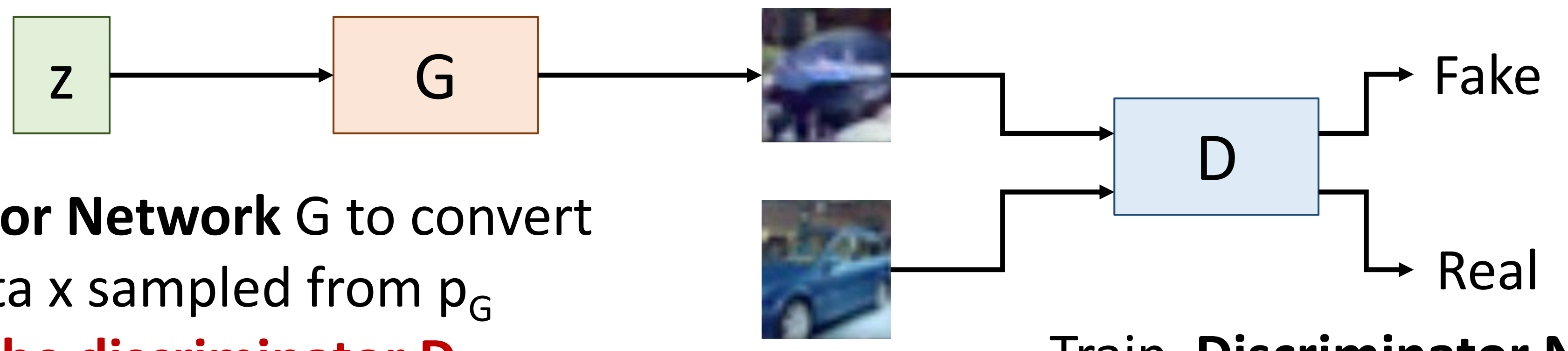

Train **Generator Network** G to convert z into fake data x sampled from p_G **by "fooling" the discriminator D**

Train **Discriminator Network** D to classify data as real or fake (1/0)

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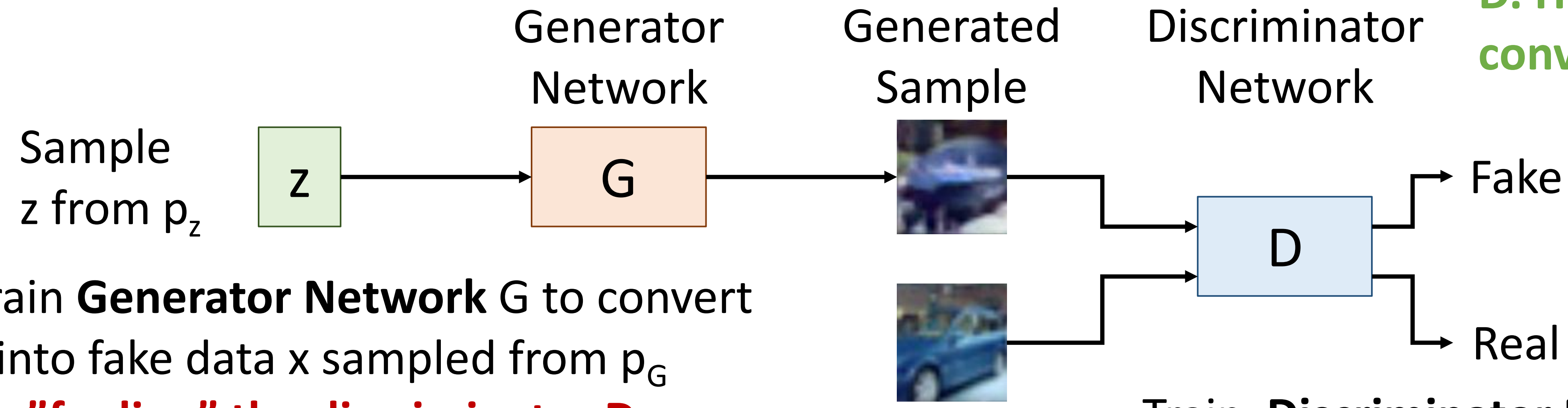
Sample $z \sim p(z)$ and pass to a **Generator Network** x = G(z)
Then x is a sample from the **Generator distribution** p_G. Want $p_G = p_{data}$!

Jointly train G and D. Hopefully p_G converges to p_{data}!



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Generative Adversarial Networks: Training Objective

Jointly train generator G and discriminator D with a **minimax game**

$$\min_G \max_D \left(E_{x \sim p_{data}}[\log D(x)] + E_{z \sim p(z)}\left[\log\left(1 - D\big(G(z)\big)\right)\right] \right)$$

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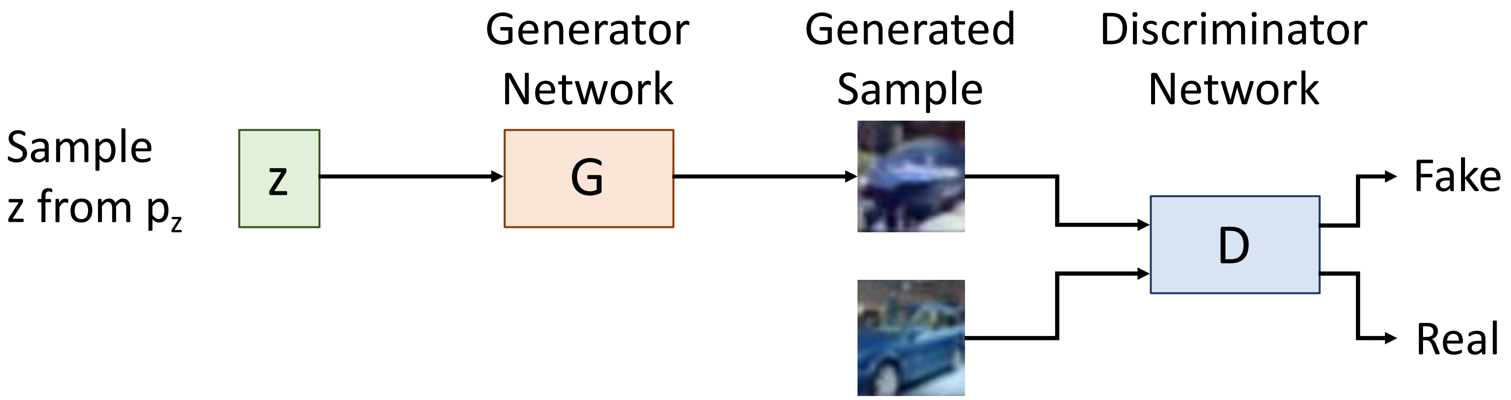

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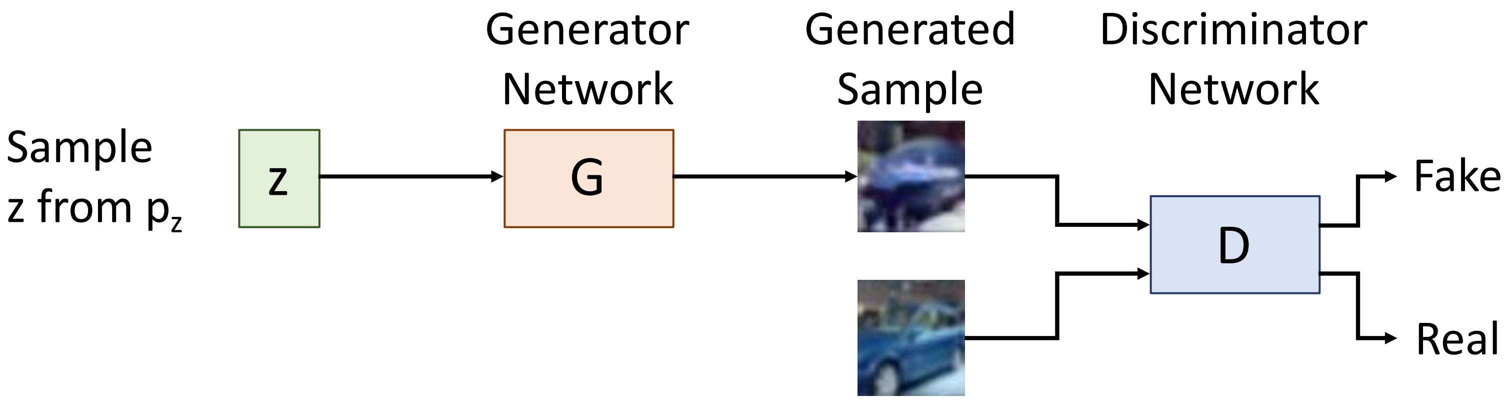

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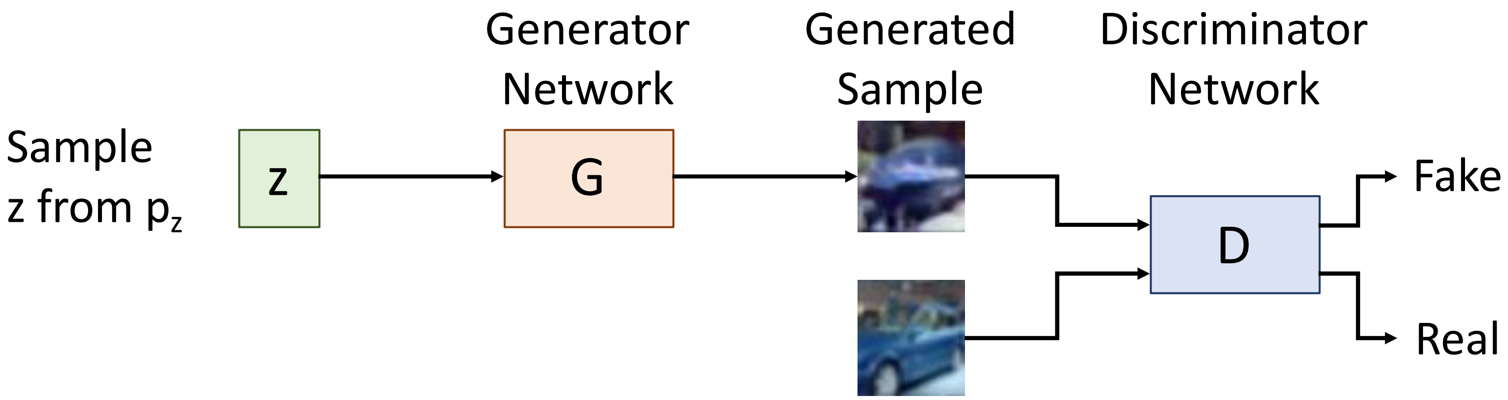

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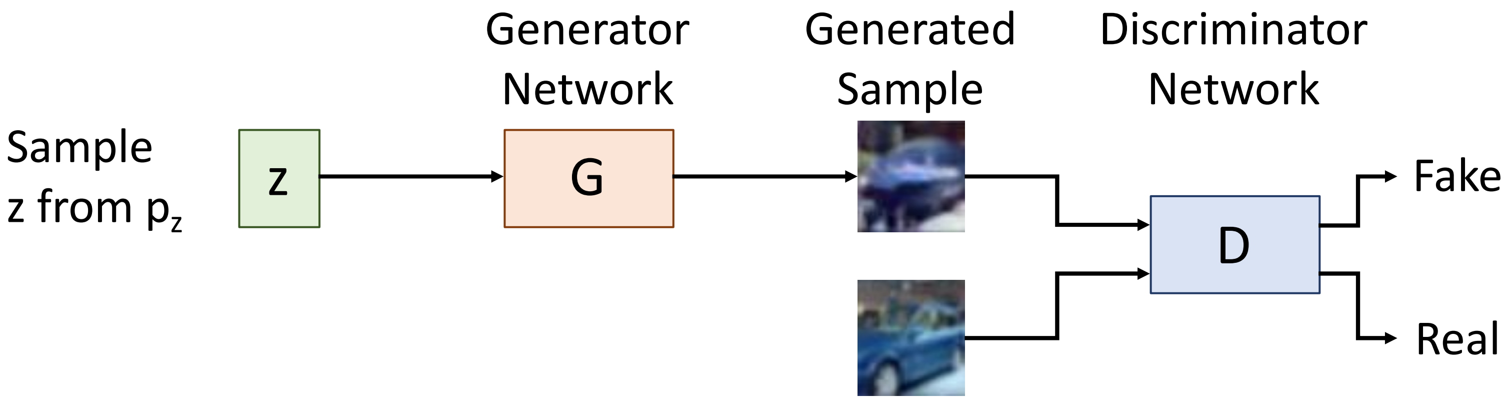

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Train G and D using alternating gradient updates

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}}[\log D(x)] + E_{z \sim p(z)} \left[\log\left(1 - D(G(z))\right)\right] \right)$$

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$$= \min_{G} \max_{D} V(G, D)$$

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For t in 1, … T:

1. (Update D) $D = D + \alpha_D \frac{\partial V}{\partial D}$
2. (Update G) $G = G - \alpha_G \frac{\partial V}{\partial G}$

Goodfellow et al, “Generative Adversarial Nets”, NeurIPS 2014

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1. (Update **D**) $D = D + \alpha_D \frac{\partial V}{\partial D}$
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We are not minimizing any overall loss! No training curves to look at!

Goodfellow et al, “Generative Adversarial Nets”, NeurIPS 2014

Generative Adversarial Networks: Training Objective

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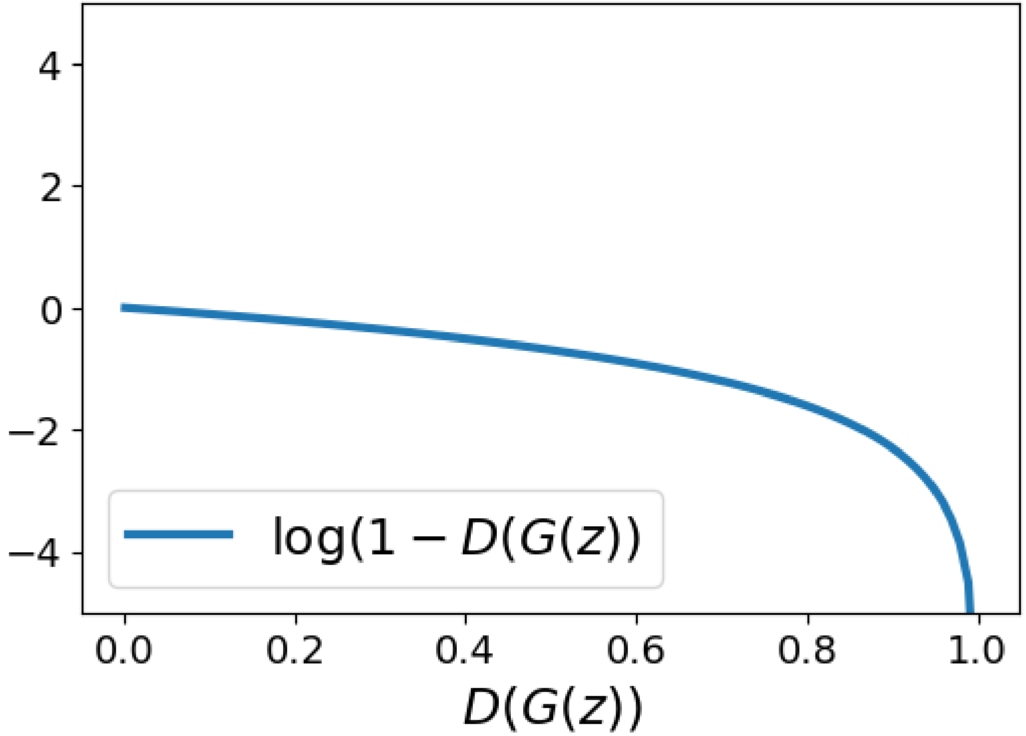
$$\min_{G}\max_{D}\left(E_{x\sim p_{data}}[\log D(x)] + E_{z\sim p(z)}\left[\log\left(1 - D(G(z))\right)\right]\right)$$

At start of training, generator is very bad and discriminator can easily tell apart real/fake, so D(G(z)) close to 0



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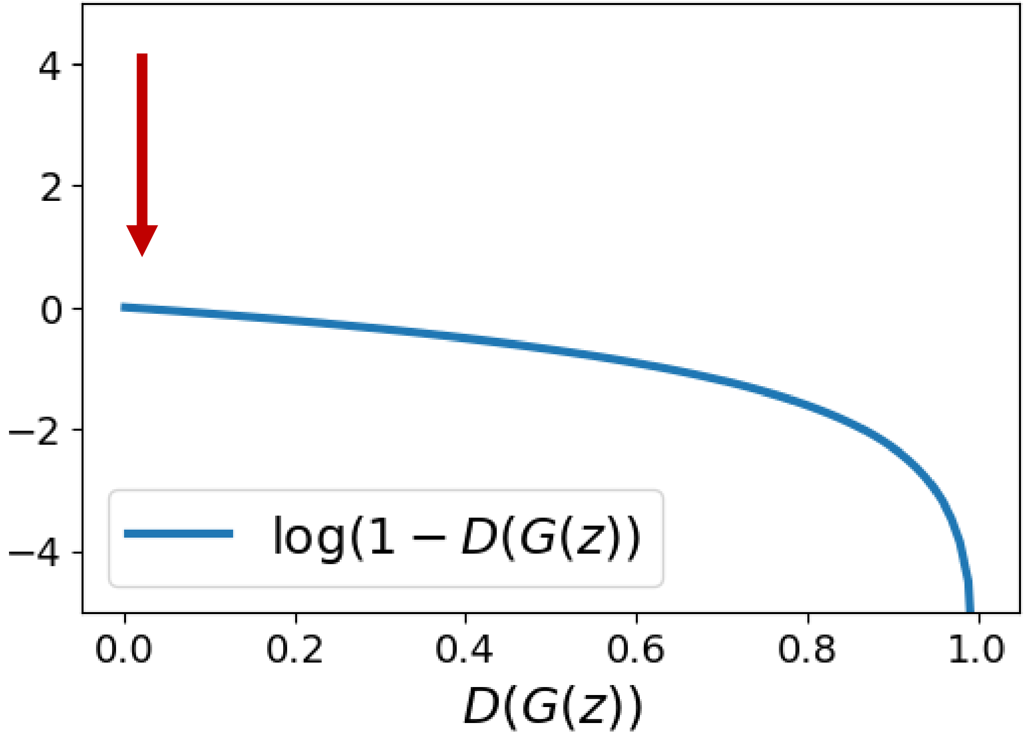
Problem: Vanishing gradients for G



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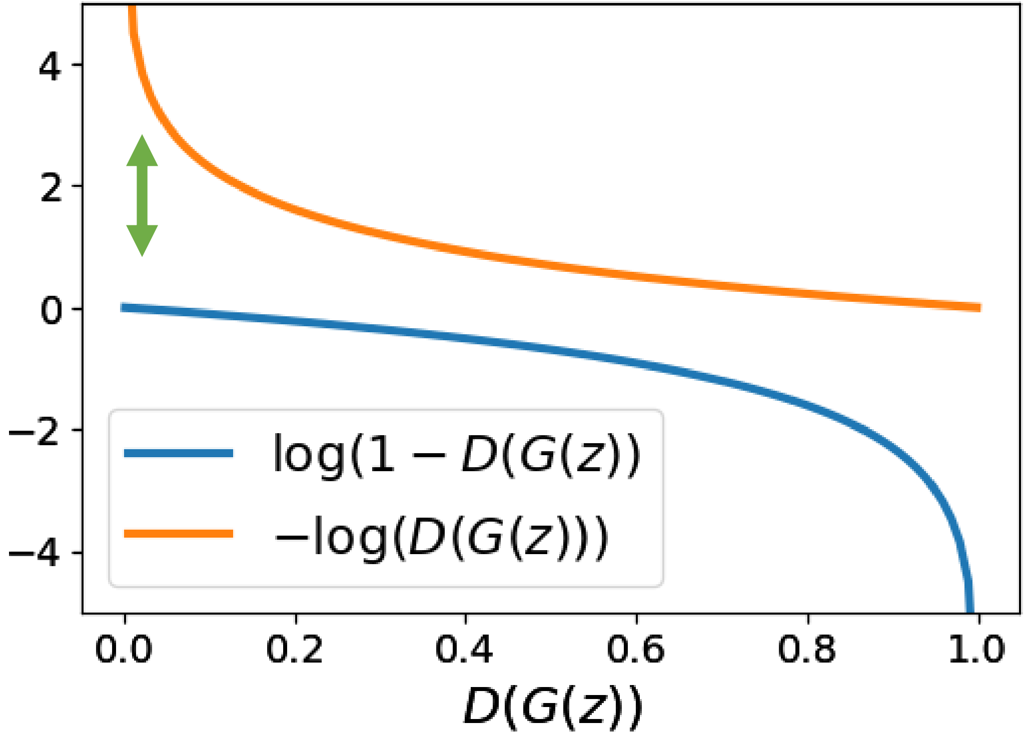
At start of training, generator is very bad and discriminator can easily tell apart real/fake, so D(G(z)) close to 0

Problem: Vanishing gradients for G

Solution: Right now G is trained to minimize log(1-D(G(z)). Instead, train G to minimize –log(D(G(z)). Then G gets strong gradients at start of training!



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Minimax Game – Optimality

Generative Adversarial Networks: Optimality

Jointly train generator G and discriminator D with a **minimax game**

Why is this particular objective a good idea?

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

This minimax game achieves its global minimum when $p_G = p_{data}$!

Goodfellow et al, “Generative Adversarial Nets”, NeurIPS 2014

Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

(Our objective so far)

Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

$$= \min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{x \sim p_G} [\log(1 - D(x))] \right)$$

(Change of variables on second term)

Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

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$$= \min_{G} \max_{D} \int_{X} \left(p_{data}(x) \log D(x) + p_G(x) \log(1 - D(x)) \right) dx$$

(Definition of expectation)

Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

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(Push $\max_D$ inside integral)

Generative Adversarial Networks: Optimality

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$$f(y) = a \log y + b \log(1 - y)$$

(Side computation to compute max)

Generative Adversarial Networks: Optimality

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$$f(y) = a \log y + b \log(1 - y)$$

$$f'(y) = \frac{a}{y} - \frac{b}{1 - y}$$

Generative Adversarial Networks: Optimality

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$$f(y) = a \log y + b \log(1 - y) \qquad f'(y) = 0 \iff y = \frac{a}{a+b} \text{ (local max)}$$

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Optimal Discriminator: $D_G^*(x) = \frac{p_{data}(x)}{p_{data}(x) + p_G(x)}$

Generative Adversarial Networks: Optimality

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Optimal Discriminator: $D_G^*(x) = \frac{p_{data}(x)}{p_{data}(x) + p_G(x)}$

Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

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$$= \min_{G} \int_{X} \left(p_{data}(x) \log D_{G}^{*}(x) + p_{G}(x) \log \left(1 - D_{G}^{*}(x) \right) \right) dx$$

$$= \min_{G} \int_{X} \left(p_{data}(x) \log \frac{p_{data}(x)}{p_{data}(x) + p_{G}(x)} + p_{G}(x) \log \frac{p_{G}(x)}{p_{data}(x) + p_{G}(x)} \right) dx$$

Optimal Discriminator: $D_{G}^{*}(x) = \dfrac{p_{data}(x)}{p_{data}(x)+p_{G}(x)}$

Generative Adversarial Networks: Optimality

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Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}}[\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

$$= \min_{G} \int_X \left(p_{data}(x) \log \frac{p_{data}(x)}{p_{data}(x) + p_G(x)} + p_G(x) \log \frac{p_G(x)}{p_{data}(x) + p_G(x)} \right) dx$$

$$= \min_{G} \left(E_{x \sim p_{data}} \left[\log \frac{p_{data}(x)}{p_{data}(x) + p_G(x)} \right] + E_{x \sim p_G} \left[\log \frac{p_G(x)}{p_{data}(x) + p_G(x)} \right] \right)$$

(Definition of expectation)

Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

$$= \min_{G} \int_X \left(p_{data}(x) \log \frac{p_{data}(x)}{p_{data}(x) + p_G(x)} + p_G(x) \log \frac{p_G(x)}{p_{data}(x) + p_G(x)} \right) dx$$

$$= \min_{G} \left(E_{x \sim p_{data}} \left[\log \frac{2}{2} \frac{p_{data}(x)}{p_{data}(x) + p_G(x)} \right] + E_{x \sim p_G} \left[\log \frac{2}{2} \frac{p_G(x)}{p_{data}(x) + p_G(x)} \right] \right)$$

(Multiply by a constant)

Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

$$= \min_{G} \int_X \left(p_{data}(x) \log \frac{p_{data}(x)}{p_{data}(x) + p_G(x)} + p_G(x) \log \frac{p_G(x)}{p_{data}(x) + p_G(x)} \right) dx$$

$$= \min_{G} \left(E_{x \sim p_{data}} \left[\log \frac{2}{2} \frac{p_{data}(x)}{p_{data}(x) + p_G(x)} \right] + E_{x \sim p_G} \left[\log \frac{2}{2} \frac{p_G(x)}{p_{data}(x) + p_G(x)} \right] \right)$$

$$= \min_{G} \left(E_{x \sim p_{data}} \left[\log \frac{2 * p_{data}(x)}{p_{data}(x) + p_G(x)} \right] + E_{x \sim p_G} \left[\log \frac{2 * p_G(x)}{p_{data}(x) + p_G(x)} \right] - \log 4 \right)$$

Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

$$= \min_{G} \left(E_{x \sim p_{data}} \left[\log \frac{2 * p_{data}(x)}{p_{data}(x) + p_G(x)} \right] + E_{x \sim p_G} \left[\log \frac{2 * p_G(x)}{p_{data}(x) + p_G(x)} \right] - \log 4 \right)$$

Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

$$= \min_{G} \left(E_{x \sim p_{data}} \left[\log \frac{2 * p_{data}(x)}{p_{data}(x) + p_G(x)} \right] + E_{x \sim p_G} \left[\log \frac{2 * p_G(x)}{p_{data}(x) + p_G(x)} \right] - \log 4 \right)$$

Kullback-Leibler Divergence:

$$KL(p, q) = E_{x \sim p} \left[\log \frac{p(x)}{q(x)} \right]$$

Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D\left(G(z)\right) \right) \right] \right)$$

$$= \min_{G} \left(E_{x \sim p_{data}} \left[\log \frac{2 * p_{data}(x)}{p_{data}(x) + p_G(x)} \right] + E_{x \sim p_G} \left[\log \frac{2 * p_G(x)}{p_{data}(x) + p_G(x)} \right] - \log 4 \right)$$

Kullback-Leibler Divergence:

$$KL(p, q) = E_{x \sim p} \left[\log \frac{p(x)}{q(x)} \right]$$

Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D\left(G(z) \right) \right) \right] \right)$$

$$= \min_{G} \left(E_{x \sim p_{data}} \left[\log \frac{2 * p_{data}(x)}{p_{data}(x) + p_G(x)} \right] + E_{x \sim p_G} \left[\log \frac{2 * p_G(x)}{p_{data}(x) + p_G(x)} \right] - \log 4 \right)$$

$$= \min_{G} \left(KL\left(p_{data}, \frac{p_{data} + p_G}{2} \right) + KL\left(p_G, \frac{p_{data} + p_G}{2} \right) - \log 4 \right)$$

Kullback-Leibler Divergence:

$$KL(p, q) = E_{x \sim p} \left[\log \frac{p(x)}{q(x)} \right]$$

Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x\sim p_{data}}[\log D(x)] + E_{z\sim p(z)}\left[\log\left(1 - D\left(G(z)\right)\right)\right]\right)$$

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Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

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$$= \min_{G} \left(KL \left(p_{data}, \frac{p_{data} + p_G}{2} \right) + KL \left(p_G, \frac{p_{data} + p_G}{2} \right) - \log 4 \right)$$

Jensen-Shannon Divergence:

$$JSD(p, q) = \frac{1}{2} KL \left(p, \frac{p+q}{2} \right) + \frac{1}{2} KL \left(q, \frac{p+q}{2} \right)$$

Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

$$= \min_{G} \left(E_{x \sim p_{data}} \left[\log \frac{2 * p_{data}(x)}{p_{data}(x) + p_G(x)} \right] + E_{x \sim p_G} \left[\log \frac{2 * p_G(x)}{p_{data}(x) + p_G(x)} \right] - \log 4 \right)$$

$$= \min_{G} \left(KL\left(p_{data}, \frac{p_{data} + p_G}{2} \right) + KL\left(p_G, \frac{p_{data} + p_G}{2} \right) - \log 4 \right)$$

Jensen-Shannon Divergence:

$$JSD(p, q) = \frac{1}{2} KL\left(p, \frac{p + q}{2} \right) + \frac{1}{2} KL\left(q, \frac{p + q}{2} \right)$$

Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

$$= \min_{G} \left(E_{x \sim p_{data}} \left[\log \frac{2 * p_{data}(x)}{p_{data}(x) + p_G(x)} \right] + E_{x \sim p_G} \left[\log \frac{2 * p_G(x)}{p_{data}(x) + p_G(x)} \right] - \log 4 \right)$$

$$= \min_{G} \left(KL\left(p_{data}, \frac{p_{data} + p_G}{2} \right) + KL\left(p_G, \frac{p_{data} + p_G}{2} \right) - \log 4 \right)$$

$$= \min_{G} (2 * JSD(p_{data}, p_G) - \log 4)$$

Jensen-Shannon Divergence:

$$JSD(p, q) = \frac{1}{2} KL\left(p, \frac{p + q}{2} \right) + \frac{1}{2} KL\left(q, \frac{p + q}{2} \right)$$

Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D\left(G(z) \right) \right) \right] \right)$$

$$= \min_{G} \left(E_{x \sim p_{data}} \left[\log \frac{2 * p_{data}(x)}{p_{data}(x) + p_{G}(x)} \right] + E_{x \sim p_{G}} \left[\log \frac{2 * p_{G}(x)}{p_{data}(x) + p_{G}(x)} \right] - \log 4 \right)$$

$$= \min_{G} \left(KL \left(p_{data}, \frac{p_{data} + p_{G}}{2} \right) + KL \left(p_{G}, \frac{p_{data} + p_{G}}{2} \right) - \log 4 \right)$$

$$= \min_{G} (2 * JSD(p_{data}, p_{G}) - \log 4)$$

JSD is always nonnegative, and zero only when the two distributions are equal! Thus $p_{data} = p_G$ is the global min, QED

Jensen-Shannon Divergence:

$$JSD(p, q) = \frac{1}{2} KL \left(p, \frac{p + q}{2} \right) + \frac{1}{2} KL \left(q, \frac{p + q}{2} \right)$$

Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

$$= \min_{G} (2 * JSD(p_{data}, p_{G}) - \log 4)$$

Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

$$= \min_{G} (2 * JSD(p_{data}, p_G) - \log 4)$$

Summary: The global minimum of the minimax game happens when:

1. $D_G^*(x) = \frac{p_{data}(x)}{p_{data}(x) + p_G(x)}$ (Optimal discriminator for any G)
2. $p_G(x) = p_{data}(x)$ (Optimal generator for optimal D)

Generative Adversarial Networks: Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

$$= \min_{G} (2 * JSD(p_{data}, p_G) - \log 4)$$

Summary: The global minimum of the minimax game happens when:

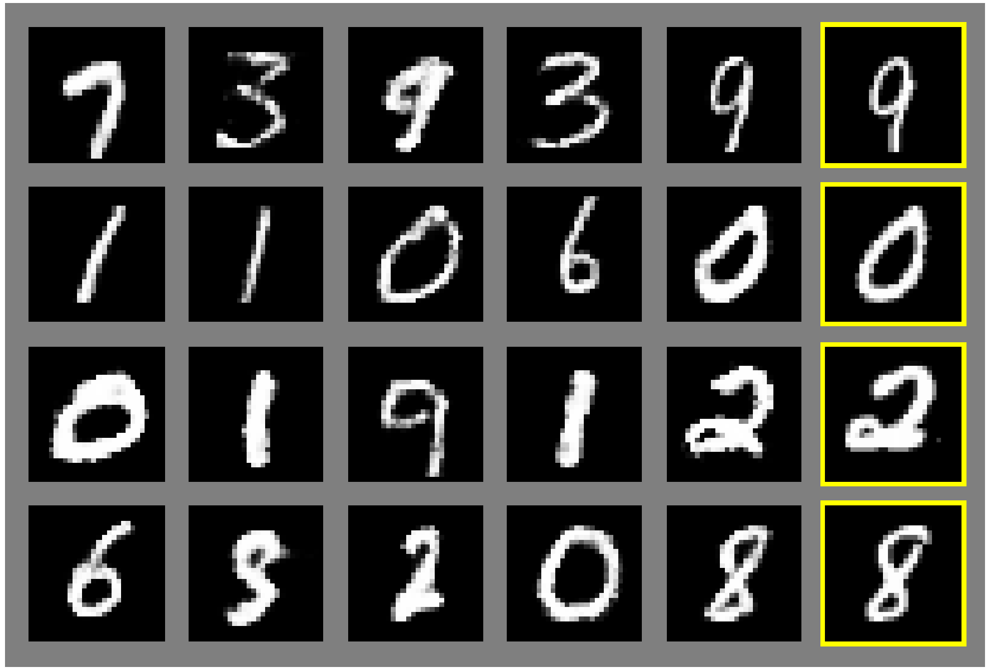
1. $D_G^*(x) = \frac{p_{data}(x)}{p_{data}(x) + p_G(x)}$ (Optimal discriminator for any G)
2. $p_G(x) = p_{data}(x)$ (Optimal generator for optimal D)

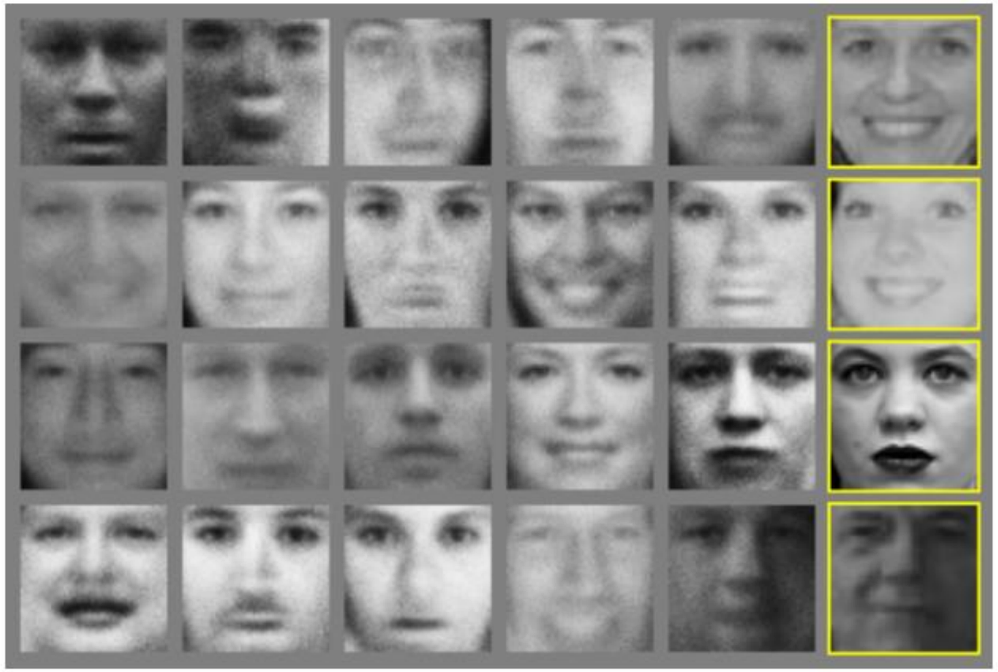
Caveats:

1. G and D are neural nets with fixed architecture. We don't know whether they can actually <u>represent</u> the optimal D and G.
2. This tells us nothing about <u>convergence</u> to the optimal solution

Generative Adversarial Networks: Results

Generated samples



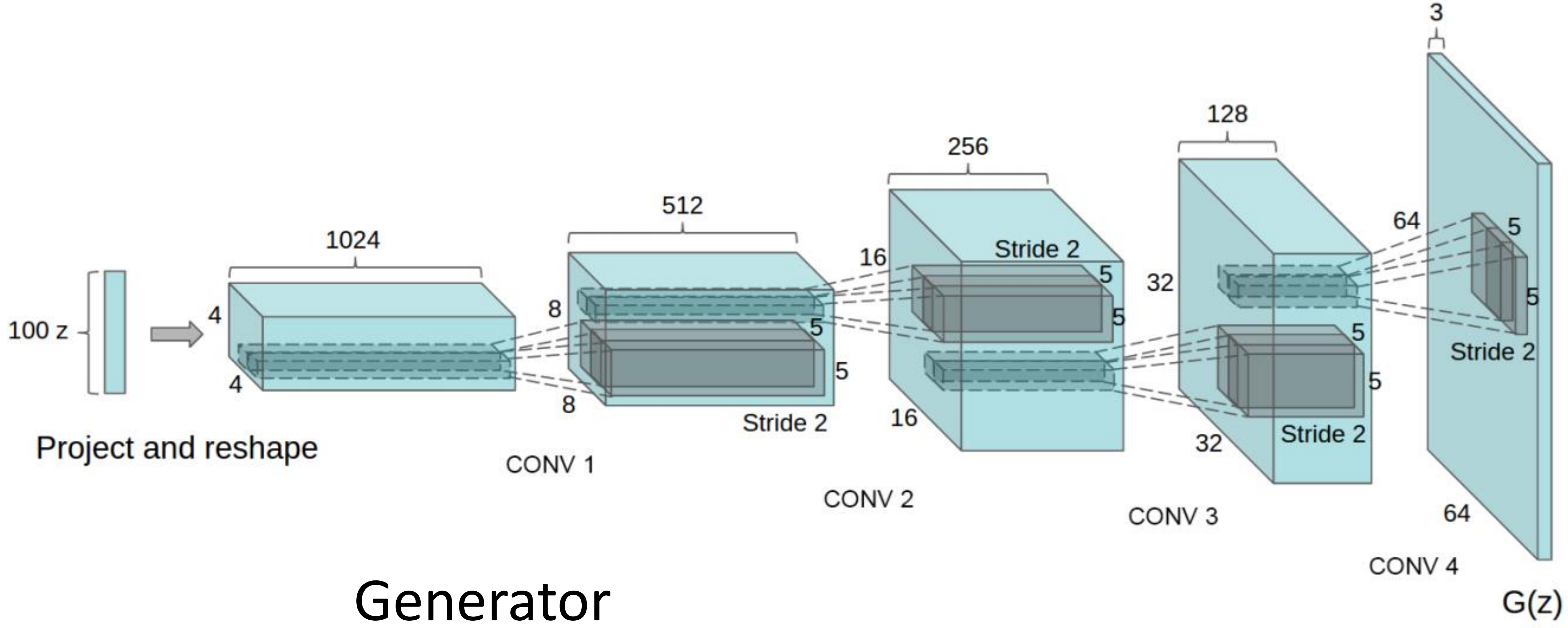


Nearest neighbor from training set

Goodfellow et al, "Generative Adversarial Nets", NeurIPS 2014

Zoo of GANs

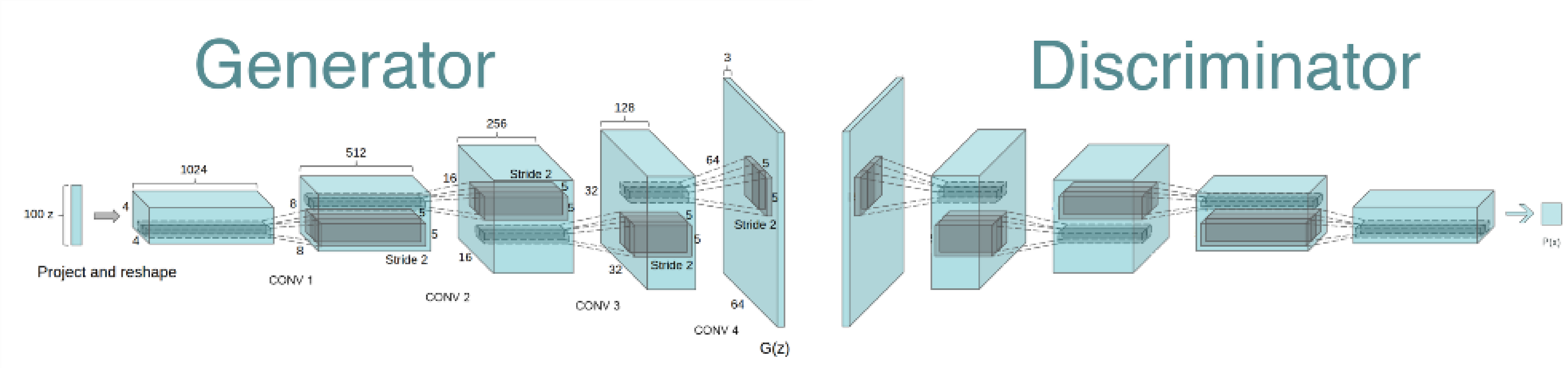
Generative Adversarial Networks: DC-GAN



Generator

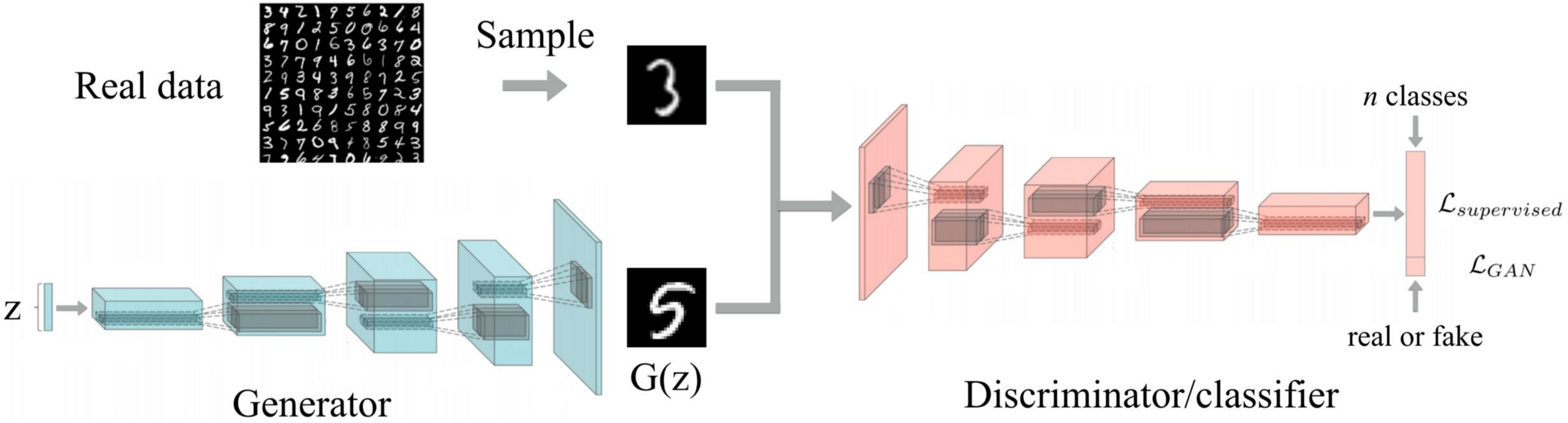
Radford et al, "Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks", ICLR 2016

Generative Adversarial Networks: DC-GAN



Radford et al, “Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks”, ICLR 2016

Generative Adversarial Networks: DC-GAN



Radford et al, "Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks", ICLR 2016

Generative Adversarial Networks: DC-GAN

Samples from the model look much better!

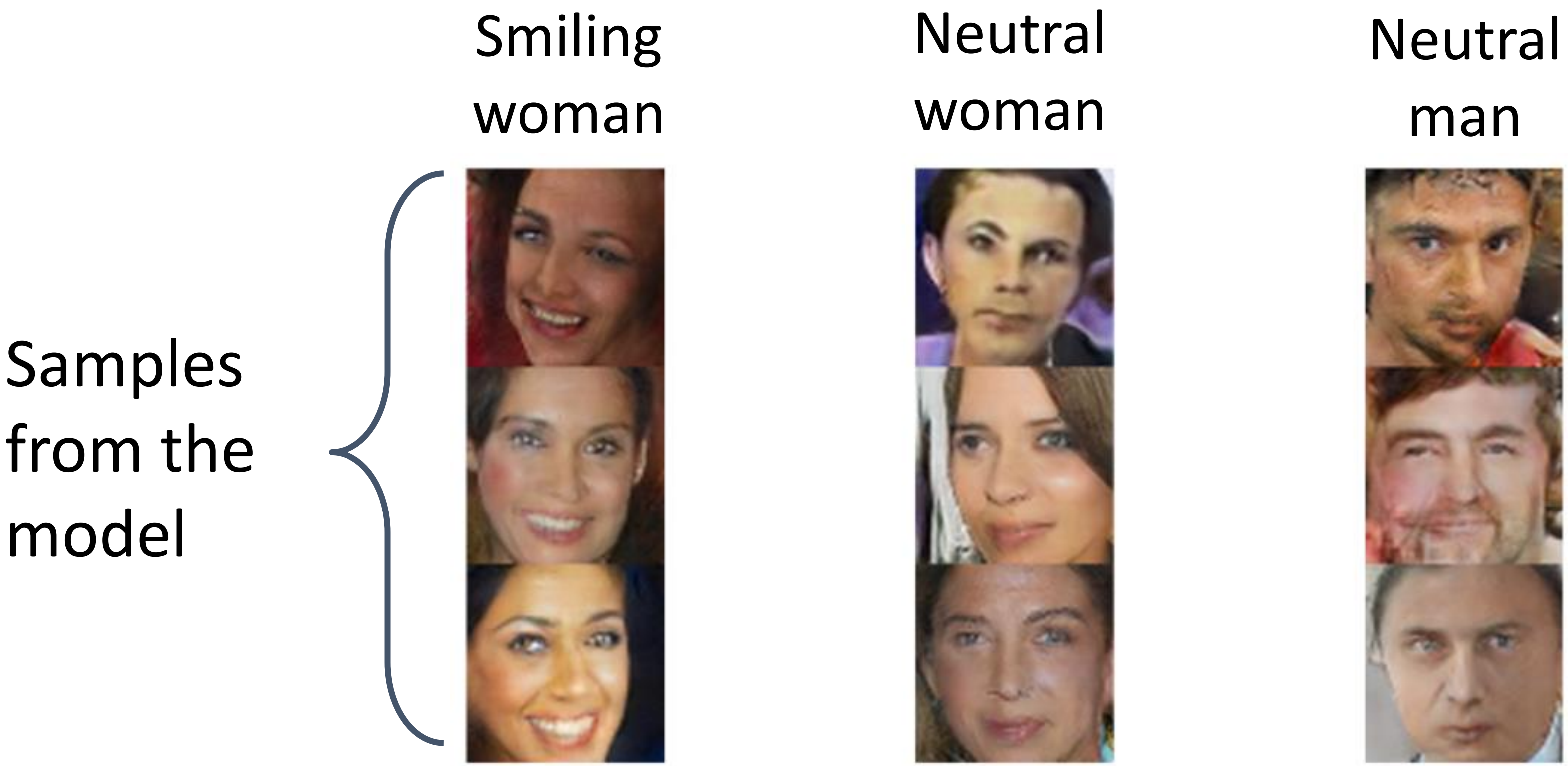
Radford et al, ICLR 2016

Generative Adversarial Networks: Interpolation

Interpolating between points in latent z space

Radford et al, ICLR 2016

Generative Adversarial Networks: Vector Math

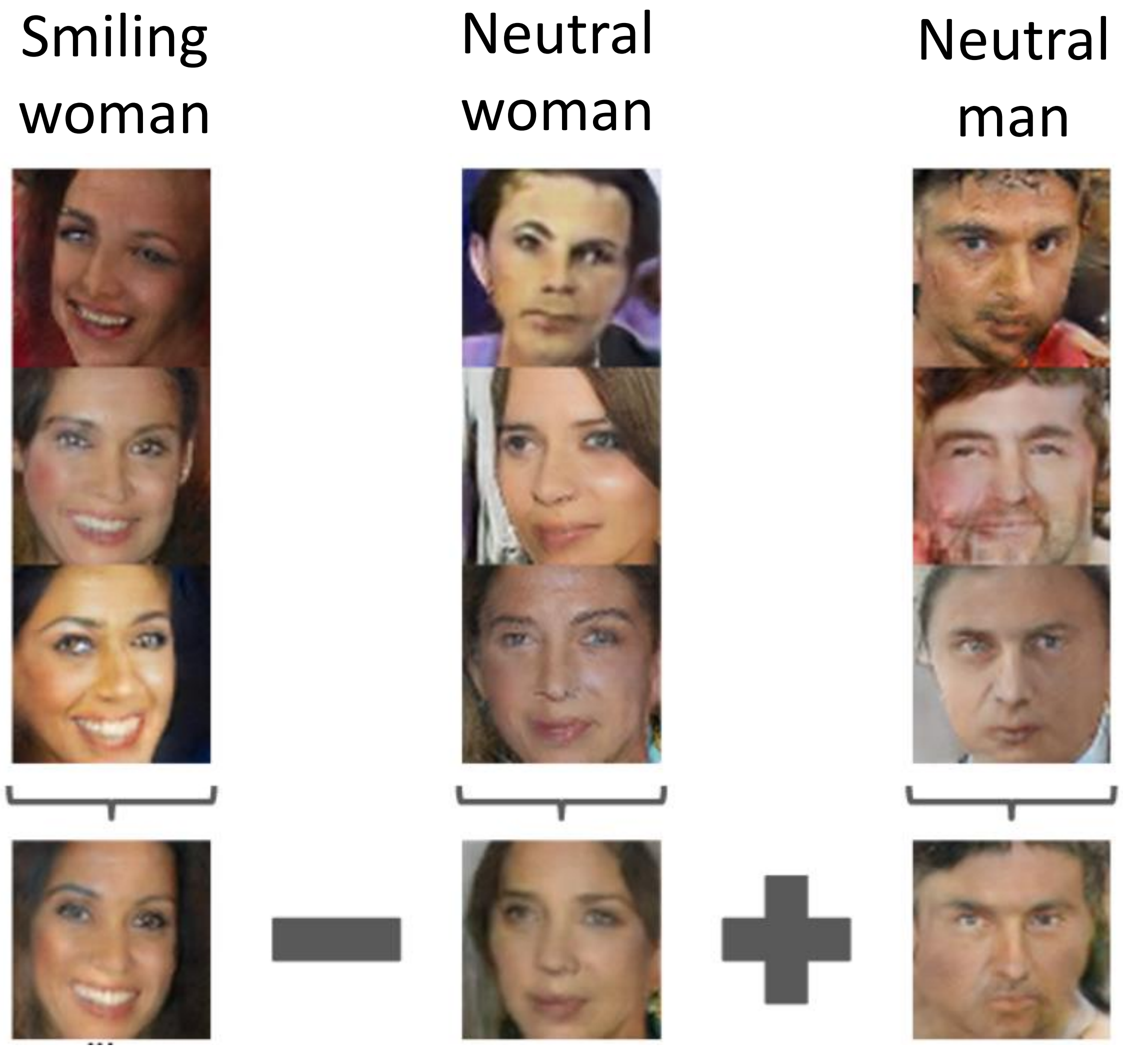


Radford et al, ICLR 2016

Generative Adversarial Networks: Vector Math

Samples from the model

Average Z vectors, do arithmetic



Radford et al, ICLR 2016

Generative Adversarial Networks: Vector Math

Smiling woman

Neutral woman

Neutral man

Samples from the model

Smiling Man

Average Z vectors, do arithmetic

Radford et al, ICLR 2016

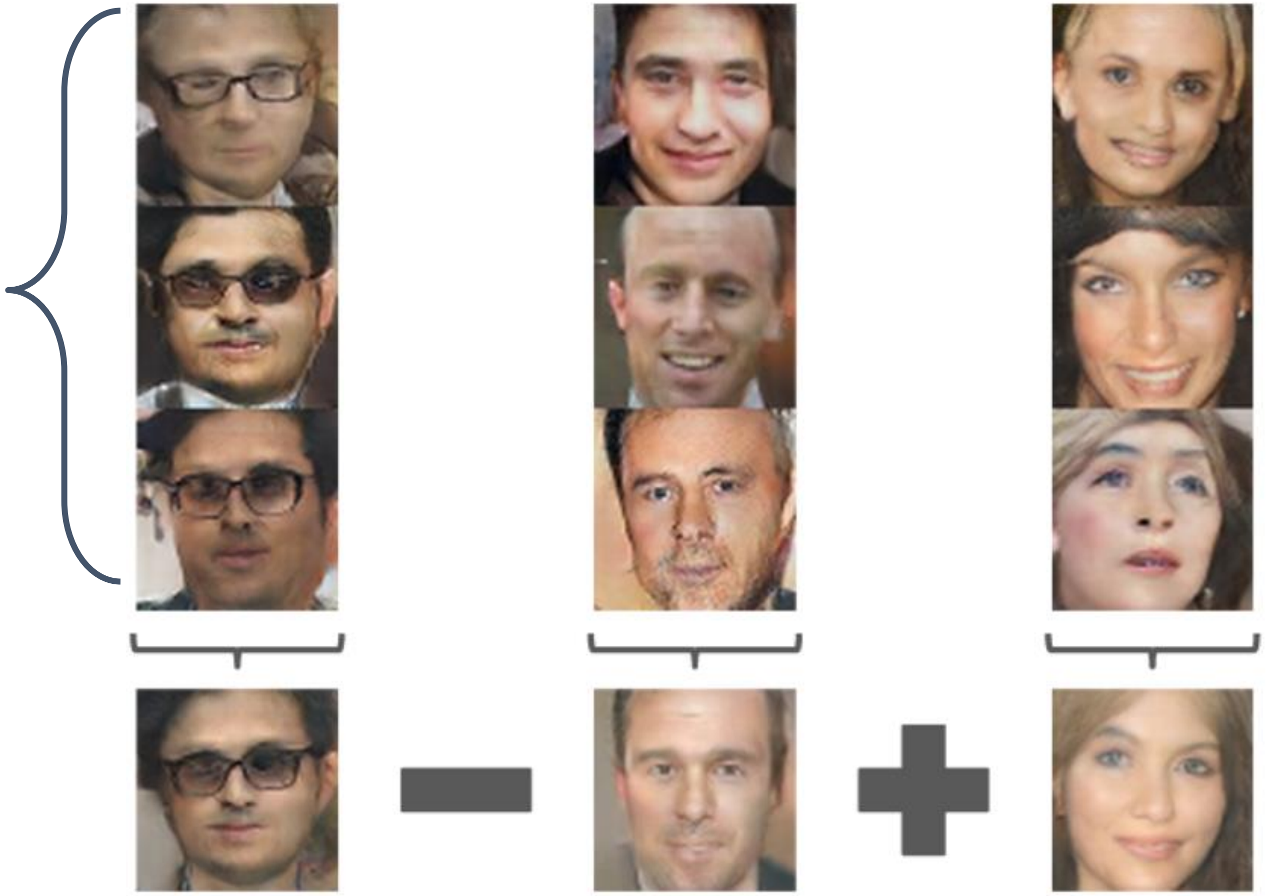
Generative Adversarial Networks: Vector Math

Man with glasses

Man w/o glasses

Woman w/o glasses

Samples from the model



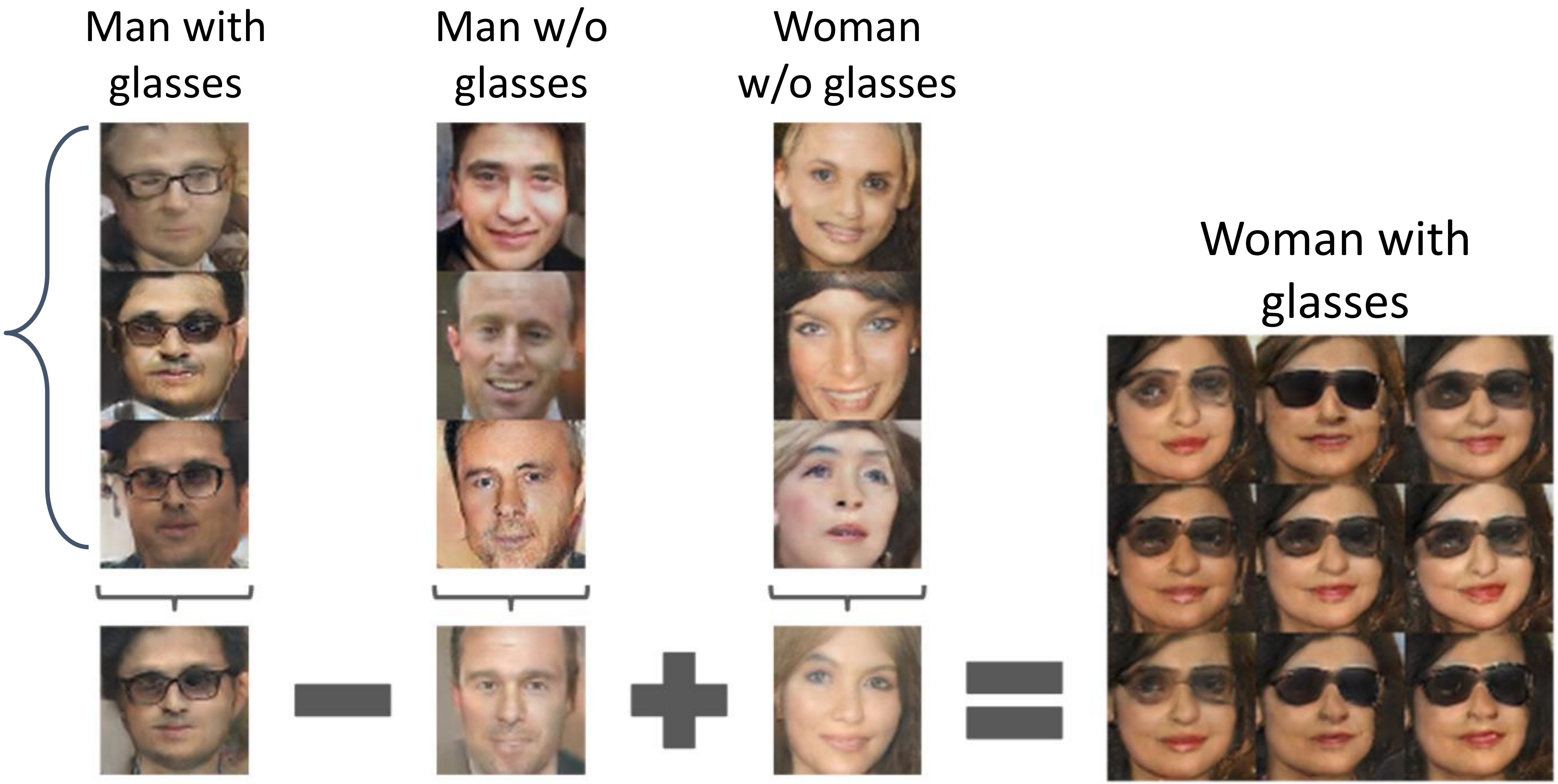
Average Z vectors, do arithmetic

Radford et al, ICLR 2016

Generative Adversarial Networks: Vector Math

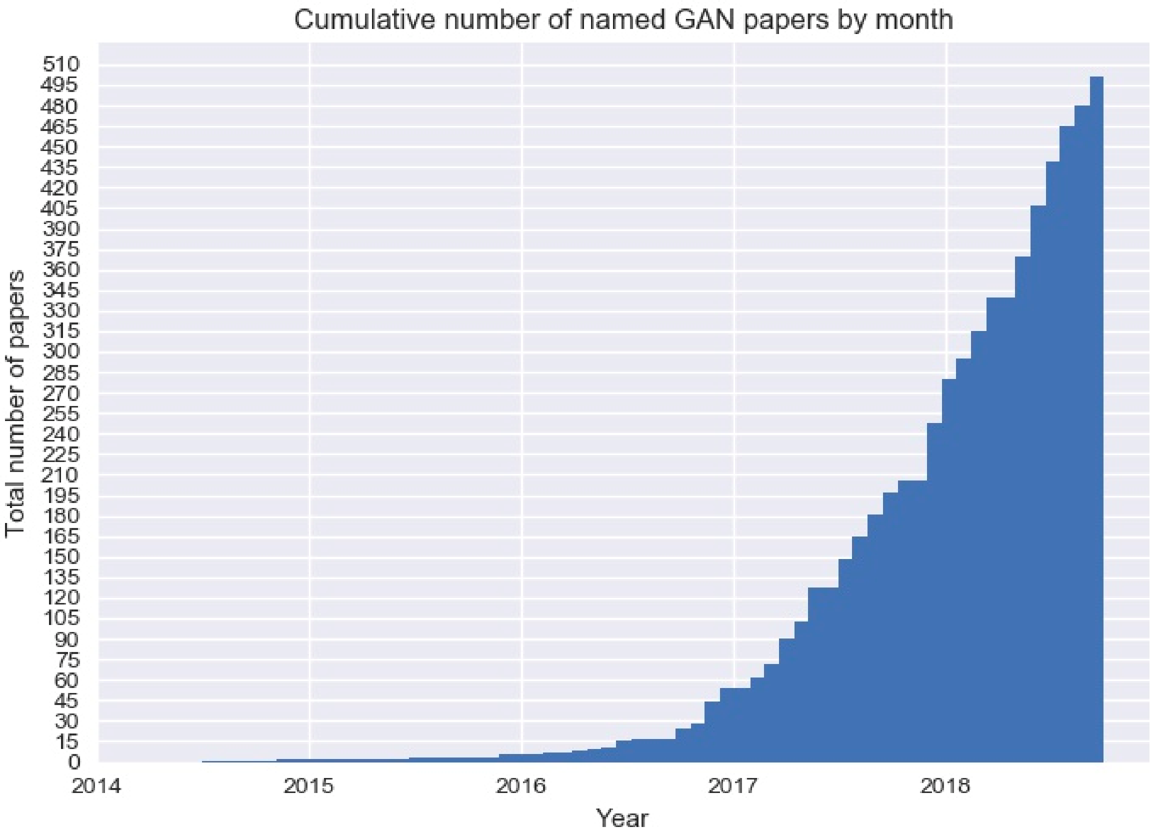
Samples from the model

Average Z vectors, do arithmetic



Radford et al, ICLR 2016

2017 to present: Explosion of GANs

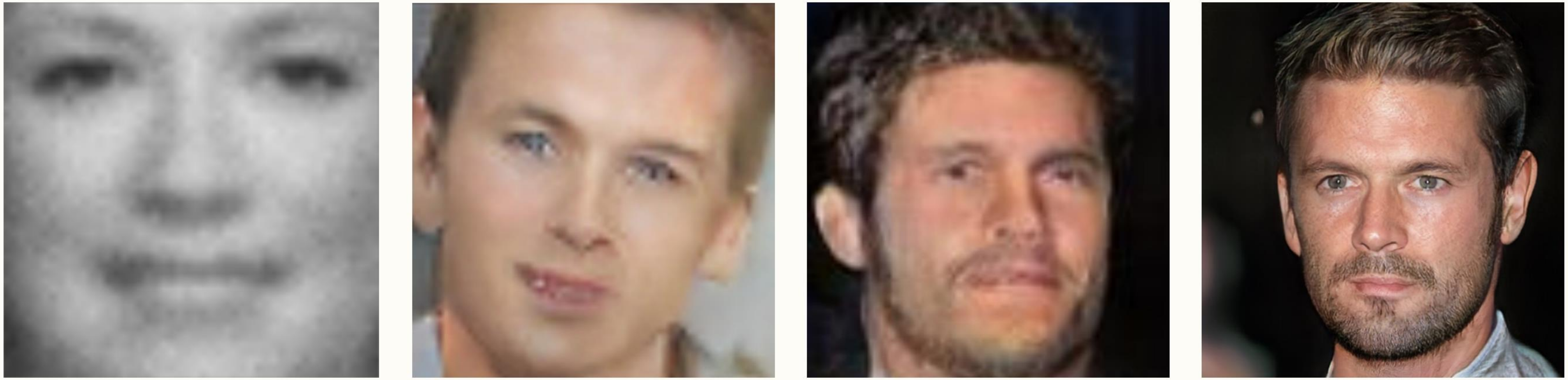


https://github.com/hindupuravinash/the-gan-zoo

- 3D-ED-GAN - Shape Inpainting using 3D Generative Adversarial Network and Recurrent Convolutional Networks
- 3D-GAN - Learning a Probabilistic Latent Space of Object Shapes via 3D Generative-Adversarial Modeling (github)
- 3D-IWGAN - Improved Adversarial Systems for 3D Object Generation and Reconstruction (github)
- 3D-PhysNet - 3D-PhysNet: Learning the Intuitive Physics of Non-Rigid Object Deformations
- 3D-RecGAN - 3D Object Reconstruction from a Single Depth View with Adversarial Learning (github)
- ABC-GAN - ABC-GAN: Adaptive Blur and Control for improved training stability of Generative Adversarial Networks (github)
- ABC-GAN - GANs for LIFE: Generative Adversarial Networks for Likelihood Free Inference
- AC-GAN - Conditional Image Synthesis With Auxiliary Classifier GANs
- acGAN - Face Aging With Conditional Generative Adversarial Networks
- ACGAN - Coverless Information Hiding Based on Generative adversarial networks
- acGAN - On-line Adaptative Curriculum Learning for GANs
- ACtuAL - ACtuAL: Actor-Critic Under Adversarial Learning
- AdaGAN - AdaGAN: Boosting Generative Models
- Adaptive GAN - Customizing an Adversarial Example Generator with Class-Conditional GANs
- AdvEntuRe - AdvEntuRe: Adversarial Training for Textual Entailment with Knowledge-Guided Examples
- AdvGAN - Generating adversarial examples with adversarial networks
- AE-GAN - AE-GAN: adversarial eliminating with GAN
- AE-OT - Latent Space Optimal Transport for Generative Models
- AEGAN - Learning Inverse Mapping by Autoencoder based Generative Adversarial Nets
- AF-DCGAN - AF-DCGAN: Amplitude Feature Deep Convolutional GAN for Fingerprint Construction in Indoor Localization System
- AffGAN - Amortised MAP Inference for Image Super-resolution
- AIM - Generating Informative and Diverse Conversational Responses via Adversarial Information Maximization
- AL-CGAN - Learning to Generate Images of Outdoor Scenes from Attributes and Semantic Layouts
- ALI - Adversarially Learned Inference (github)
- AlignGAN - AlignGAN: Learning to Align Cross-Domain Images with Conditional Generative Adversarial Networks
- AlphaGAN - AlphaGAN: Generative adversarial networks for natural image matting
- AM-GAN - Activation Maximization Generative Adversarial Nets
- AmbientGAN - AmbientGAN: Generative models from lossy measurements (github)
- AMC-GAN - Video Prediction with Appearance and Motion Conditions
- AnoGAN - Unsupervised Anomaly Detection with Generative Adversarial Networks to Guide Marker Discovery
- APD - Adversarial Distillation of Bayesian Neural Network Posteriors
- APE-GAN - APE-GAN: Adversarial Perturbation Elimination with GAN
- ARAE - Adversarially Regularized Autoencoders for Generating Discrete Structures (github)
- ARDA - Adversarial Representation Learning for Domain Adaptation
- ARIGAN - ARIGAN: Synthetic Arabidopsis Plants using Generative Adversarial Network
- ArtGAN - ArtGAN: Artwork Synthesis with Conditional Categorial GANs
- ASDL-GAN - Automatic Steganographic Distortion Learning Using a Generative Adversarial Network
- ATA-GAN - Attention-Aware Generative Adversarial Networks (ATA-GANs)
- Attention-GAN - Attention-GAN for Object Transfiguration in Wild Images
- AttGAN - Arbitrary Facial Attribute Editing: Only Change What You Want (github)
- AttnGAN - AttnGAN: Fine-Grained Text to Image Generation with Attentional Generative Adversarial Networks (github)
- AVID - AVID: Adversarial Visual Irregularity Detection
- B-DCGAN - B-DCGAN:Evaluation of Binarized DCGAN for FPGA
- b-GAN - Generative Adversarial Nets from a Density Ratio Estimation Perspective
- BAGAN - BAGAN: Data Augmentation with Balancing GAN
- Bayesian GAN - Deep and Hierarchical Implicit Models
- Bayesian GAN - Bayesian GAN (github)
- BCGAN - Bayesian Conditional Generative Adverserial Networks
- BCGAN - Bidirectional Conditional Generative Adversarial networks
- BEAM - Boltzmann Encoded Adversarial Machines
- BEGAN - BEGAN: Boundary Equilibrium Generative Adversarial Networks
- BEGAN-CS - Escaping from Collapsing Modes in a Constrained Space
- Bellman GAN - Distributional Multivariate Policy Evaluation and Exploration with the Bellman

- BGAN - Binary Generative Adversarial Networks for Image Retrieval (github)
- Bi-GAN - Autonomously and Simultaneously Refining Deep Neural Network Parameters by a Bi-Generative Adversarial Network Aided Genetic Algorithm
- BicycleGAN - Toward Multimodal Image-to-Image Translation (github)
- BiGAN - Adversarial Feature Learning
- BinGAN - BinGAN: Learning Compact Binary Descriptors with a Regularized GAN
- BourGAN - BourGAN: Generative Networks with Metric Embeddings
- BranchGAN - Branched Generative Adversarial Networks for Multi-Scale Image Manifold Learning
- BRE - Improving GAN Training via Binarized Representation Entropy (BRE) Regularization (github)
- BridgeGAN - Generative Adversarial Frontal View to Bird View Synthesis
- BS-GAN - Boundary-Seeking Generative Adversarial Networks
- BubGAN - BubGAN: Bubble Generative Adversarial Networks for Synthesizing Realistic Bubbly Flow Images
- BWGAN - Banach Wasserstein GAN
- C-GAN - Face Aging with Contextual Generative Adversarial Nets
- C-RNN-GAN - C-RNN-GAN: Continuous recurrent neural networks with adversarial training (github)
- CA-GAN - Composition-aided Sketch-realistic Portrait Generation
- CaloGAN - CaloGAN: Simulating 3D High Energy Particle Showers in Multi-Layer Electromagnetic Calorimeters with Generative Adversarial Networks (github)
- CAN - CAN: Creative Adversarial Networks, Generating Art by Learning About Styles and Deviating from Style Norms
- CapsGAN - CapsGAN: Using Dynamic Routing for Generative Adversarial Networks
- CapsuleGAN - CapsuleGAN: Generative Adversarial Capsule Network
- CatGAN - Unsupervised and Semi-supervised Learning with Categorical Generative Adversarial Networks
- CatGAN - CatGAN: Coupled Adversarial Transfer for Domain Generation
- CausalGAN - CausalGAN: Learning Causal Implicit Generative Models with Adversarial Training
- CC-GAN - Semi-Supervised Learning with Context-Conditional Generative Adversarial Networks (github)
- cd-GAN - Conditional Image-to-Image Translation
- CDcGAN - Simultaneously Color-Depth Super-Resolution with Conditional Generative Adversarial Network
- CE-GAN - Deep Learning for Imbalance Data Classification using Class Expert Generative Adversarial Network
- CFG-GAN - Composite Functional Gradient Learning of Generative Adversarial Models
- CGAN - Conditional Generative Adversarial Nets
- CGAN - Controllable Generative Adversarial Network
- Chekhov GAN - An Online Learning Approach to Generative Adversarial Networks
- ciGAN - Conditional Infilling GANs for Data Augmentation in Mammogram Classification
- CinCGAN - Unsupervised Image Super-Resolution using Cycle-in-Cycle Generative Adversarial Networks
- CipherGAN - Unsupervised Cipher Cracking Using Discrete GANs
- ClusterGAN - ClusterGAN : Latent Space Clustering in Generative Adversarial Networks
- CM-GAN - CM-GANs: Cross-modal Generative Adversarial Networks for Common Representation Learning
- CoAtt-GAN - Are You Talking to Me? Reasoned Visual Dialog Generation through Adversarial Learning
- CoGAN - Coupled Generative Adversarial Networks
- ComboGAN - ComboGAN: Unrestrained Scalability for Image Domain Translation (github)
- ConceptGAN - Learning Compositional Visual Concepts with Mutual Consistency
- Conditional cycleGAN - Conditional CycleGAN for Attribute Guided Face Image Generation
- constrast-GAN - Generative Semantic Manipulation with Contrasting GAN
- Context-RNN-GAN - Contextual RNN-GANs for Abstract Reasoning Diagram Generation
- CorrGAN - Correlated discrete data generation using adversarial training
- Coulomb GAN - Coulomb GANs: Provably Optimal Nash Equilibria via Potential Fields
- Cover-GAN - Generative Steganography with Kerckhoffs' Principle based on Generative Adversarial Networks
- cowboy - Defending Against Adversarial Attacks by Leveraging an Entire GAN
- CR-GAN - CR-GAN: Learning Complete Representations for Multi-view Generation
- Cramèr GAN - The Cramer Distance as a Solution to Biased Wasserstein Gradients
- Cross-GAN - Crossing Generative Adversarial Networks for Cross-View Person Re-identification
- crVAE-GAN - Channel-Recurrent Variational Autoencoders
- CS-GAN - Improving Neural Machine Translation with Conditional Sequence Generative Adversarial Nets
- CSG - Speech-Driven Expressive Talking Lips with Conditional Sequential Generative Adversarial Networks
- CT-GAN - CT-GAN: Conditional Transformation Generative Adversarial Network for Image Attribute Modification
- CVAE-GAN - CVAE-GAN: Fine-Grained Image Generation through Asymmetric Training
- CycleGAN - Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks

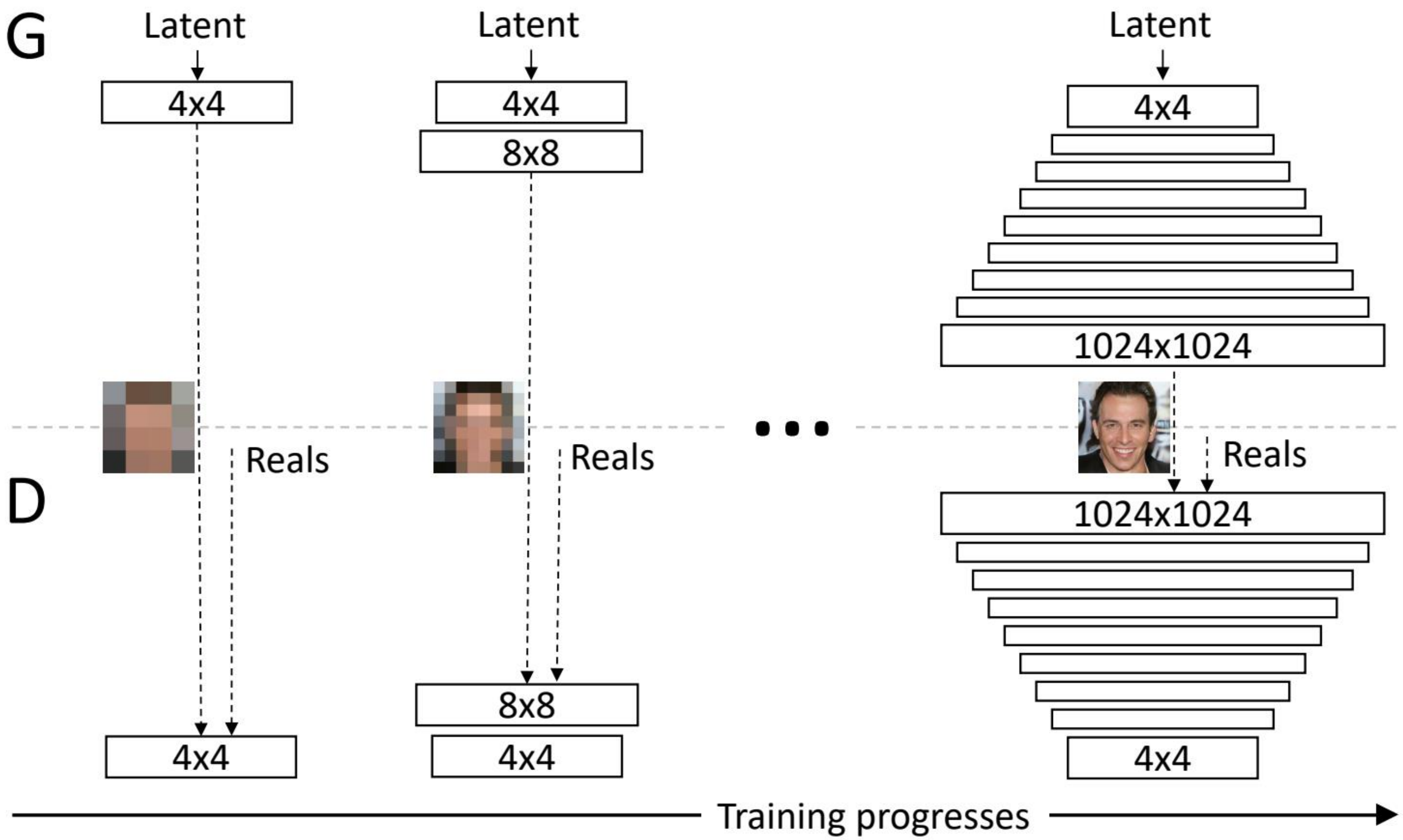
3.5 Years of Progress on Faces

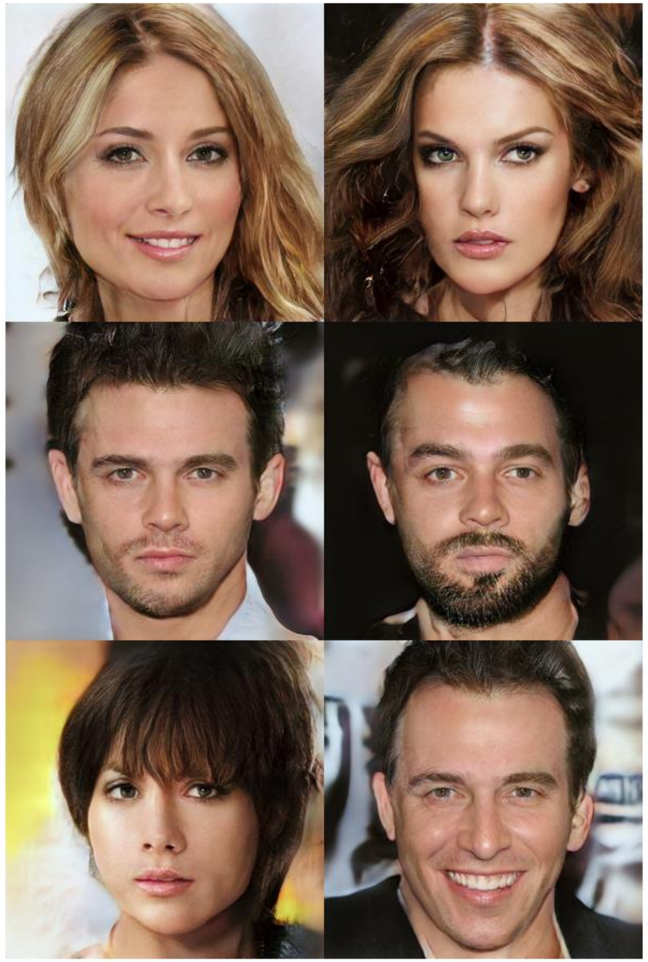


2014 2015 2016 2017

(Brundage et al, 2018)

GAN Improvements: Progressive GAN





Karras et al, "Progressive Growing of GANs for Improved Quality, Stability, and Variation", ICLR 2018

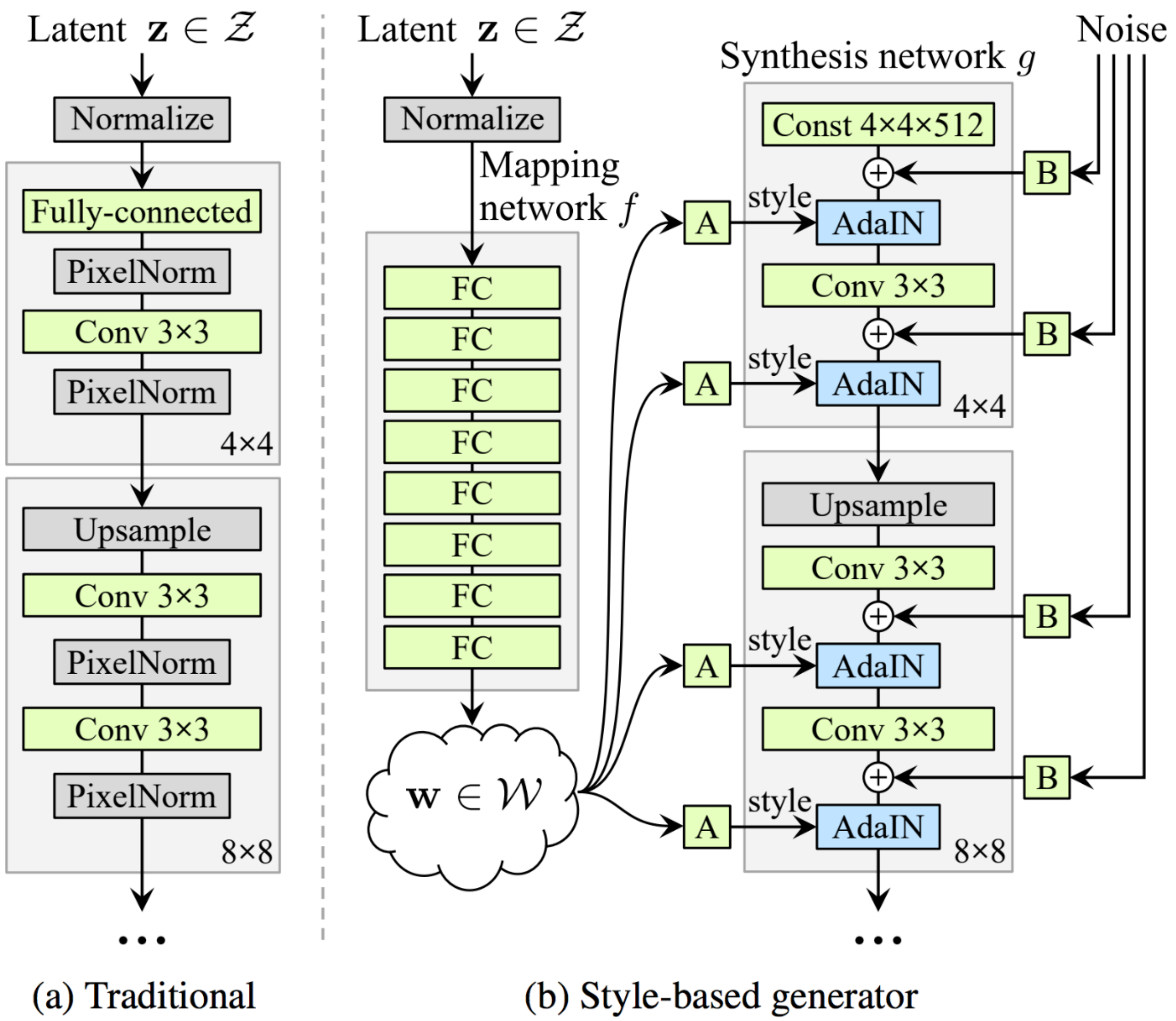
GAN Improvements: Higher Resolution

256 x 256 bedrooms

1024 x 1024 faces

Karras et al, “Progressive Growing of GANs for Improved Quality, Stability, and Variation”, ICLR 2018

GAN Improvements: StyleGAN



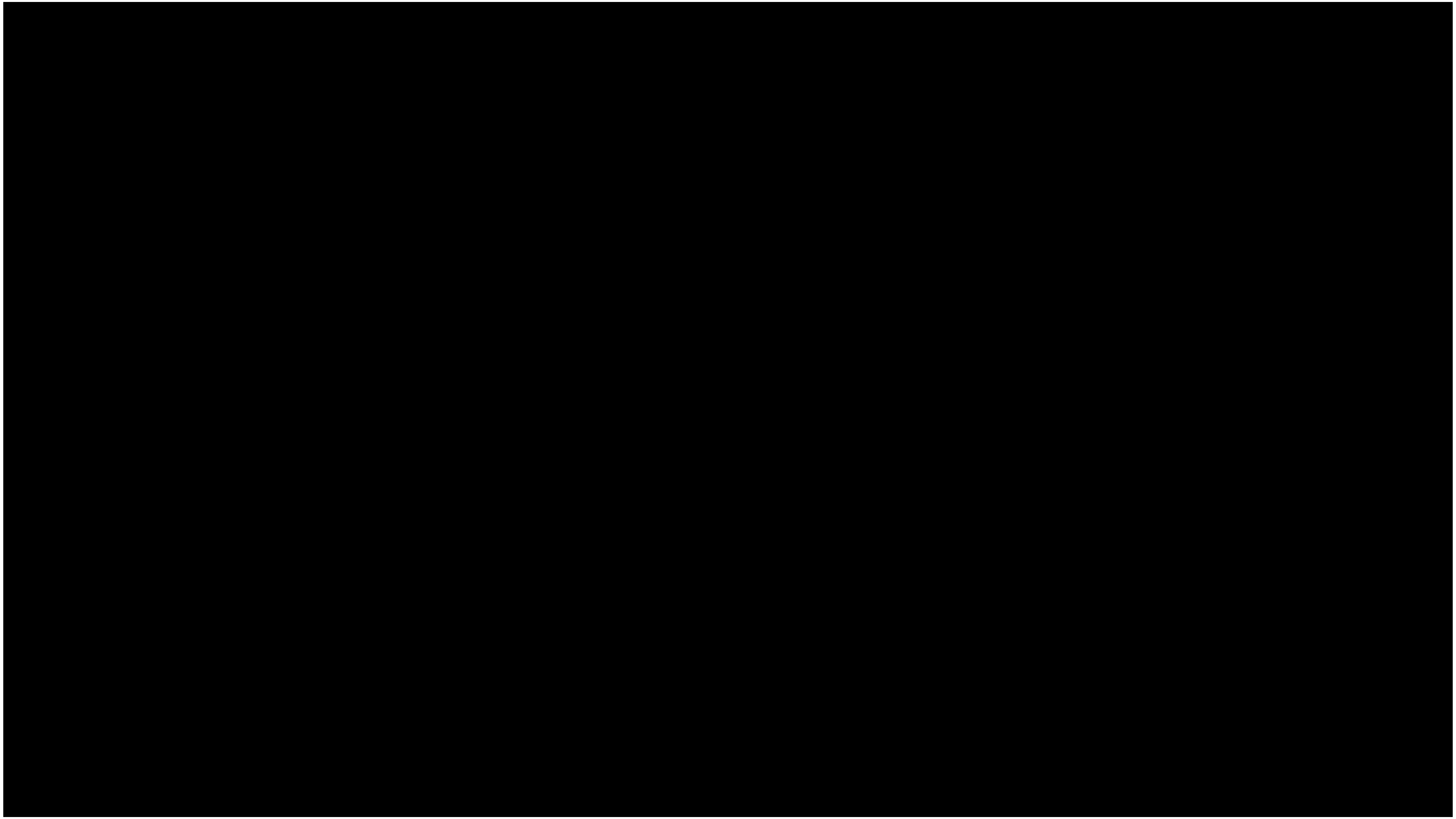
Karras et al, "A Style-Based Generator Architecture for Generative Adversarial Networks", CVPR 2019

GAN Improvements: Higher Resolution

512 x 384 cars

1024 x 1024 faces

Karras et al, "A Style-Based Generator Architecture for Generative Adversarial Networks", CVPR 2019



Karras et al, “A Style-Based Generator Architecture for Generative Adversarial Networks”, CVPR 2019

Problems of GANs

Recall: Generative Adversarial Networks - Optimality

$$\min_{G} \max_{D} \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

$$= \min_{G} (2 * JSD(p_{data}, p_G) - \log 4)$$

Summary: The global minimum of the minimax game happens when:

1. $D_G^*(x) = \frac{p_{data}(x)}{p_{data}(x) + p_G(x)}$ (Optimal discriminator for any G)
2. $p_G(x) = p_{data}(x)$ (Optimal generator for optimal D)

Caveats:

1. G and D are neural nets with fixed architecture. We don't know whether they can actually represent the optimal D and G.
2. This tells us nothing about convergence to the optimal solution

Problems of GANs: Mode Collapse

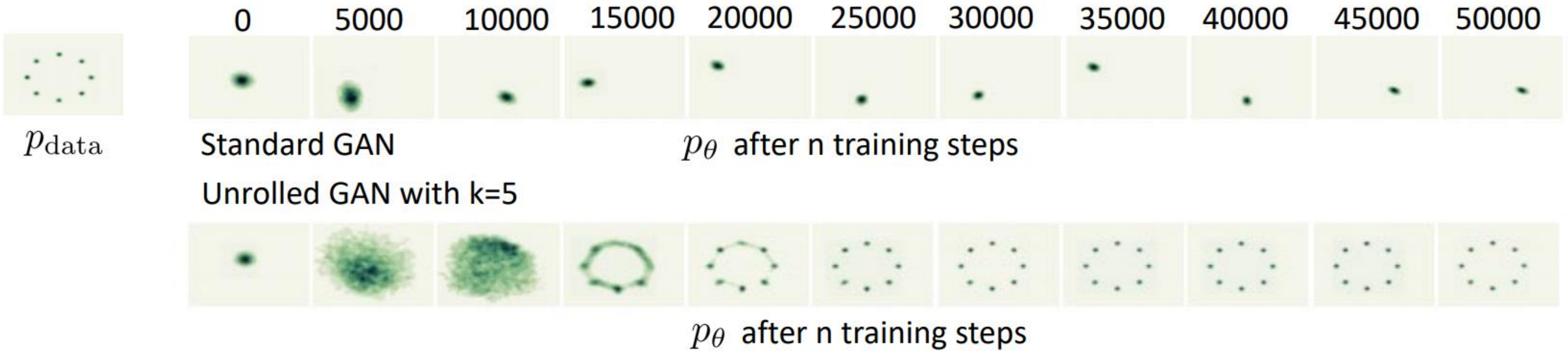
p_θ only covers one or a few modes of p_{data}

p_{data}

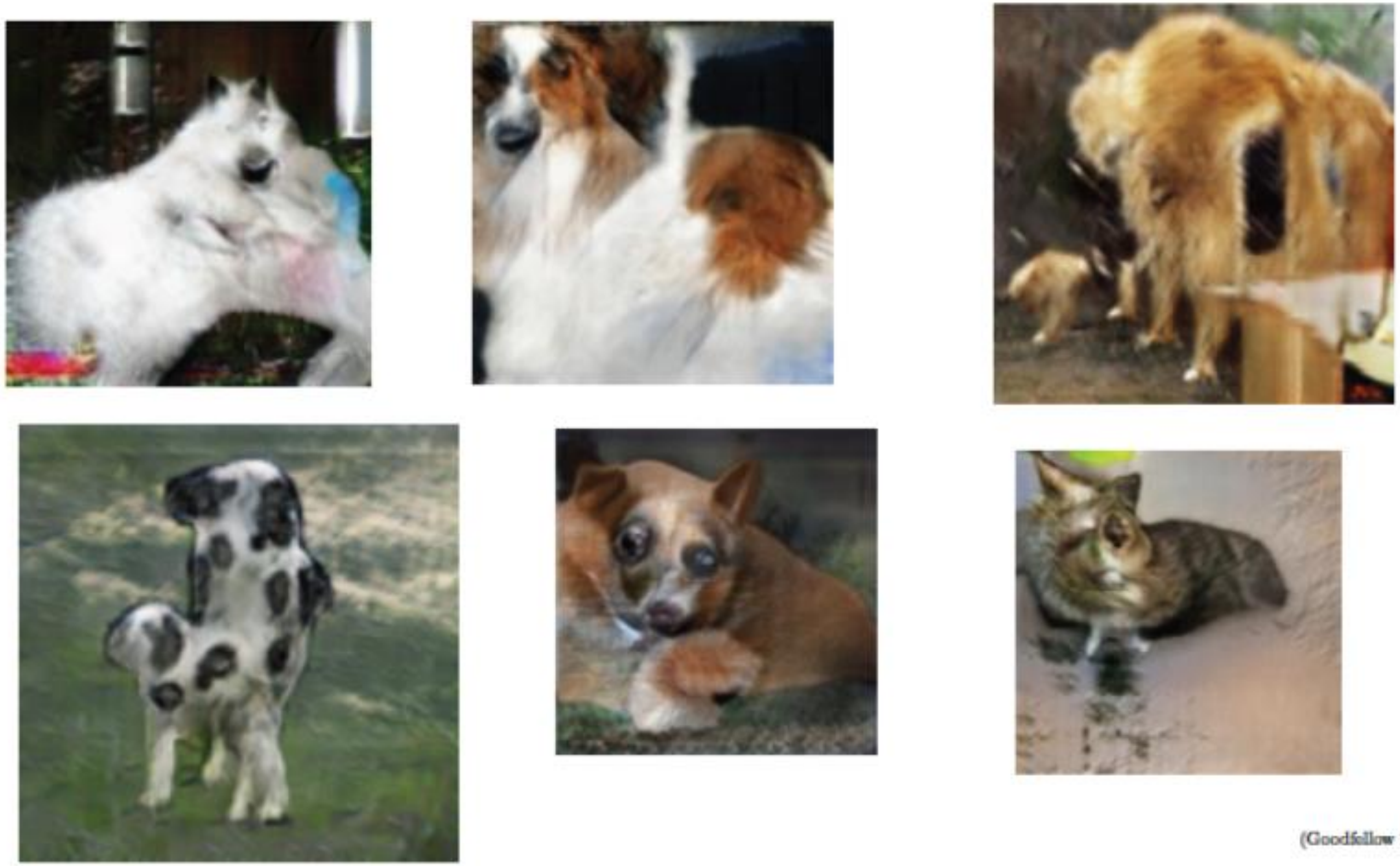
p_θ after n training steps

GAN Improvements: Unrolled GANs

- Unrolled GANs: Take k steps with the discriminator in each iteration, and backpropagate through all of them to update the generator

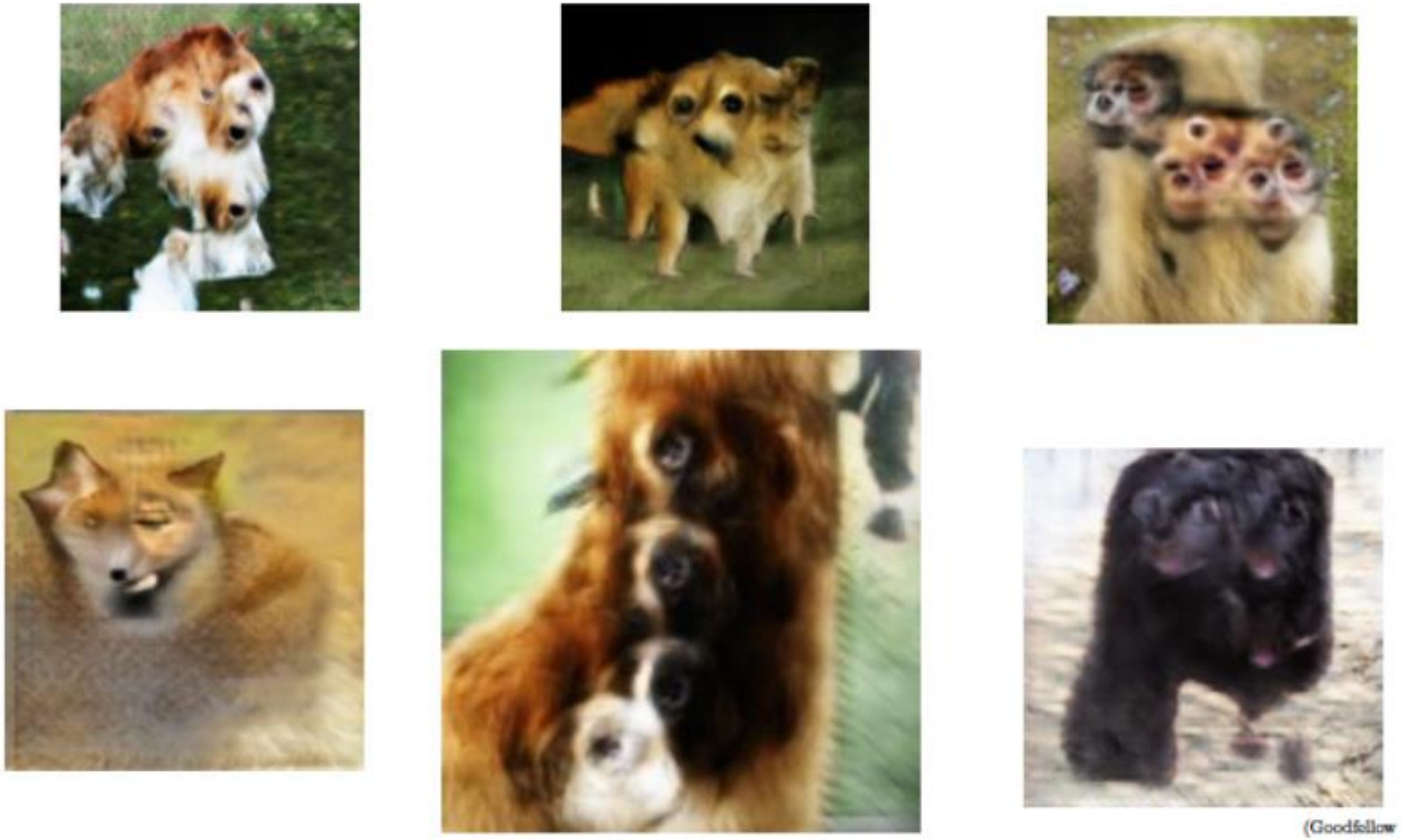


Problems of GANs: Global Structure



(Goodfellow 2016)

Problems of GANs: Multiple Objects



(Goodfellow 2016)

GAN Improvements: Self-Attention

Goldfish

Indigo bunting

Redshank

Saint Bernard

Zhang et al, "Self-Attention Generative Adversarial Networks", ICML 2019

128x128 images on ImageNet

GAN Improvements

Tricks

- Label smoothing
- Historical batches
- ...

New objectives

- EBGAN
- LSGAN
- WGAN
- BEGAN
- fGAN
- ...

Surrogate or auxiliary objective

- UnrolledGAN
- WGAN-GP
- DRAGAN
- ...

Network architecture

- LAPGAN
- ...

GAN Improvements: Wasserstein GAN

GAN: minimize Jensen-Shannon divergence between p_X and $p_{G(Z)}$

$$JS(p_X||p_{G(Z)}) = KL(p_X||\frac{p_X + p_{G(Z)}}{2}) + KL(p_{G(Z)}||\frac{p_X + p_{G(Z)}}{2})$$

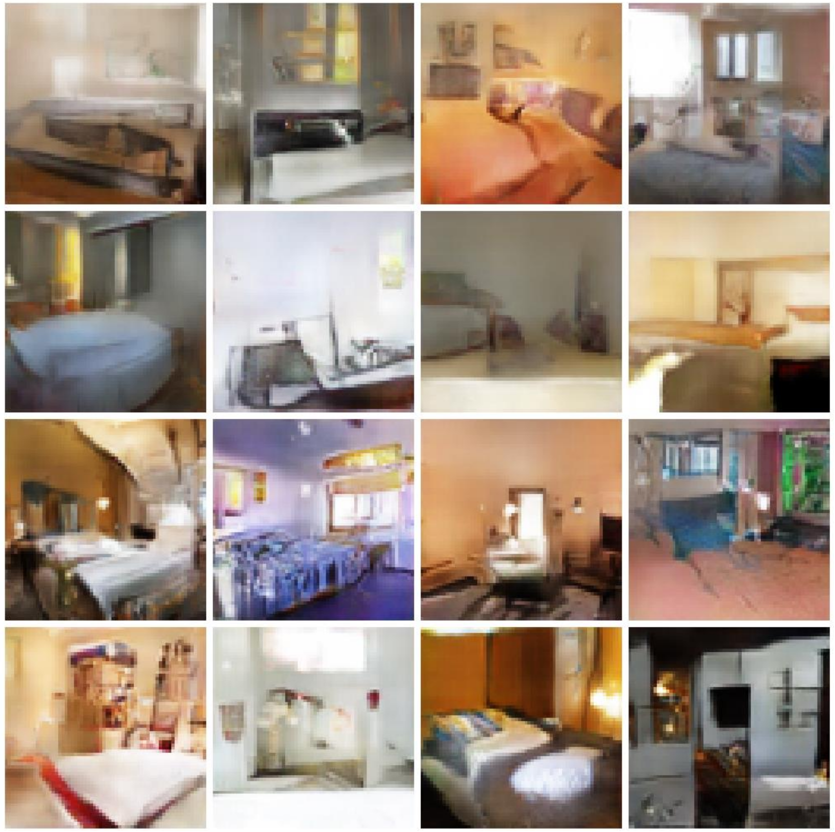
WGAN: minimize earth mover distance between p_X and $p_{G(Z)}$

$$EM(p_X, p_{G(Z)}) = \inf_{\gamma \in \prod(p_X, p_{G(Z)})} E_{(x,y)\sim\gamma}[||x - y||]$$

Arjovsky, Chintala, and Bouttou, “Wasserstein GAN”, 2017

GAN Improvements: Improved Loss Functions

Wasserstein GAN (WGAN)



Arjovsky, Chintala, and Bouttou, “Wasserstein GAN”, 2017

WGAN with Gradient Penalty (WGAN-GP)

Gulrajani et al, “Improved Training of Wasserstein GANs”, NeurIPS 2017

Conditional GAN

Conditional GANs

Recall: Conditional Generative Models learn p(x|y) instead of p(x)
Make generator and discriminator both take label y as an additional input!

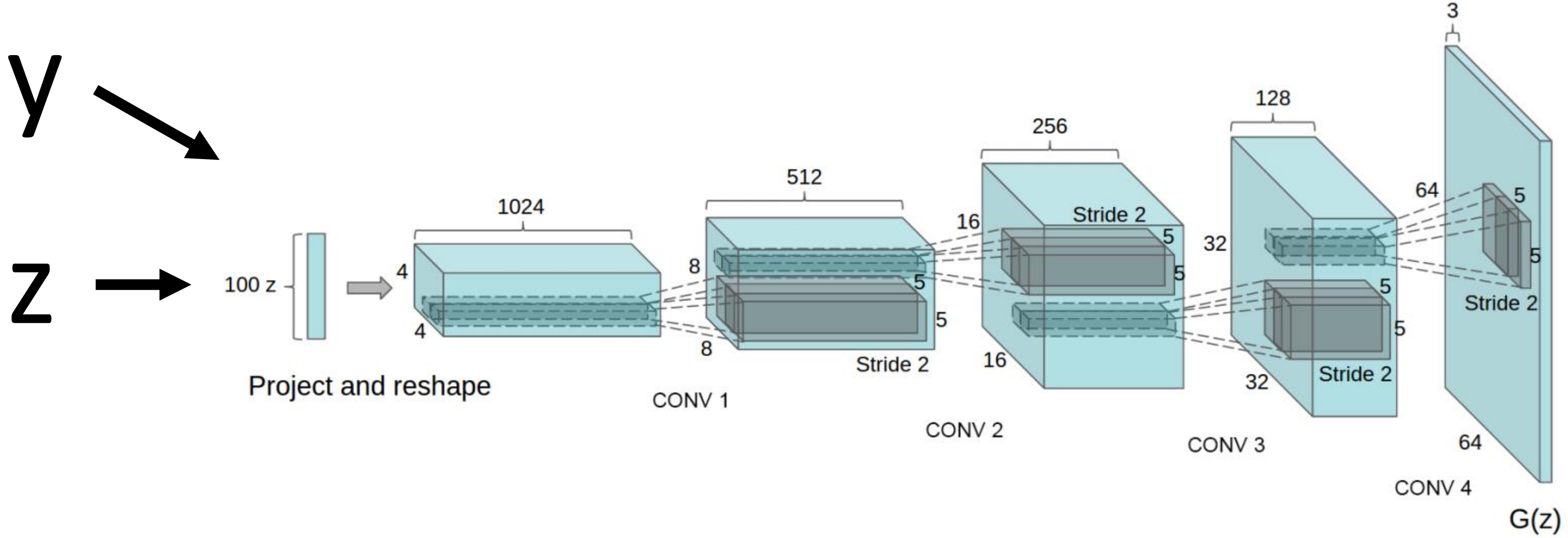


Figure credit: Radford et al, "Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks", ICLR 2016

Conditional GANs: Conditional Batch Normalization

Batch Normalization

$$\mu_j = \frac{1}{N}\sum_{i=1}^{N} x_{i,j}$$

$$\sigma_j^2 = \frac{1}{N}\sum_{i=1}^{N}\left(x_{i,j} - \mu_j\right)^2$$

$$\hat{x}_{i,j} = \frac{x_{i,j} - \mu_j}{\sqrt{\sigma_j^2 + \epsilon}}$$

$$y_{i,j} = \gamma_j \hat{x}_{i,j} + \beta_j$$

Learn a separate scale and shift for each different label y

Conditional Batch Normalization

$$\mu_j = \frac{1}{N}\sum_{i=1}^{N} x_{i,j}$$

$$\sigma_j^2 = \frac{1}{N}\sum_{i=1}^{N}\left(x_{i,j} - \mu_j\right)^2$$

$$\hat{x}_{i,j} = \frac{x_{i,j} - \mu_j}{\sqrt{\sigma_j^2 + \epsilon}}$$

$$y_{i,j} = \boldsymbol{\gamma_j^y} \hat{x}_{i,j} + \boldsymbol{\beta_j^y}$$

Dumoulin et al, "A learned representation for artistic style", ICLR 2017

Conditional GANs: Spectral Normalization

Miyato et al, “Spectral Normalization for Generative Adversarial Networks”, ICLR 2018

128x128 images on ImageNet

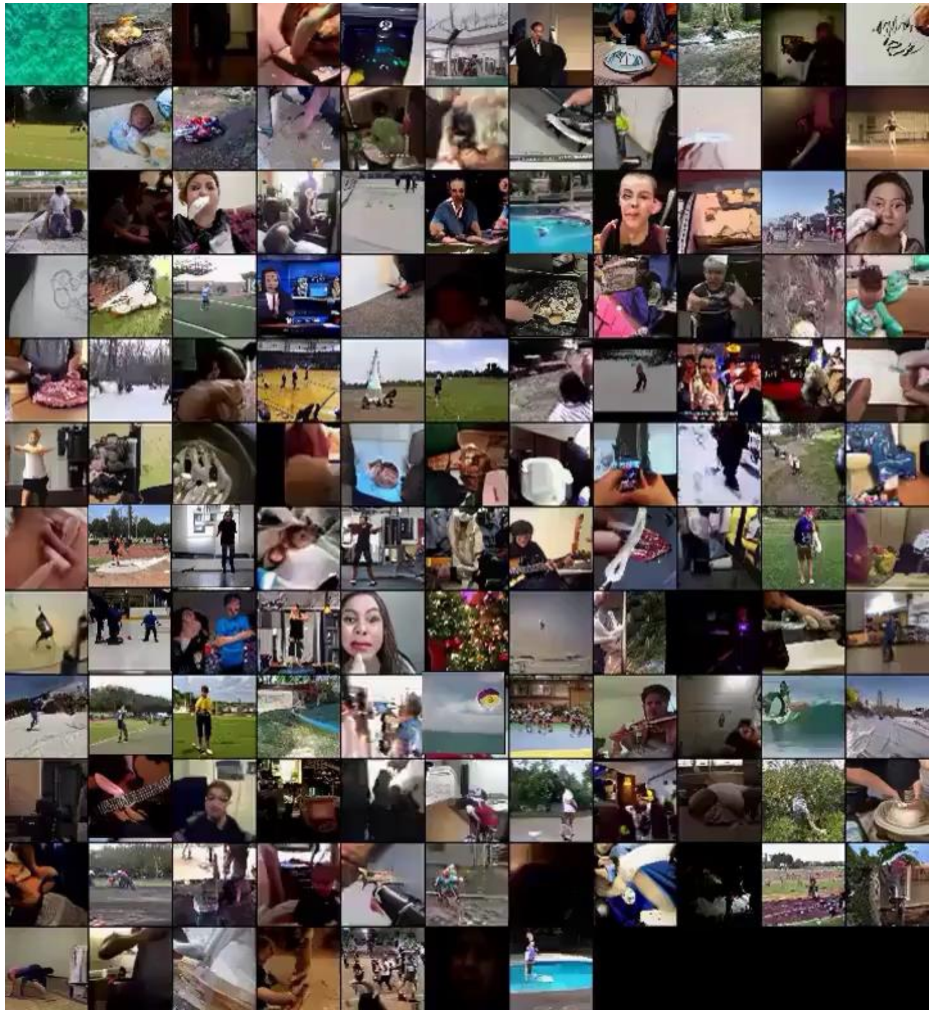
Conditional GANs: BigGAN

Brock et al, “Large Scale GAN Training for High Fidelity Natural Image Synthesis”, ICLR 2019

512x512 images on ImageNet

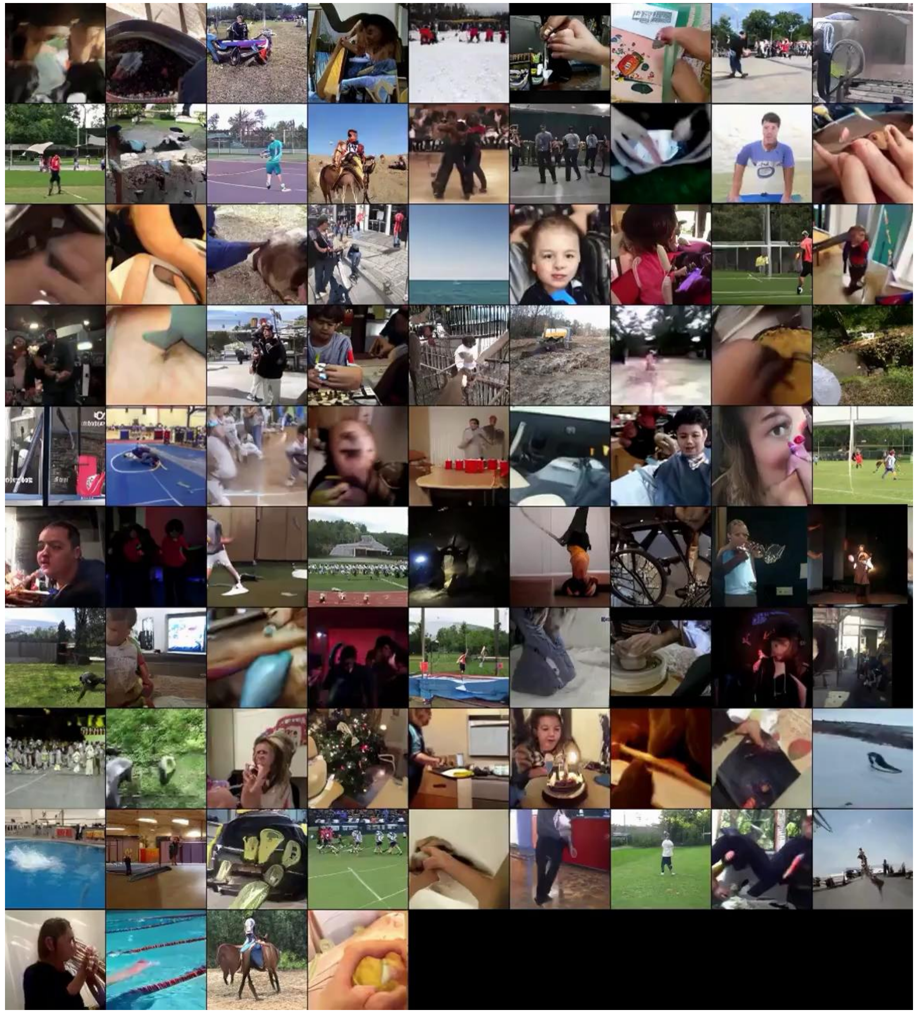
Generating Videos with GANs

Clark et al, “Adversarial Video Generation on Complex Datasets”, arXiv 2019



64x64 images, 48 frames

https://drive.google.com/file/d/1FjOQYdUuxPXvS8yeOhXdPQMapUQaklLi/view



128x128 images, 12 frames

https://drive.google.com/file/d/165Yxuvvu3viOy-39LhhSDGtczbWphj_i/view

Conditioning on more than labels! Text to Image

This bird is red and brown in color, with a stubby beak

The bird is short and stubby with yellow on its body

A bird with a medium orange bill white body gray wings and webbed feet

This small black bird has a short, slightly curved bill and long legs

A picture of a very clean living room

A group of people on skis stand in the snow

Eggs fruit candy nuts and meat served on white dish

A street sign on a stoplight pole in the middle of a day

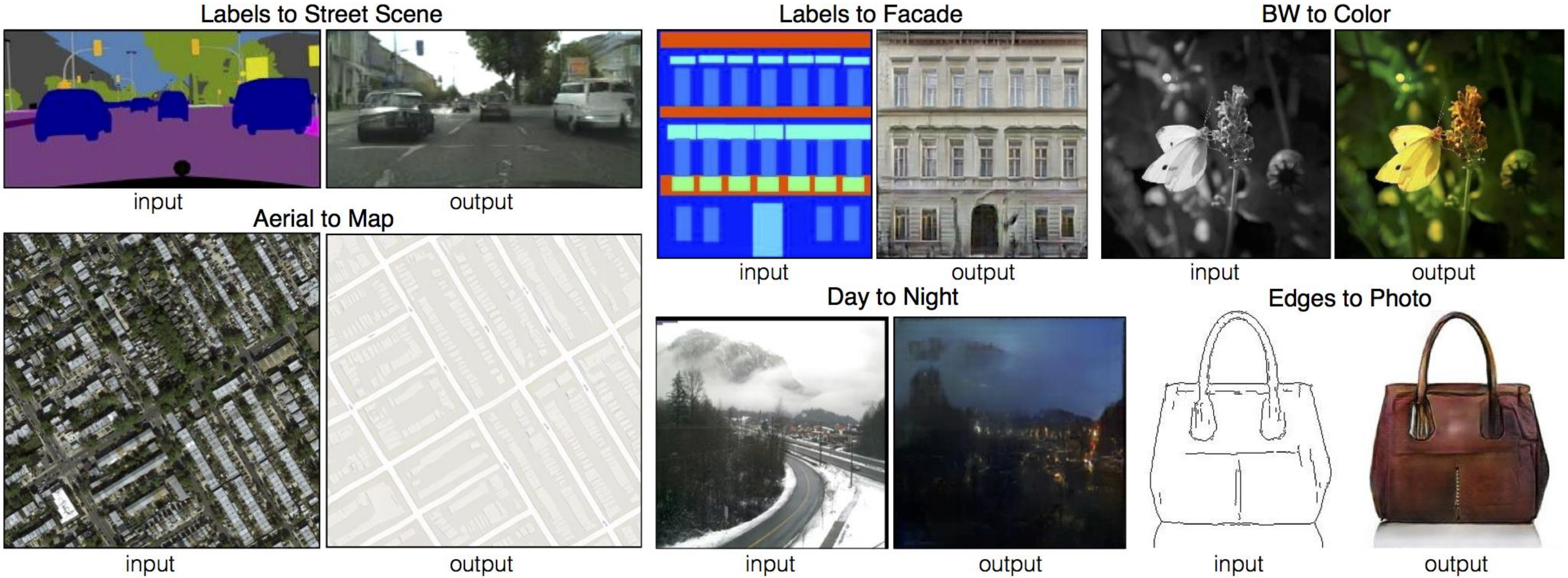
Zhang et al, “StackGAN++: Realistic Image Synthesis with Stacked Generative Adversarial Networks.”, TPAMI 2018
Zhang et al, “StackGAN: Text to Photo-realistic Image Synthesis with Stacked Generative Adversarial Networks.”, ICCV 2017
Reed et al, “Generative Adversarial Text-to-Image Synthesis”, ICML 2016

Image Super-Resolution: Low-Res to High-Res

bicubic	SRResNet	SRGAN	original
(21.59dB/0.6423)	(23.53dB/0.7832)	(21.15dB/0.6868)	

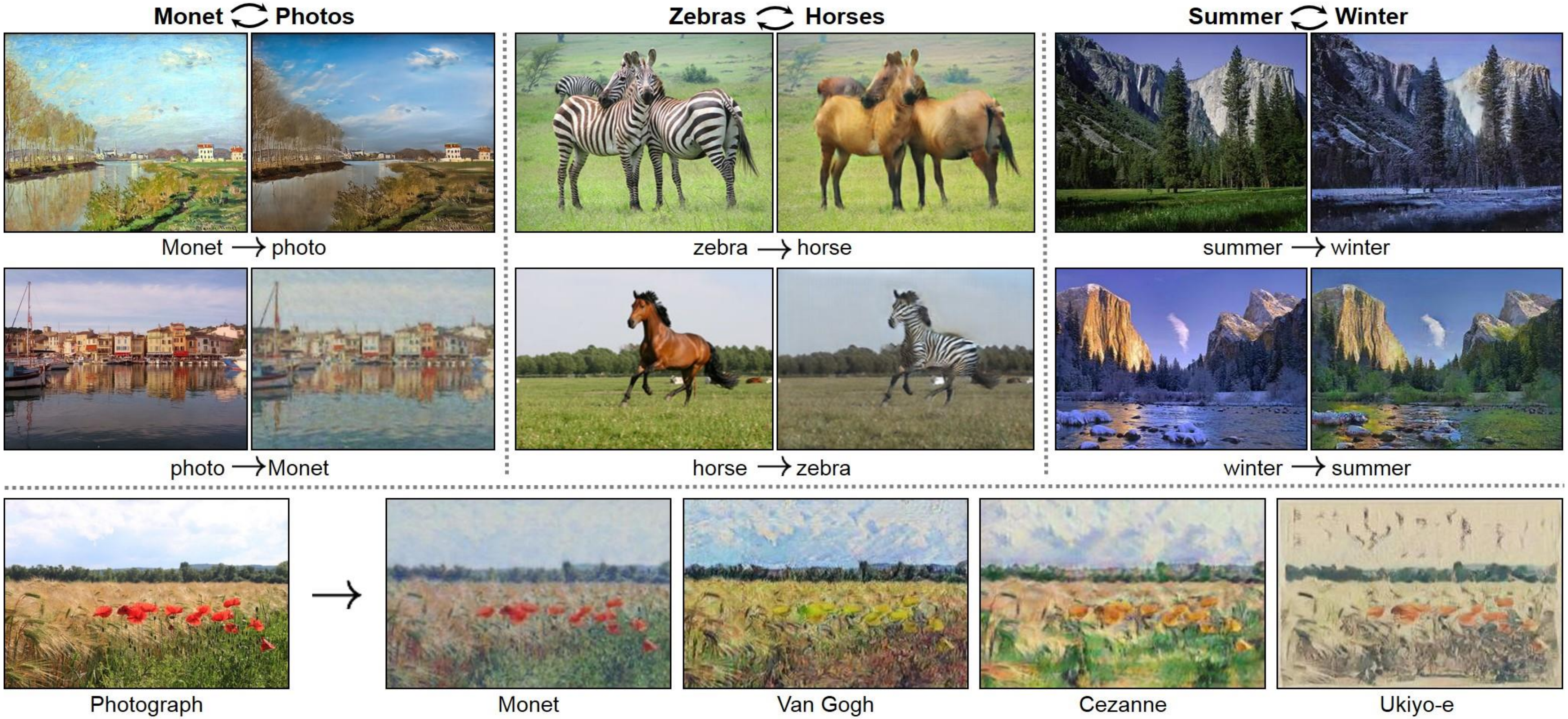
Ledig et al, “Photo-Realistic Single Image Super-Resolution Using a Generative Adversarial Network”, CVPR 2017

Image-to-Image Translation: Pix2Pix



Isola et al, "Image-to-Image Translation with Conditional Adversarial Nets", CVPR 2017

Unpaired Image-to-Image Translation: CycleGAN



Zhu et al, "Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks", ICCV 2017

Unpaired Image-to-Image Translation: CycleGAN

Input Video: Horse

Output Video: Zebra

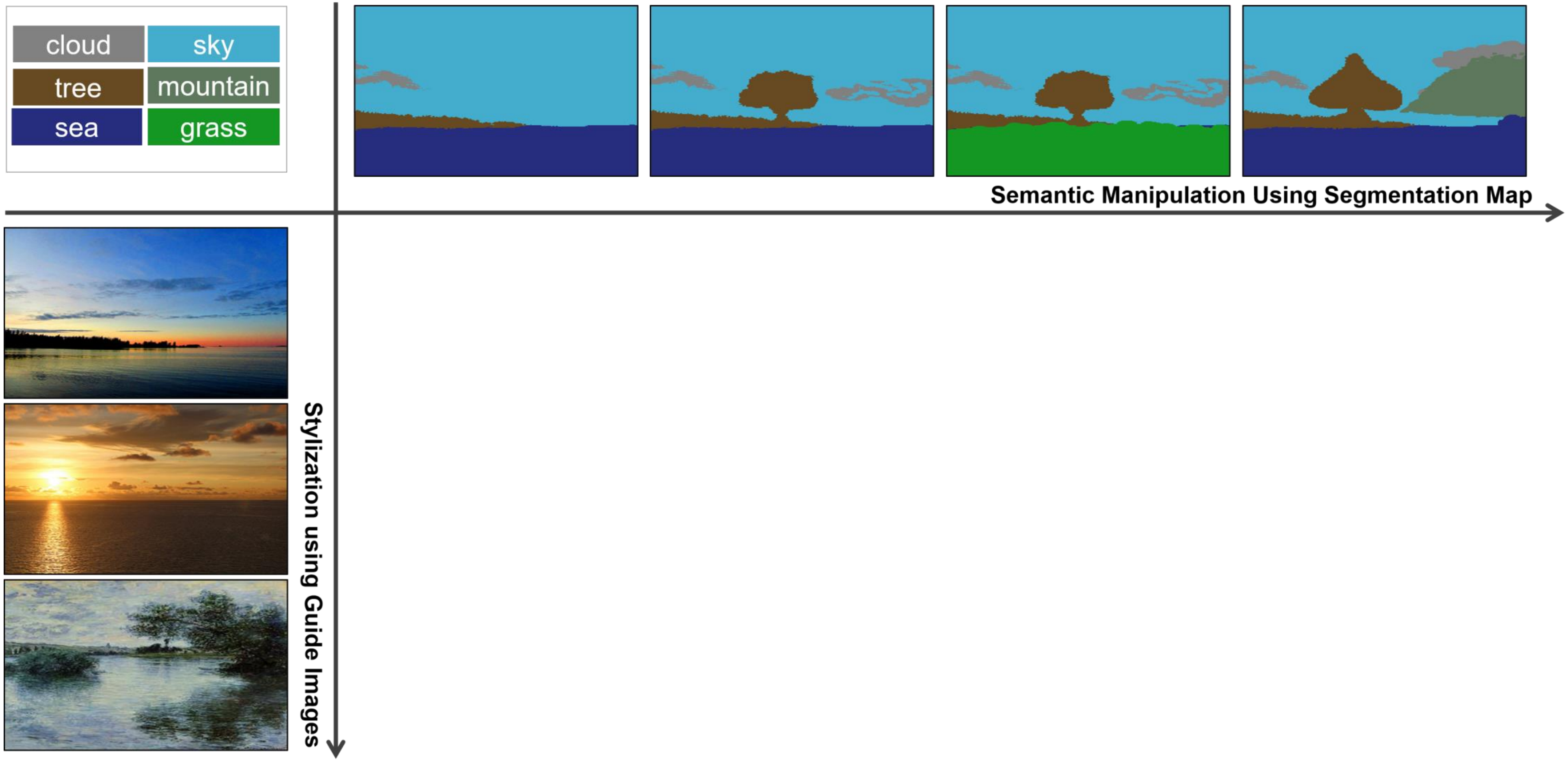
https://www.youtube.com/watch?v=9reHvktowLY

Zhu et al, "Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks", ICCV 2017

Label Map to Image

Input: Label Map

Input: Style Image

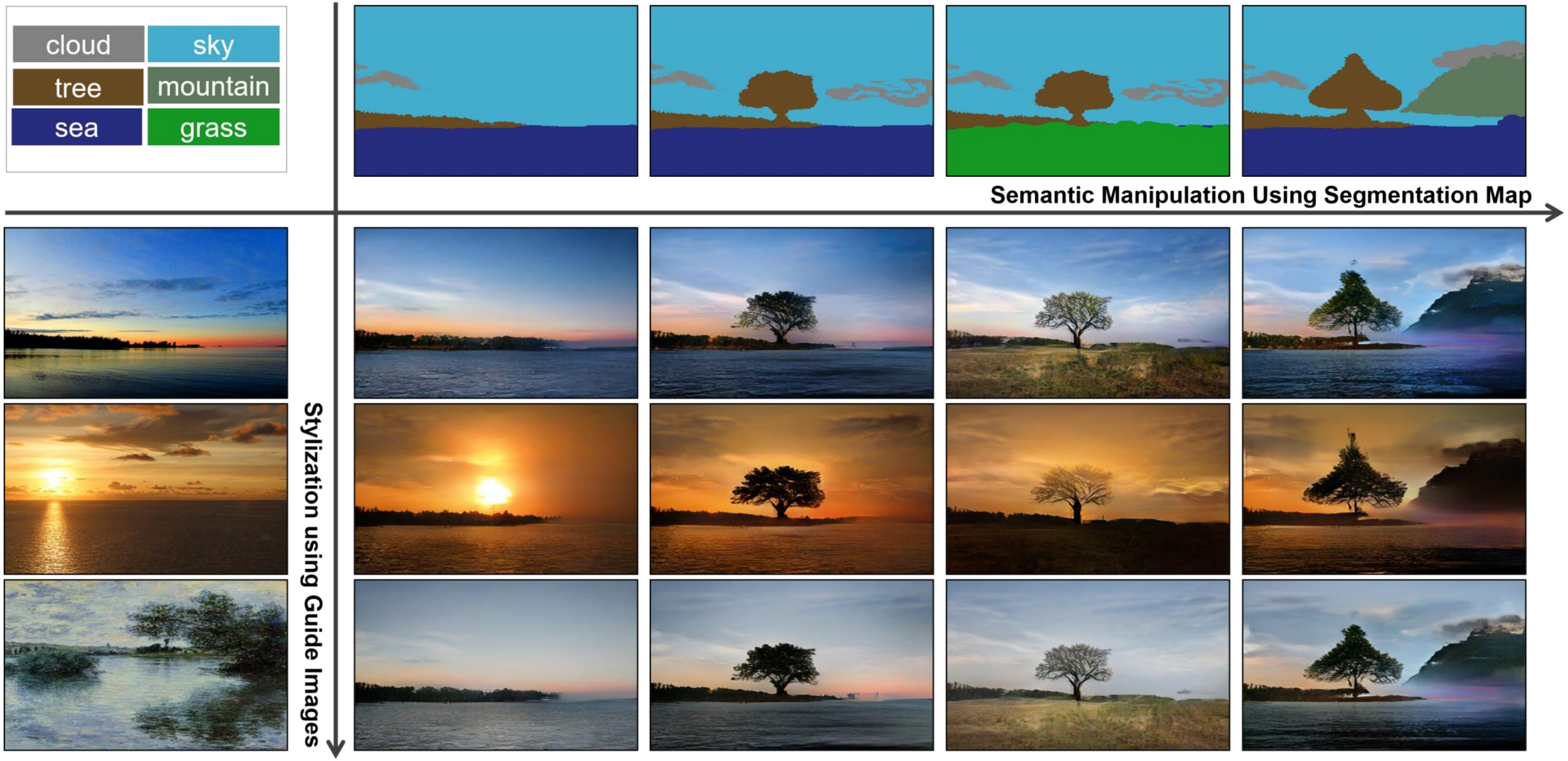


Park et al, “Semantic Image Synthesis with Spatially-Adaptive Normalization”, CVPR 2019

Label Map to Image

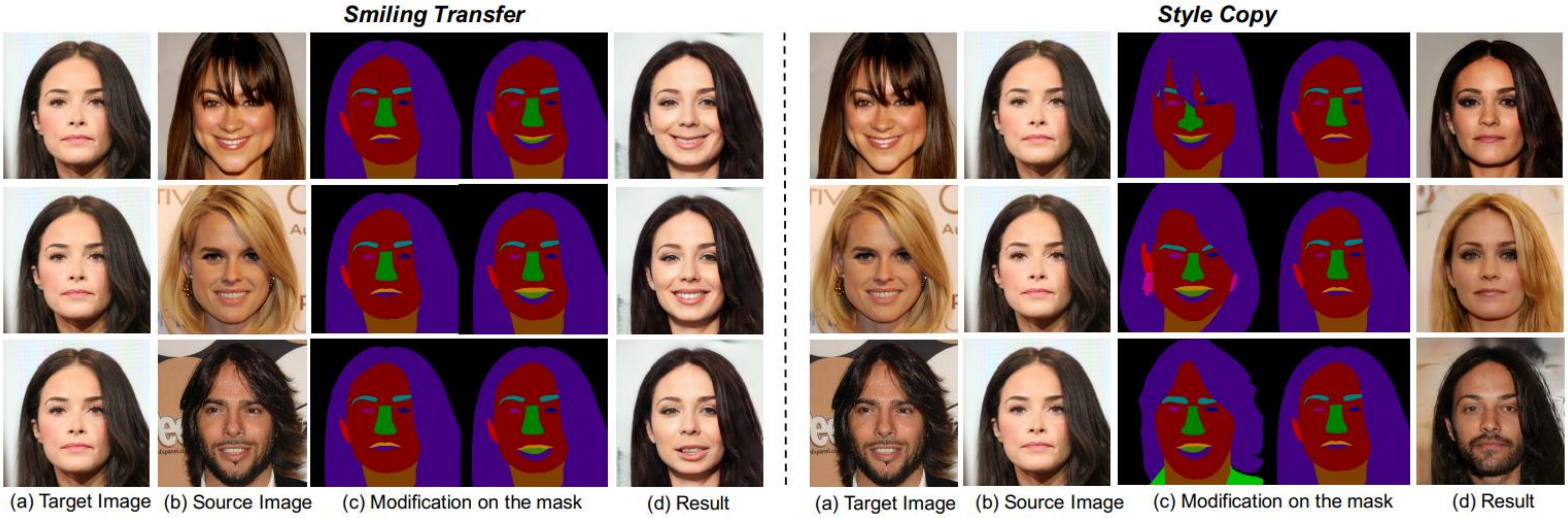
Input: Label Map

Input: Style Image



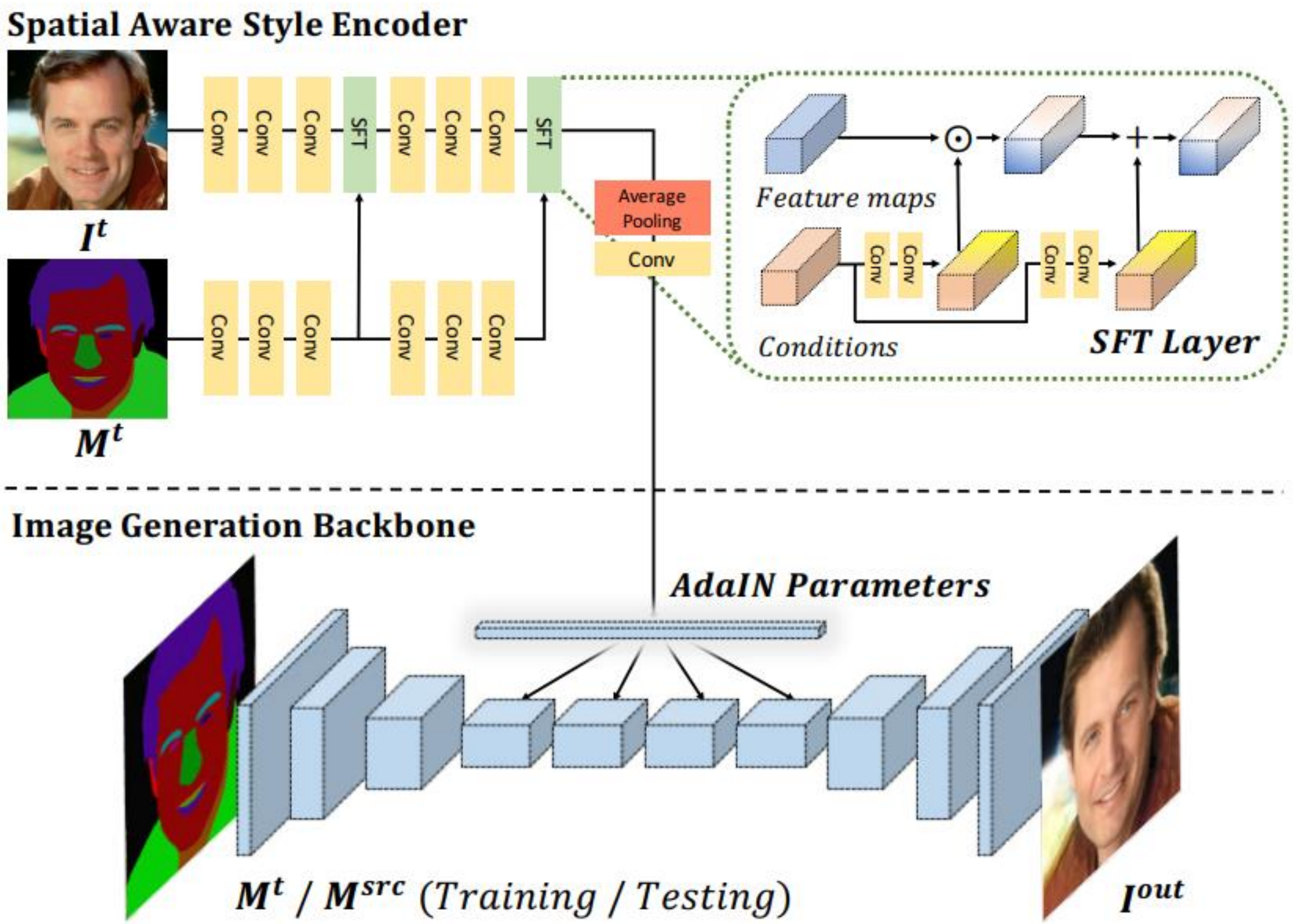
Park et al, “Semantic Image Synthesis with Spatially-Adaptive Normalization”, CVPR 2019

MaskGAN



Lee et al, "MaskGAN: Towards Diverse and Interactive Facial Image Manipulation", CVPR 2020

MaskGAN

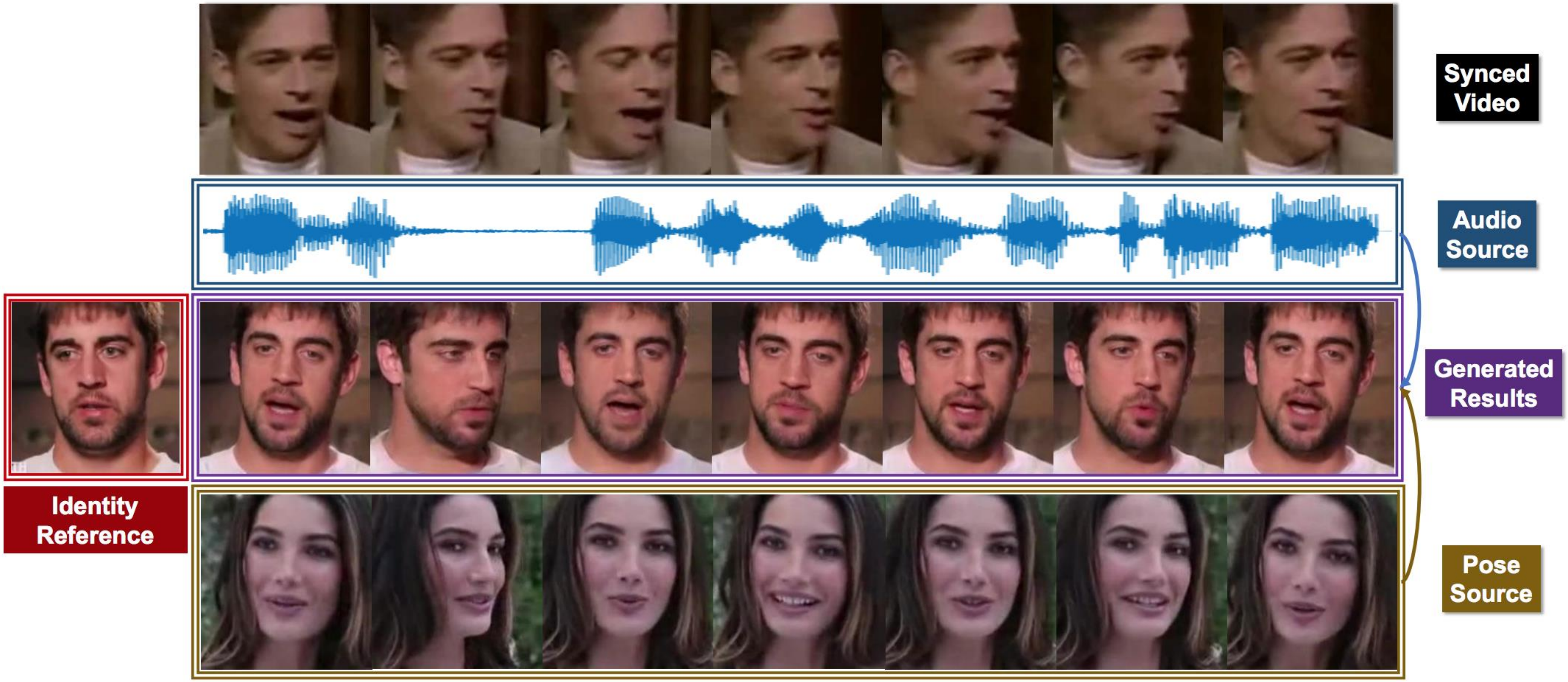


Lee et al, "MaskGAN: Towards Diverse and Interactive Facial Image Manipulation", CVPR 2020

MaskGAN

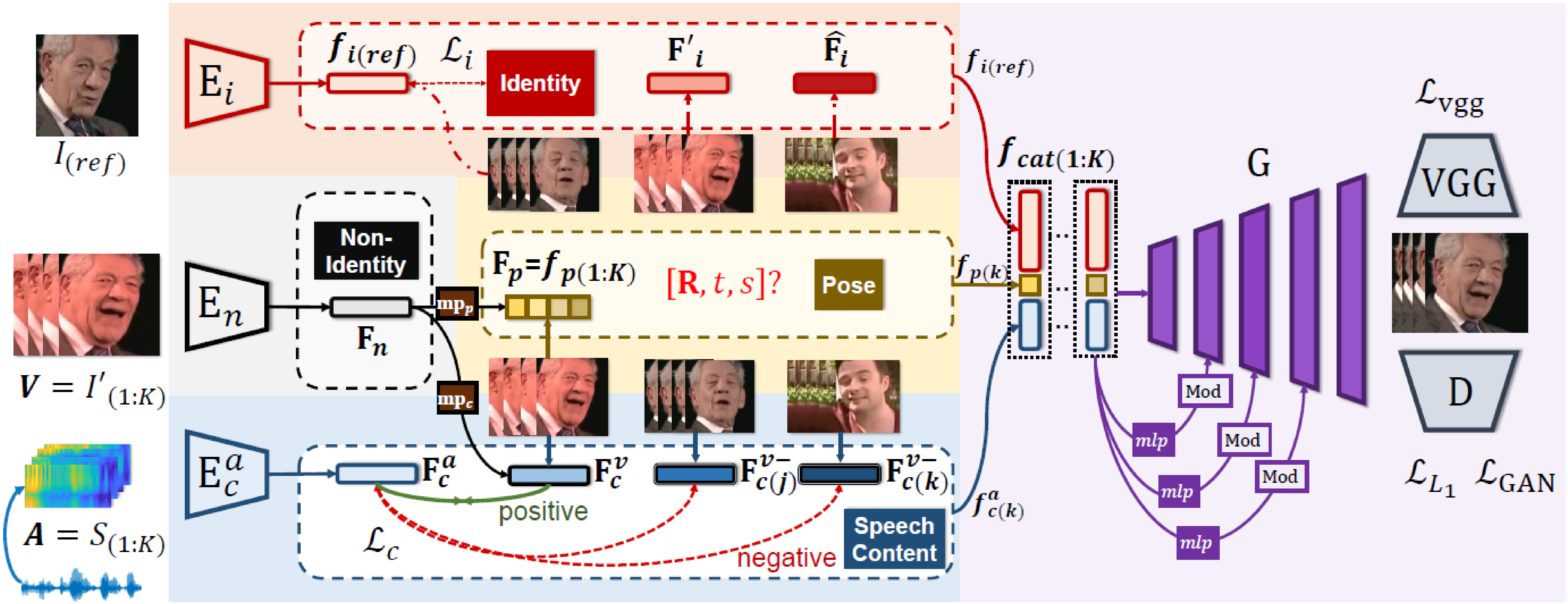
Lee et al, “MaskGAN: Towards Diverse and Interactive Facial Image Manipulation”, CVPR 2020

Talking Face Generation



Zhou et al, “Pose-Controllable Talking Face Generation by Implicitly Modularized Audio-Visual Representation”, CVPR 2021

Talking Face Generation



Zhou et al, “Pose-Controllable Talking Face Generation by Implicitly Modularized Audio-Visual Representation”, CVPR 2021

Talking Face Generation

Zhou et al, "Pose-Controllable Talking Face Generation by Implicitly Modularized Audio-Visual Representation", CVPR 2021

DeepFake Detection


synthesia


VFXCHRISUME

Democratizing Image Forensics

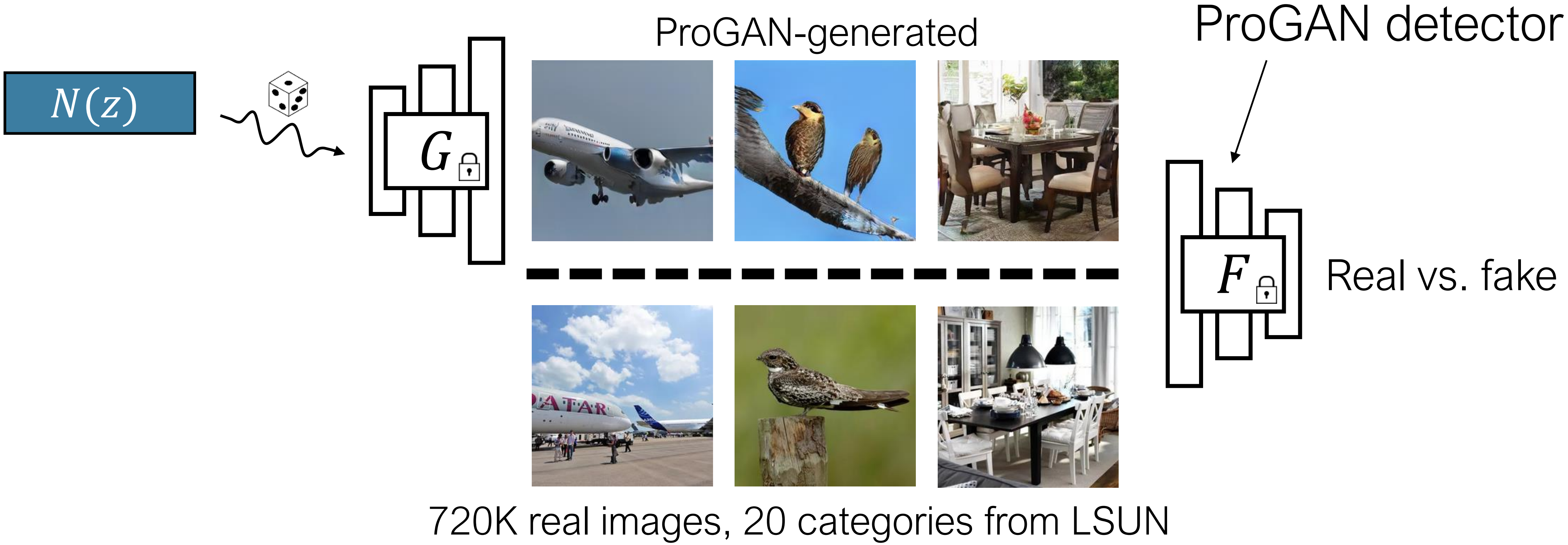
https://gizmodo.com/russian-state-tv-photoshops-an-awkward-smile-on-kim-jon-1826529277 June 2018

DeepFakes (https://github.com/deepfakes/faceswap)

Face2Face (Thies et al. 2016)

Lip-syncing Obama (Suwajanakorn et al. 2017)

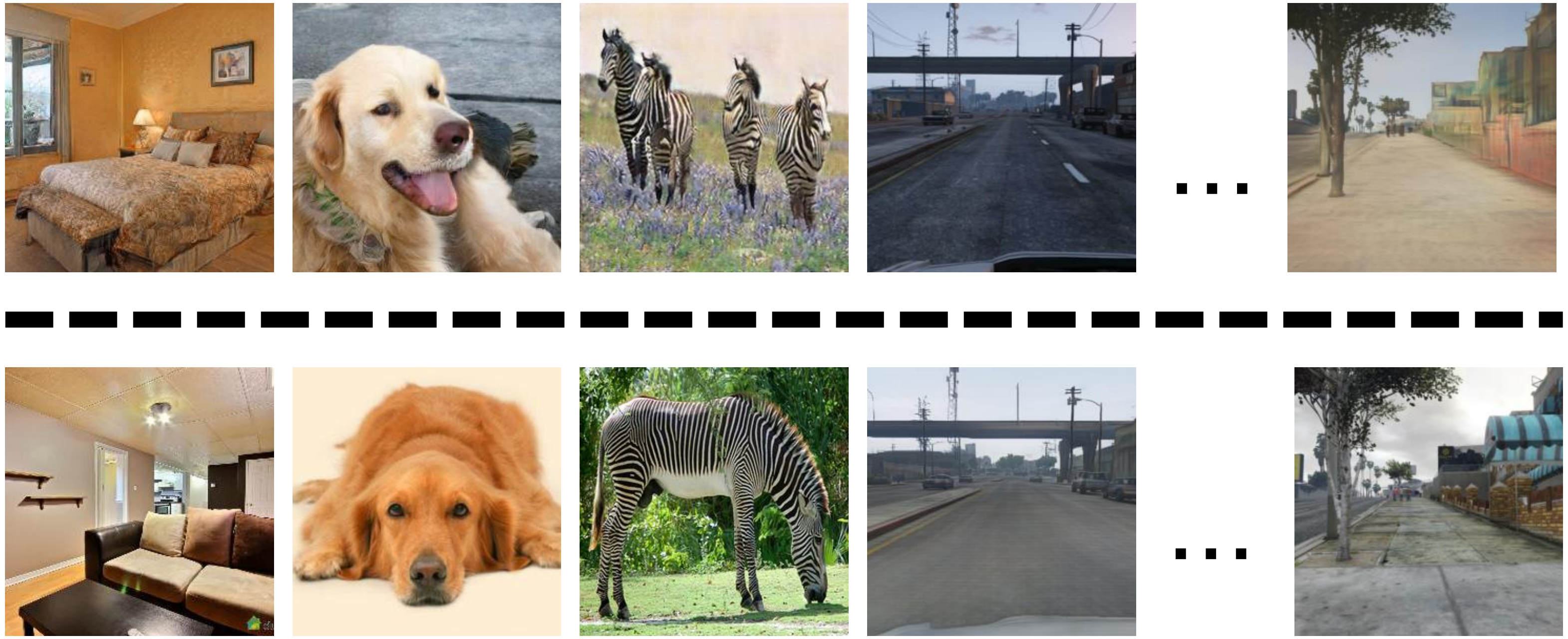
Training on ProGAN



Wang et al, “CNN-Generated Images are Surprisingly Easy to Spot...For Now”, CVPR 2020

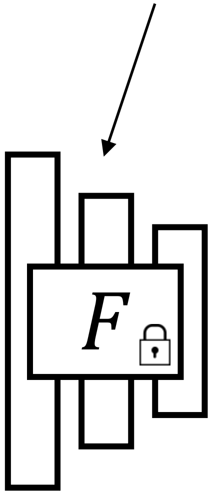
Testing across architectures

Synthesized images from **other** CNNs

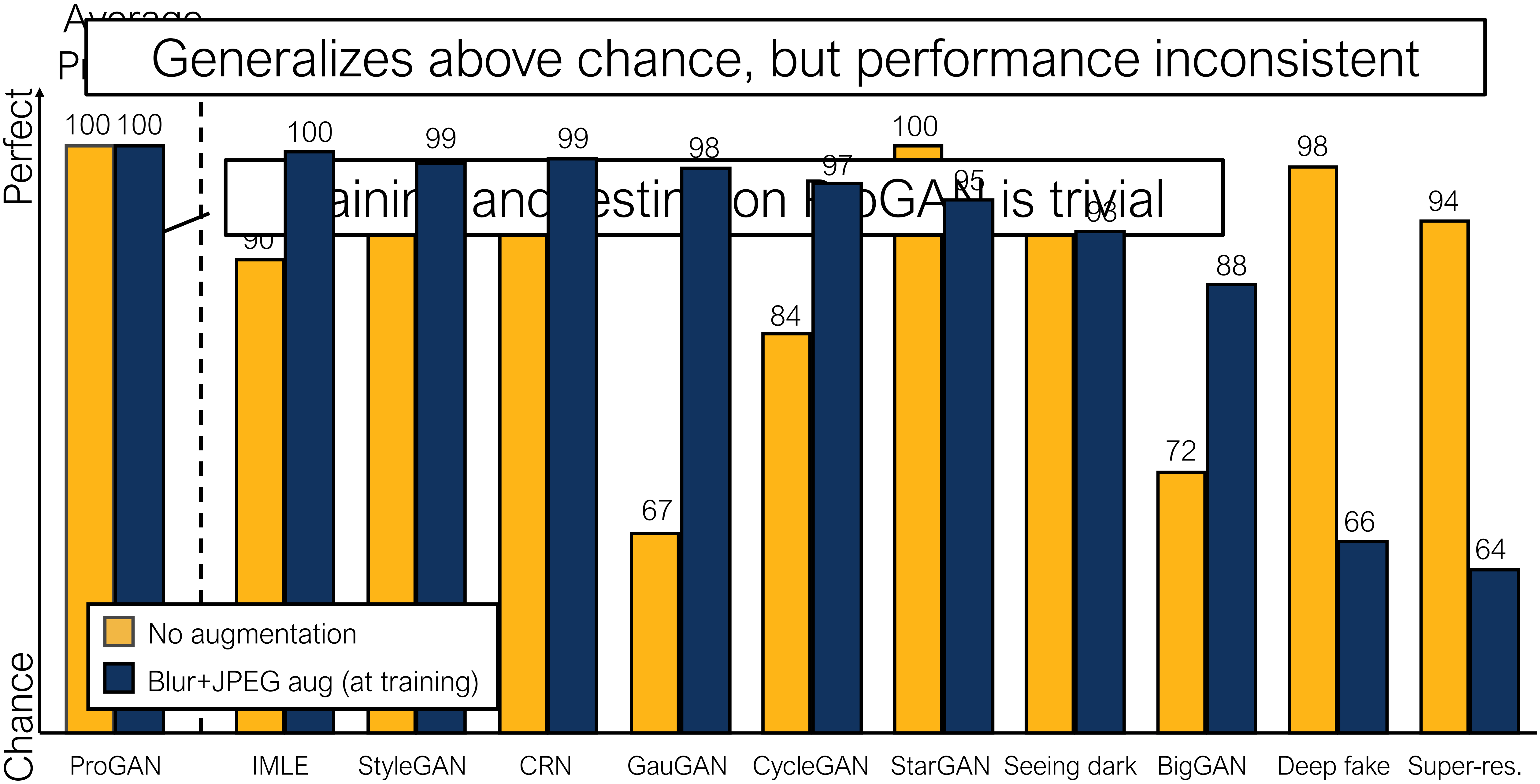


Real images

ProGAN detector



Real vs. fake


Generalizes above chance, but performance inconsistent
Perfect
Chance
100
100
90
100
99
99
67
98
84
97
100
95
72
88
98
66
94
64
No augmentation
Blur+JPEG aug (at training)
ProGAN
IMLE
StyleGAN
CRN
GauGAN
CycleGAN
StarGAN
Seeing dark
BigGAN
Deep fake
Super-res.

Discussion

- Suggests CNN-generated images have common artifacts
- Artifacts can be detected by a simple classifier!
  - StyleGAN2 (released **after** our submission): 100% AP on FFHQ
  - Maybe generalizes beyond CNNs [Chai et al. ECCV 2020]
  - **Note**: AP is computed on a collection of images;
    a real/fake decision on a per-image basis is more difficult
- Situation may not persist
  - GANs train with a discriminator
  - Future architecture changes

Large-Scale Study on DeepFake

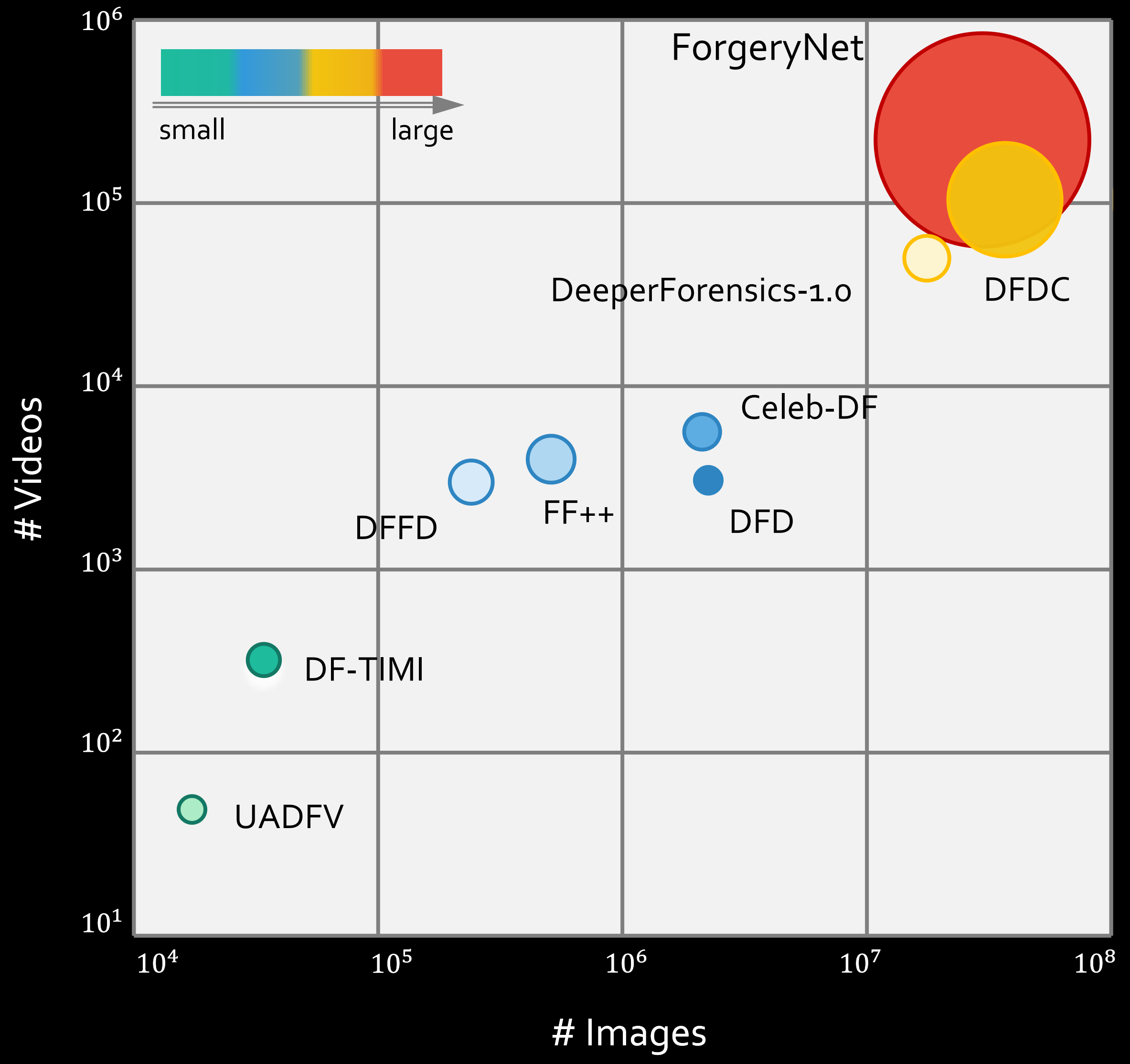
He et al, "ForgeryNet: A Versatile Benchmark for Comprehensive Forgery Analysis", CVPR 2021


Real or Fake?
NERDGEIST
EOS 7D
what is 3.45 pounds expressed in grams?
pounds → grams
For more videos, check out:
OCDE
3. add relevant detail to explain or
answers
YTN사이언스24
"북, 신형 ICBM도 첫 시험발사 가능성"
George


NERDGEIST
PART WOLF
Canon
EOS 7D
What is 3.45 pounds expressed in grams?
pounds → grams
For more videos, check out:
www.video chemistry textbook.com
Good News in Oakland
ATHLETE
OCDE
Coherence is the ability to:
1. expand your answers
2. answer the questions directly
4. connect your senten
connectors
YTN사이언스24
“북, 신형 ICBM도 첫 시험발사 가능성”
George
ulopoulos


NERDGEIST
What is 3.45 pounds expressed in grams?
pounds → grams
For more videos, check out:
www.video chemistry textbook.com
Good News in Oakland
OCDE
Coherence is the ability to:
1. expand your answers
2. answer the questions directly
YTN사이언스24
“북, 신형 ICBM도 첫 시험발사 가능성”

Current Forgery Dataset



Limited scales

Limited diversity

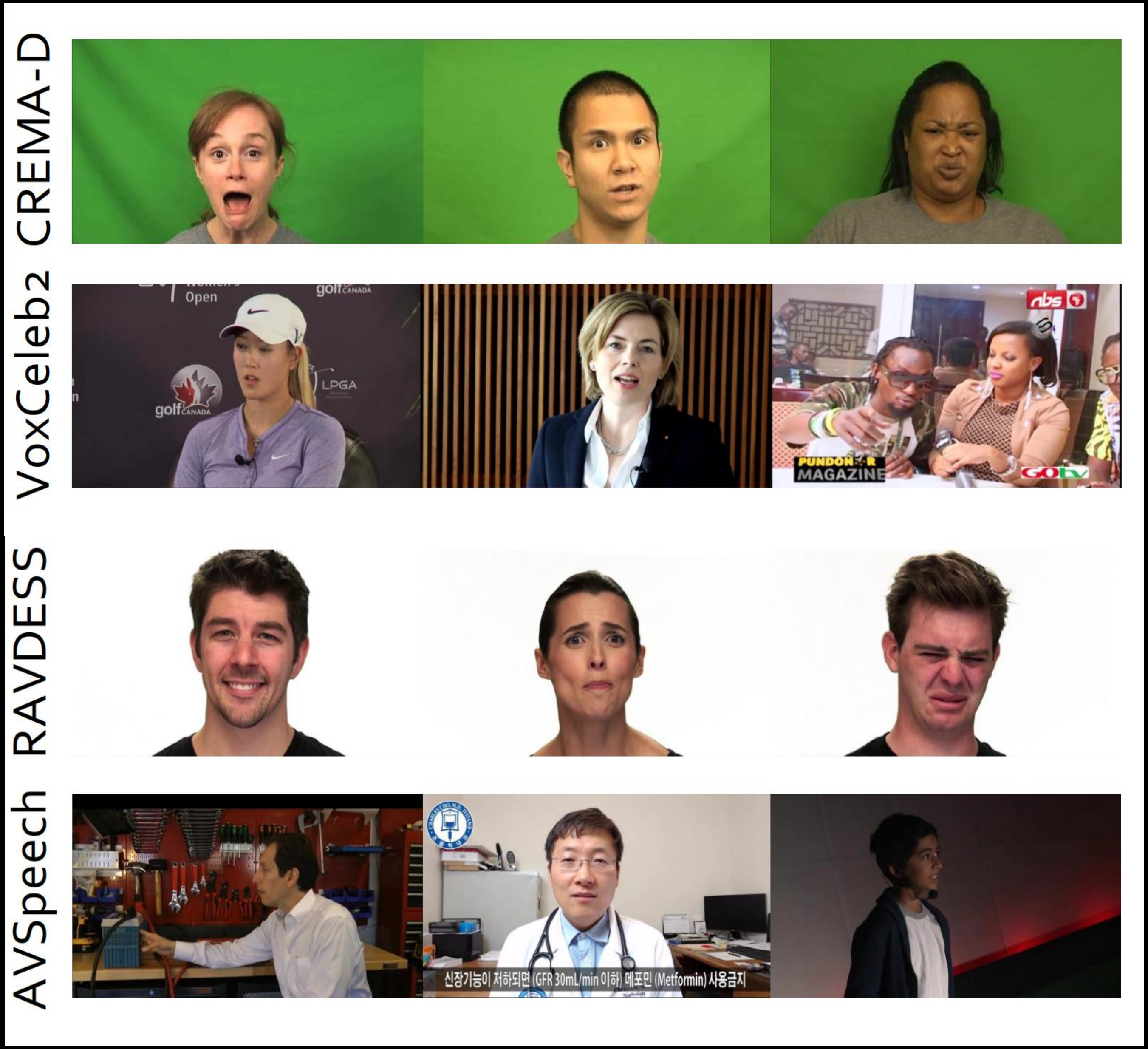
Only binary label

How about ForgeryNet ?

ForgeryNet: **Wild Original Data**

diversified dimensions

- Angle
- Expression
- Identity
- Lighting
- Scenario
- ……



ForgeryNet: **Various Forgery Approaches**

15 approaches

variety of learning-based models

2.9M still images

220k video clips

Real **Fake**

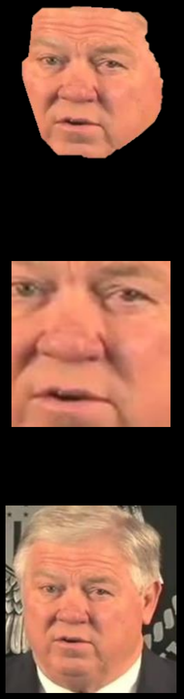
Swap Identity

DeepFakes
FSGAN
MMReplacement
FaceShifter
BlendFace
...

Modify Attribute

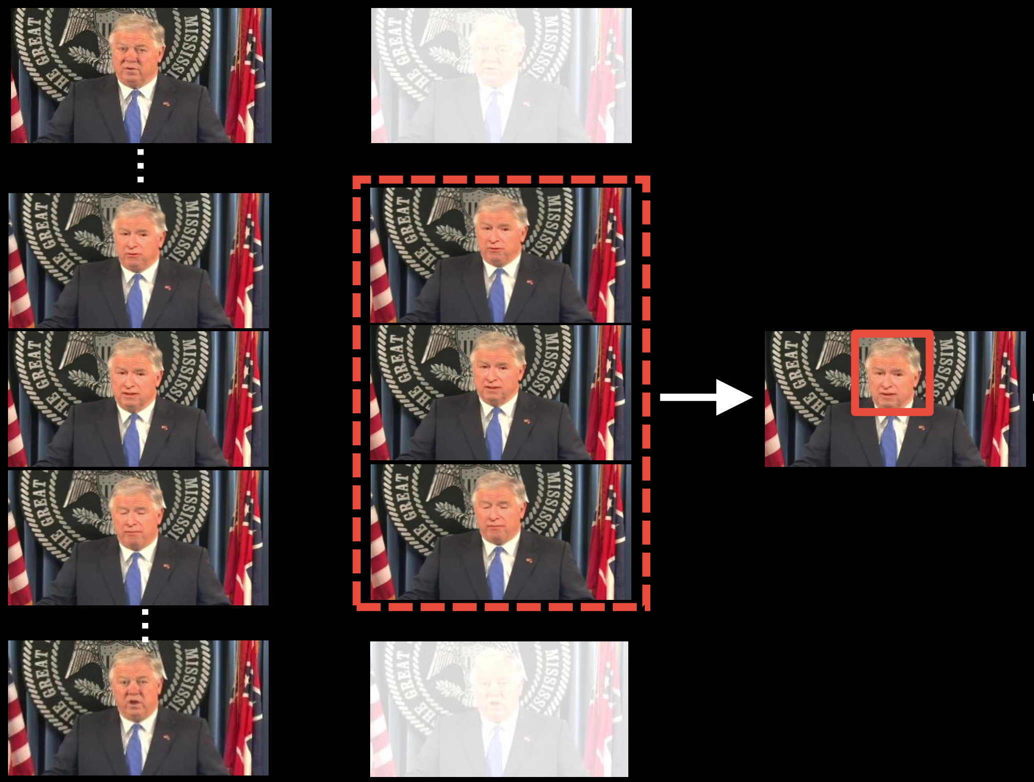
MaskGAN
StarGAN2
Audio2Lip
ATVG-Net
SC-FEGAN
...

ForgeryNet: **Diverse Re-rendering Process**



more than 36 mix-perturbations

ForgeryNet: **Comprehensive Annotations and Tasks**



Video

Task #I
Temporal Forgery Localization

Image

Real

Fake

Swap Identity

- DeepFakes
- FSGAN
- MMReplacement
- FaceShifter
- BlendFace
- …

Modify Attribute

- MaskGAN
- StarGAN2
- Audio2Lip
- ATVG-Net
- SC-FEGAN
- …

Task #II
Forgery Classification

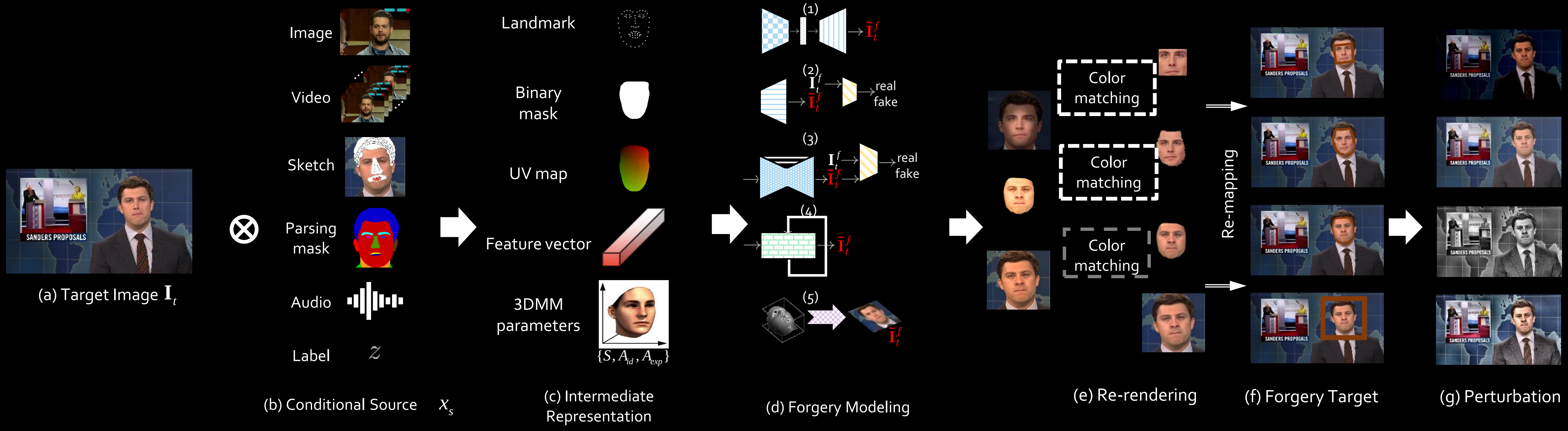
Task #III
Spatial Forgery Localization

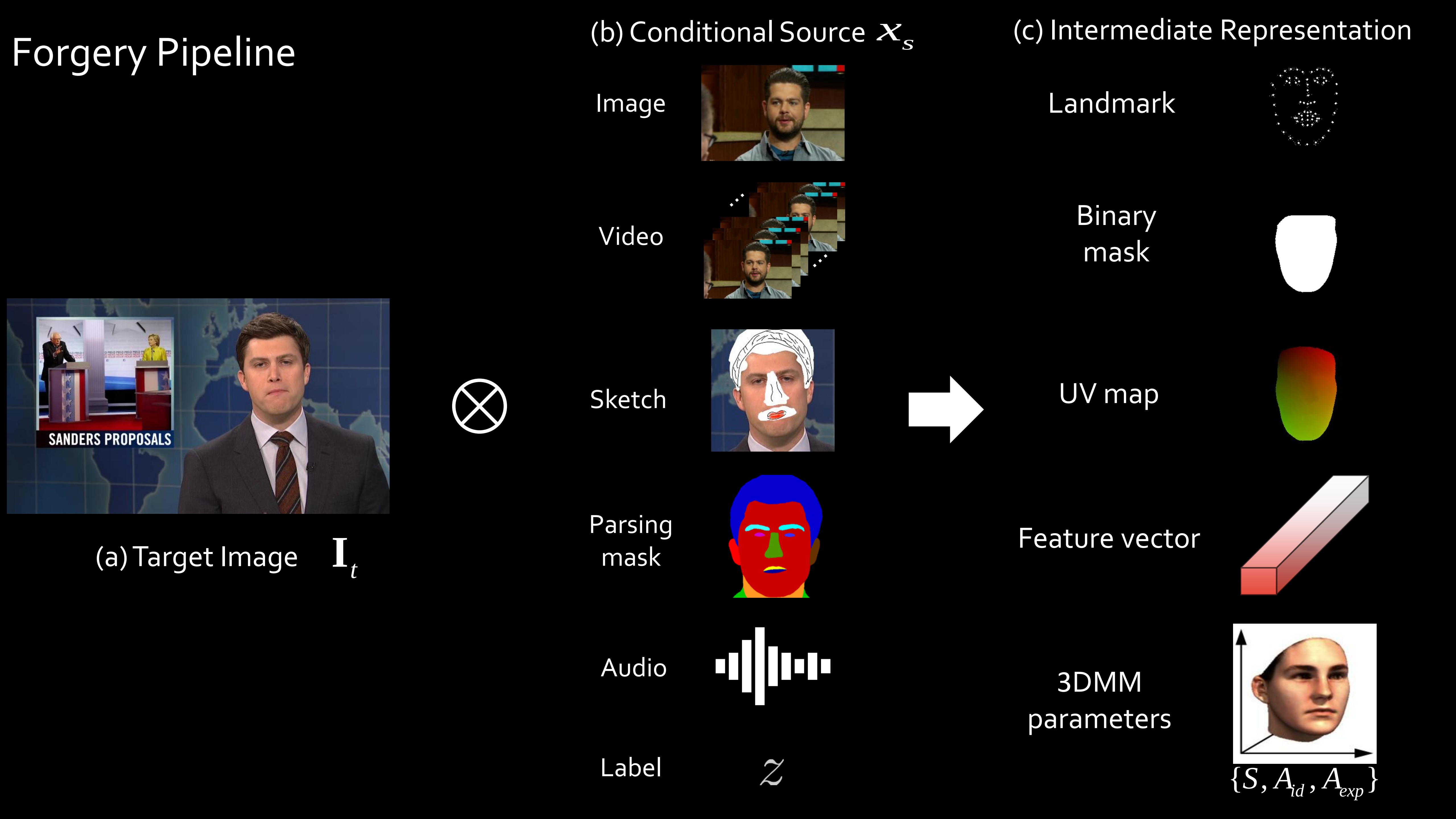
4 Tasks

9.4M annotations

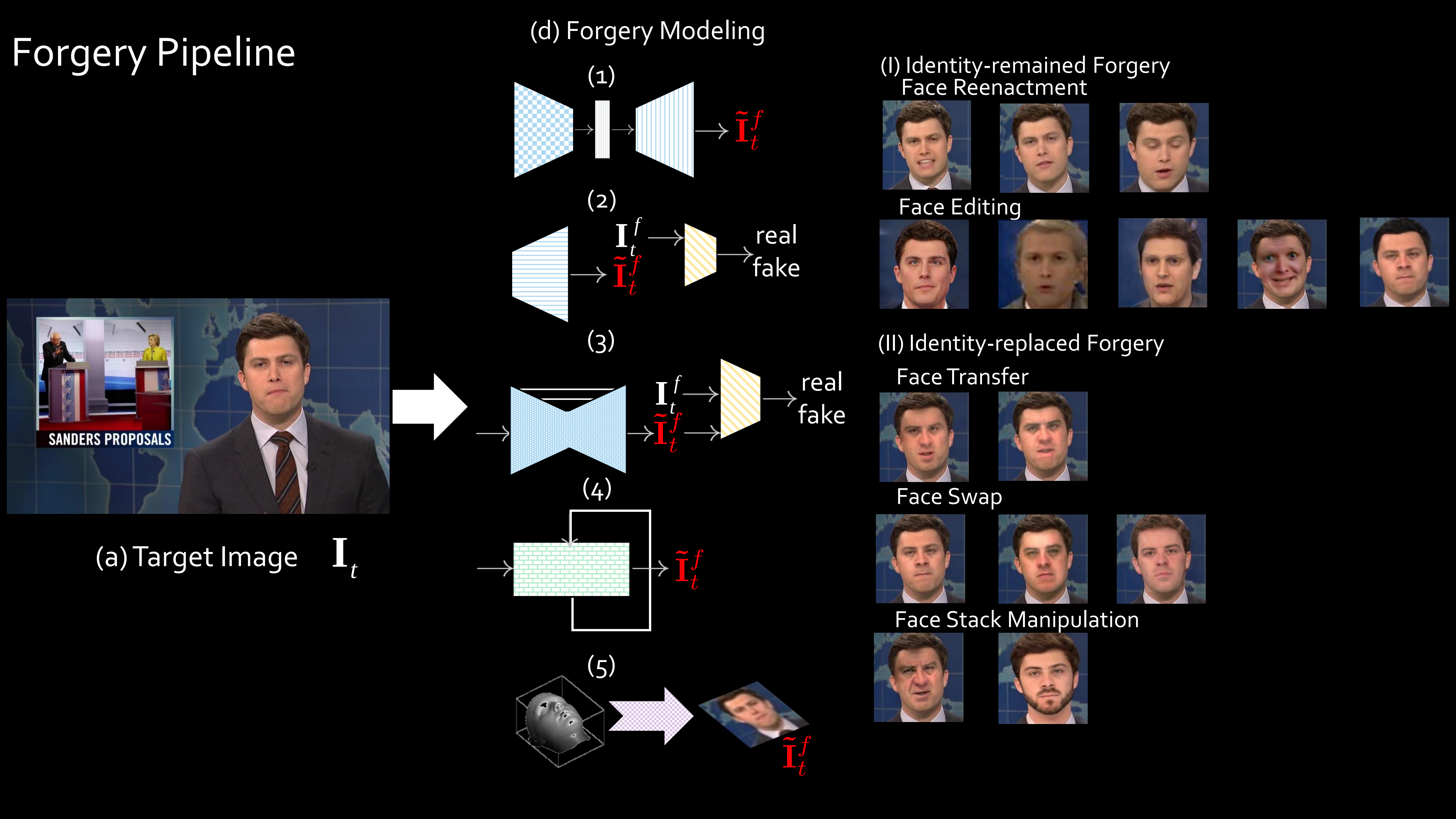
Forgery Pipeline

Forgery Pipeline

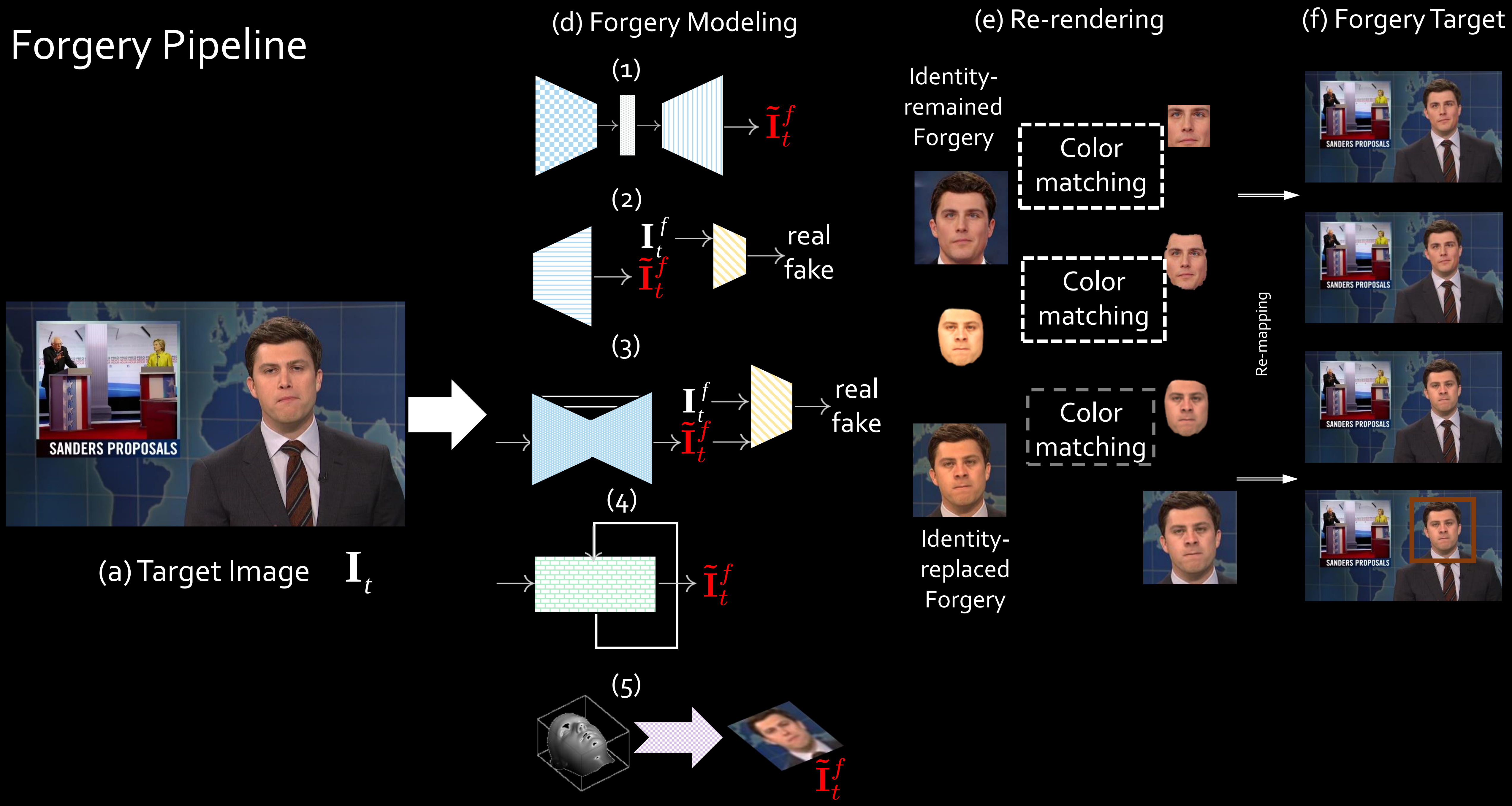




Forgery Pipeline
SANDERS PROPOSALS
(a) Target Image $\mathbf{I}_t$
(b) Conditional Source x_s
Image
Video
Sketch
Parsing mask
Audio
Label
z
(c) Intermediate Representation
Landmark
Binary mask
UV map
Feature vector
3DMM parameters
$\{S, A_{id}, A_{exp}\}$



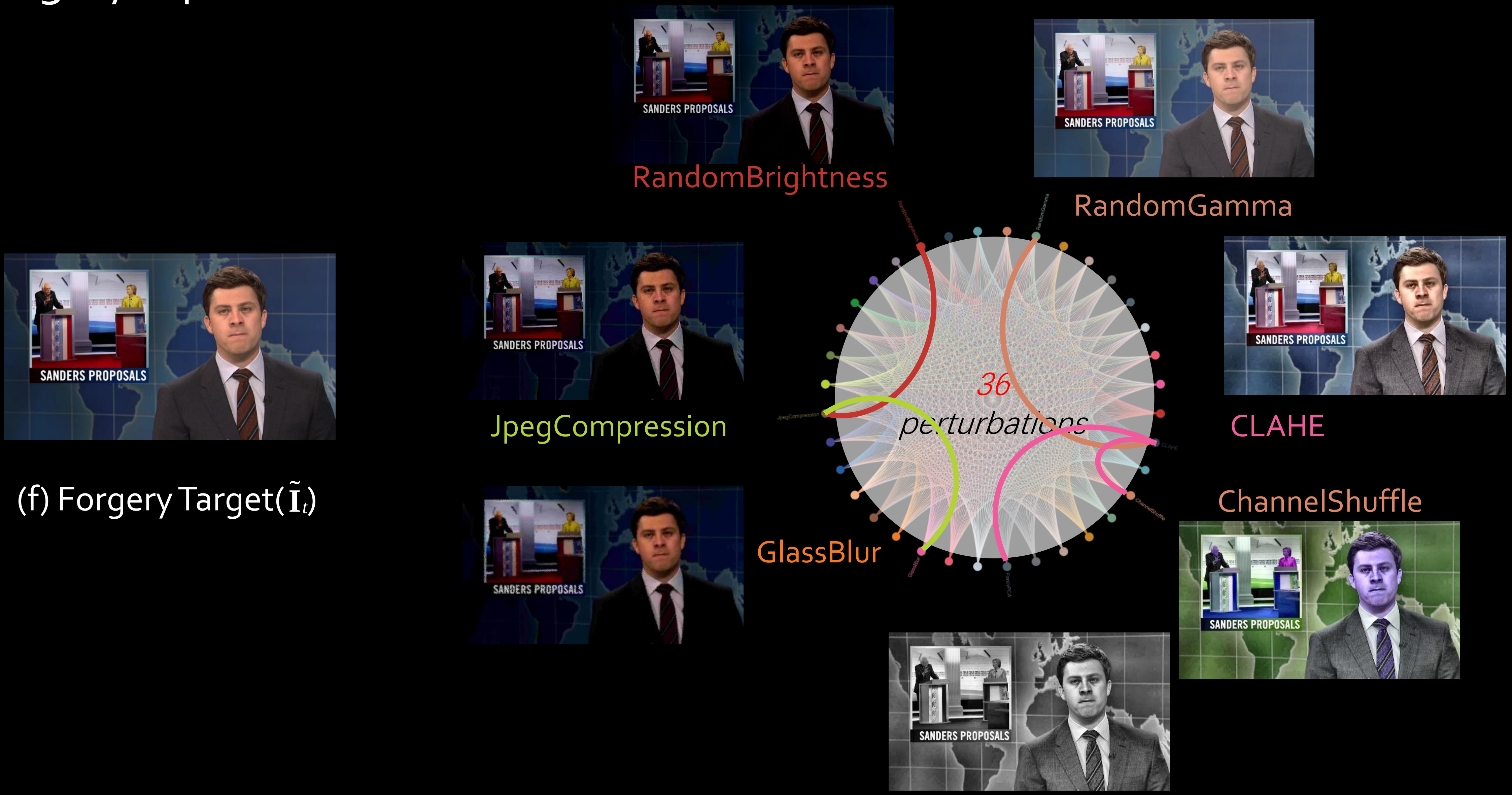
Forgery Pipeline
(d) Forgery Modeling
(1)
$\tilde{\mathbf{I}}_t^f$
(2)
$\mathbf{I}_t^f$
$\tilde{\mathbf{I}}_t^f$
real
fake
(3)
$\mathbf{I}_t^f$
$\tilde{\mathbf{I}}_t^f$
real
fake
(4)
$\tilde{\mathbf{I}}_t^f$
(5)
$\tilde{\mathbf{I}}_t^f$
SANDERS PROPOSALS
(a) Target Image $\mathbf{I}_t$
(I) Identity-remained Forgery
Face Reenactment
Face Editing
(II) Identity-replaced Forgery
Face Transfer
Face Swap
Face Stack Manipulation



Forgery Pipeline
(d) Forgery Modeling
(e) Re-rendering
(f) Forgery Target
(1)
$\tilde{\mathbf{I}}_t^f$
(2)
$\mathbf{I}_t^f$
$\tilde{\mathbf{I}}_t^f$
real
fake
(3)
$\mathbf{I}_t^f$
$\tilde{\mathbf{I}}_t^f$
real
fake
(4)
$\tilde{\mathbf{I}}_t^f$
(5)
$\tilde{\mathbf{I}}_t^f$
Identity-remained Forgery
Color matching
Color matching
Color matching
Identity-replaced Forgery
Re-mapping
SANDERS PROPOSALS
(a) Target Image $\mathbf{I}_t$

Forgery Pipeline

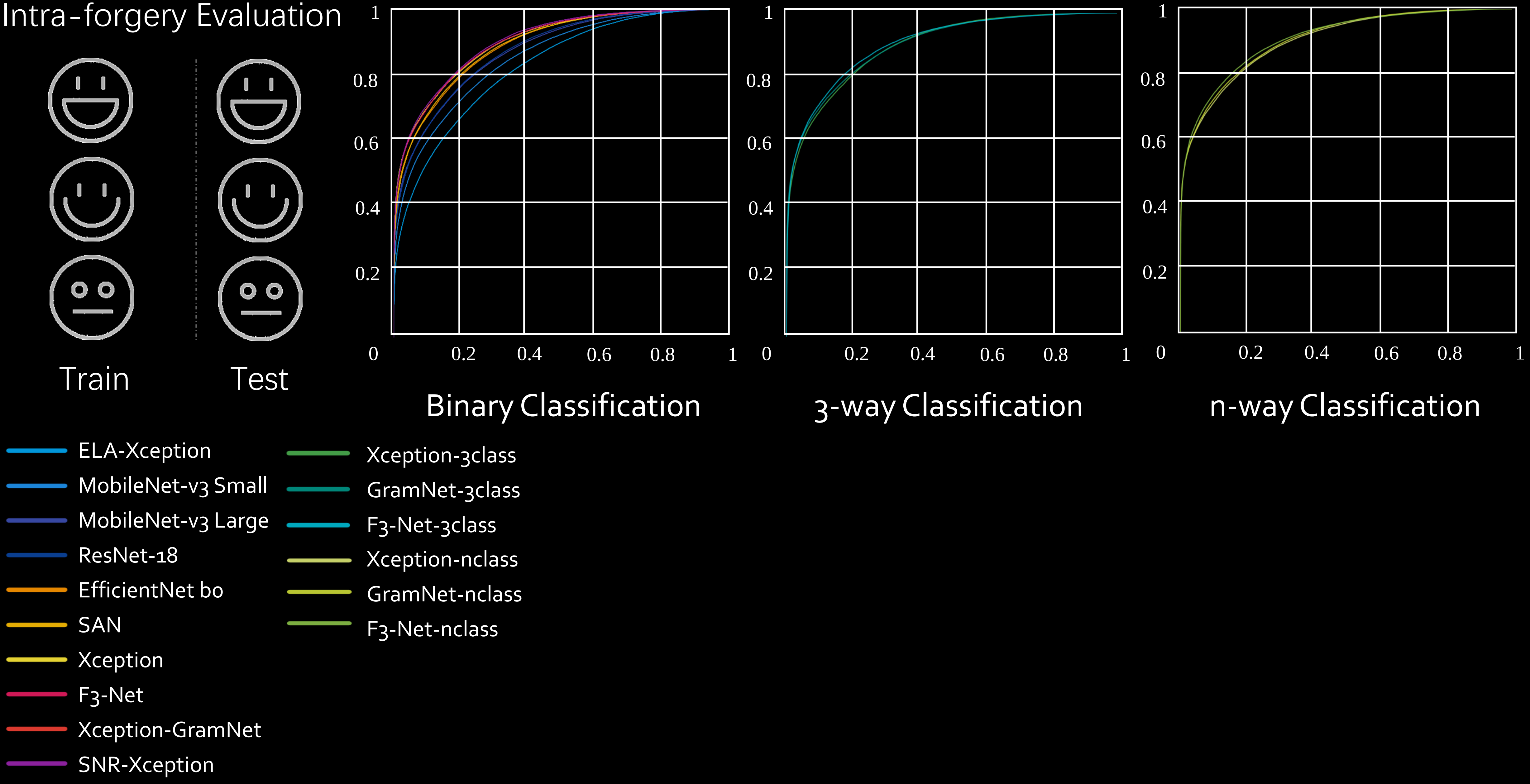
(g) Perturbation



4 Task Benchmark

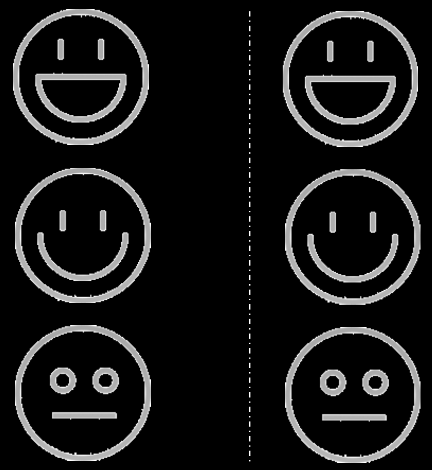
Task I: Image Forgery Classification

Intra-forgery Evaluation

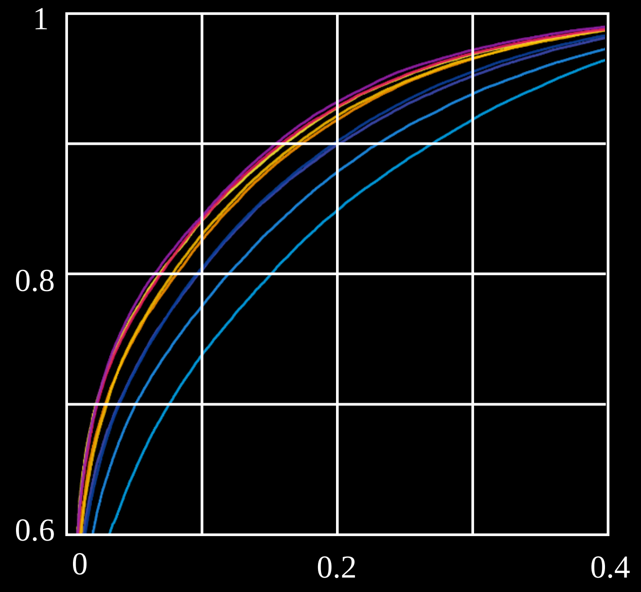


Task I: Image Forgery Classification

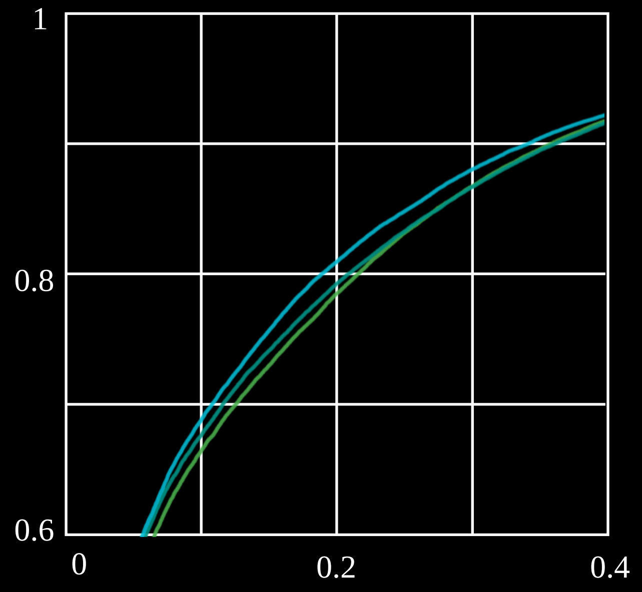
Intra-forgery Evaluation



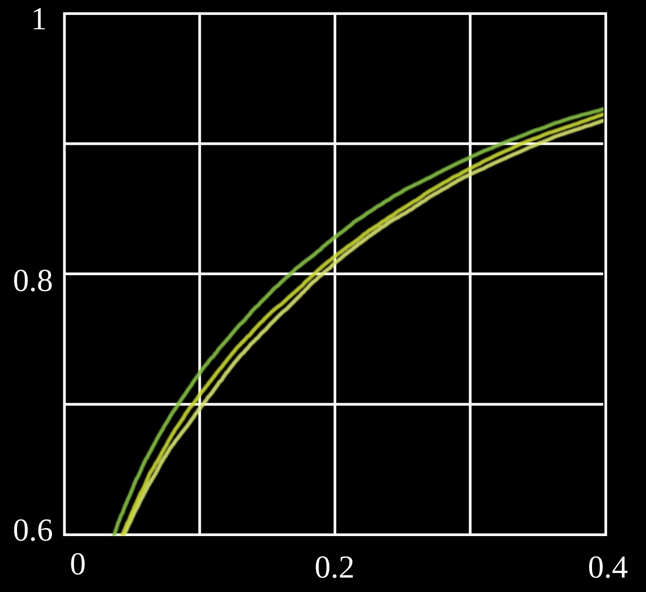
Train Test



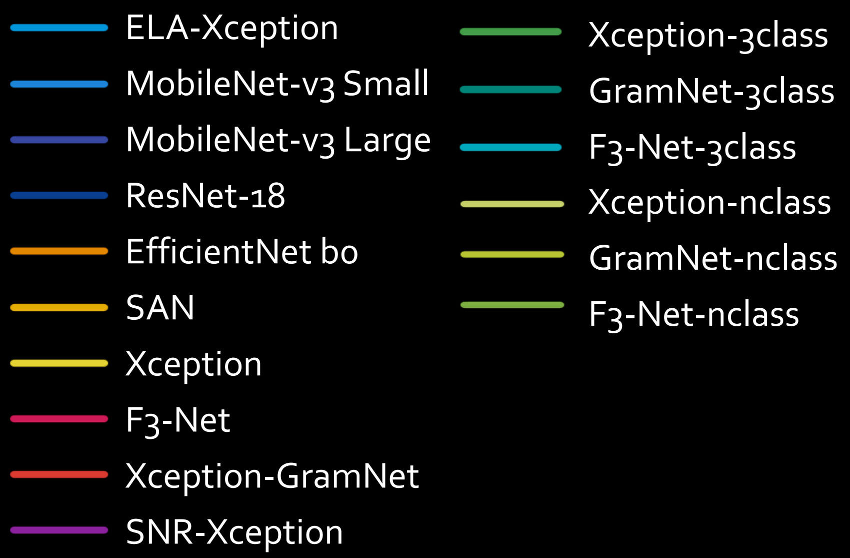
Binary Classification



3-way Classification



n-way Classification

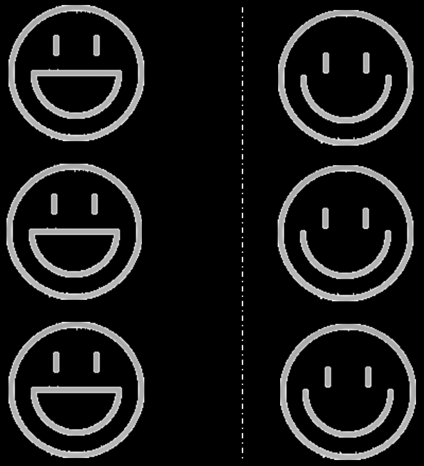


Classification becomes more difficult when the number of categories increases.

More auxiliary information potentially makes the forensics model more discriminative.

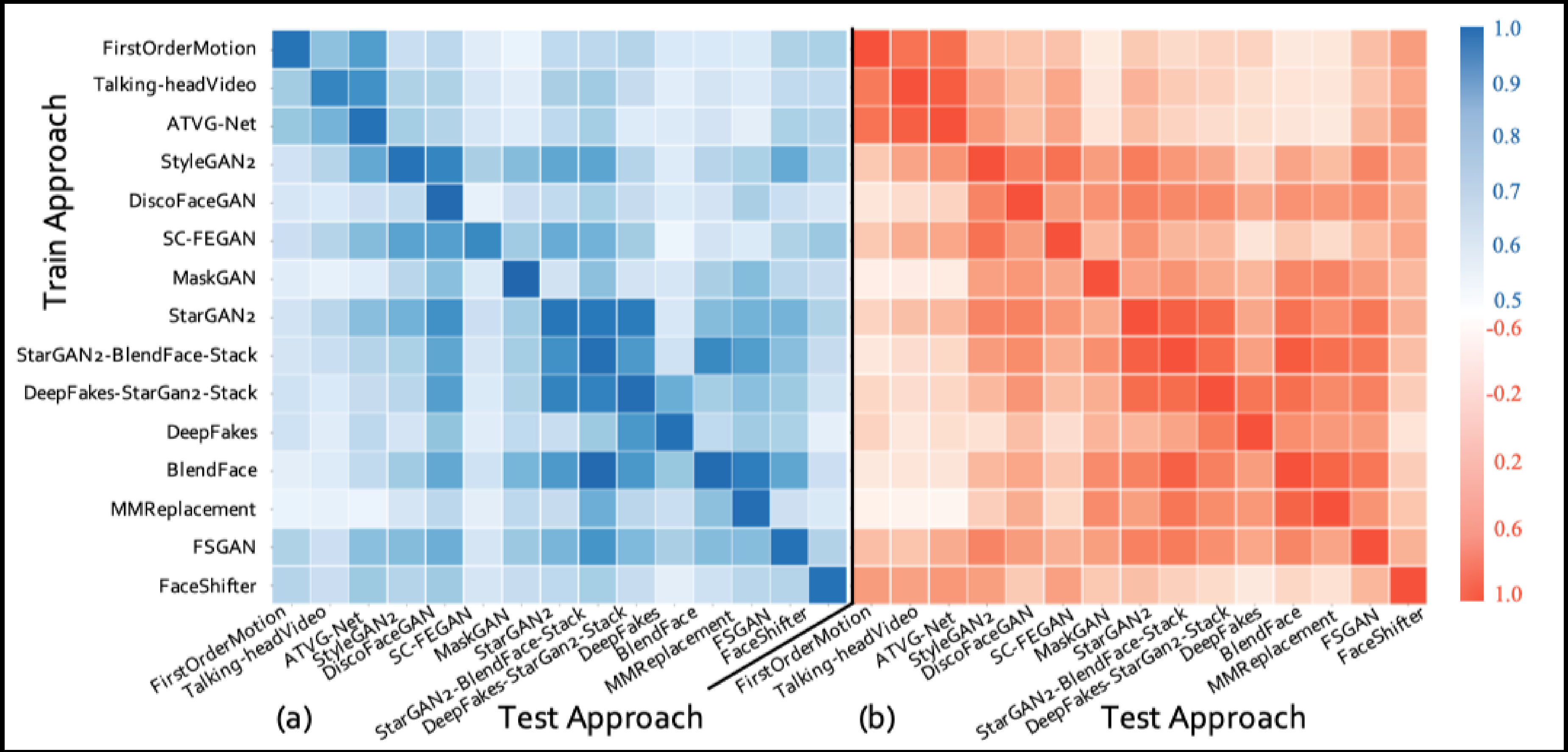
Task I: Image Forgery Classification

Cross-forgery Evaluation



Train

Test



AUC score map

Correlation map

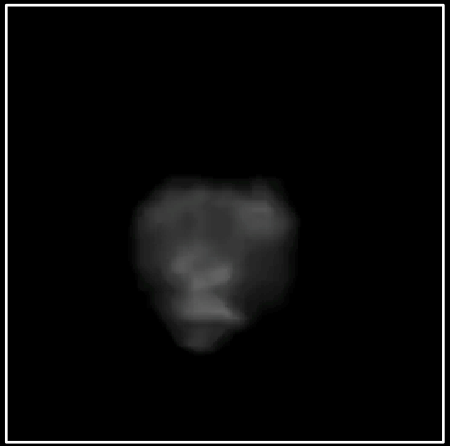
Forgery approaches belonging to the same meta-category usually have higher correlations mutually.

The generalization ability of forensics methods across forgery approaches.

Task II: Spatial Forgery Localization

Video

Ground-truth



Predicted map

Images along with forgery masks are used to train the localization model

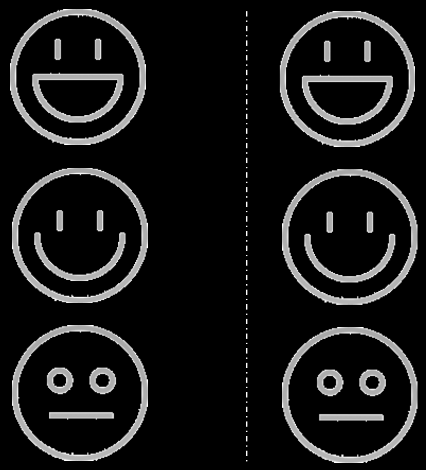
aims to specify manipulated regions

Method	IoU		IoU_{diff}			Loss_{l1}
	0.1	0.2	0.01	0.05	0.1	
Xception+Reg.	89.55	93.70	67.57	83.25	89.22	0.0131
Xeption+Unet [37]	95.99	98.76	79.71	92.70	97.13	0.0134
HRNet [42]	96.27	98.78	88.73	92.99	96.27	0.0114

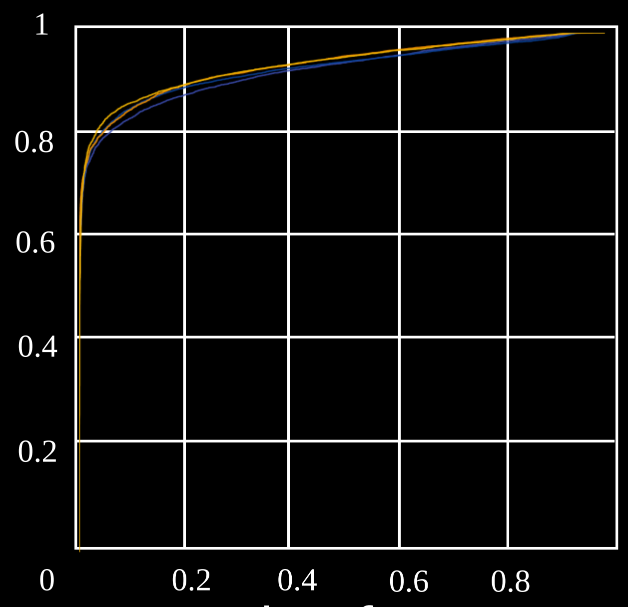
results with IOU, IOUdiff and L1 distance.

Task III: Video Forgery Classification

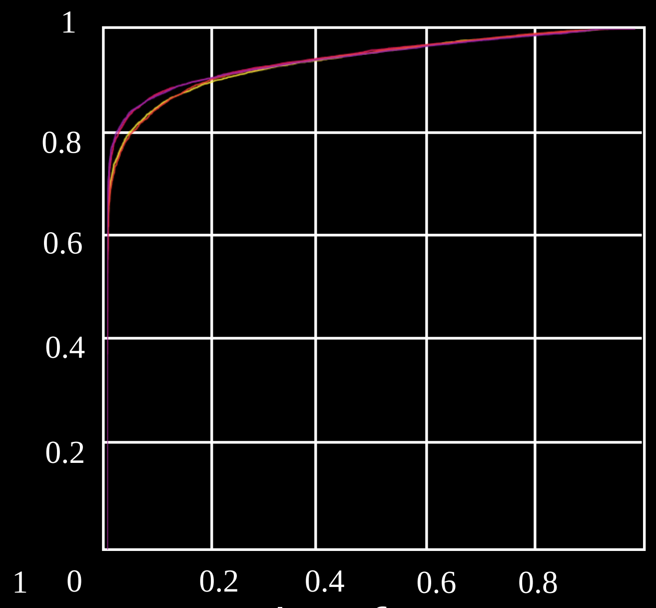
Intra-forgery Evaluation



Train Test

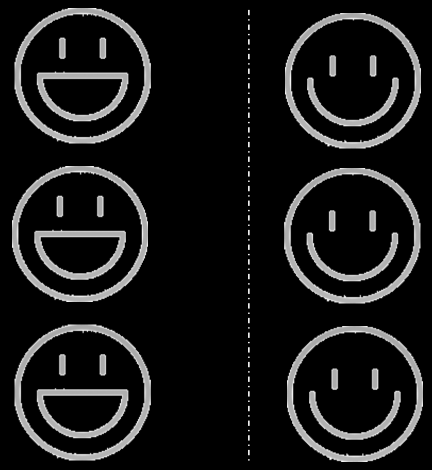


Binary Classification



n-way Classification

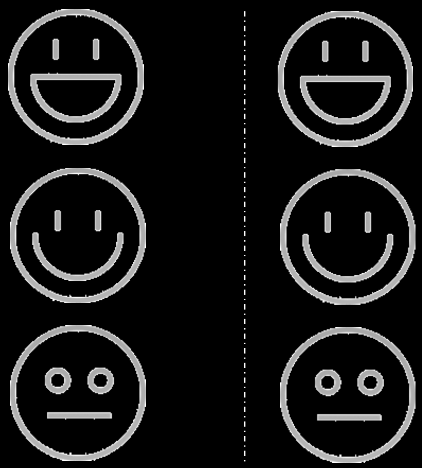
Cross-forgery Evaluation



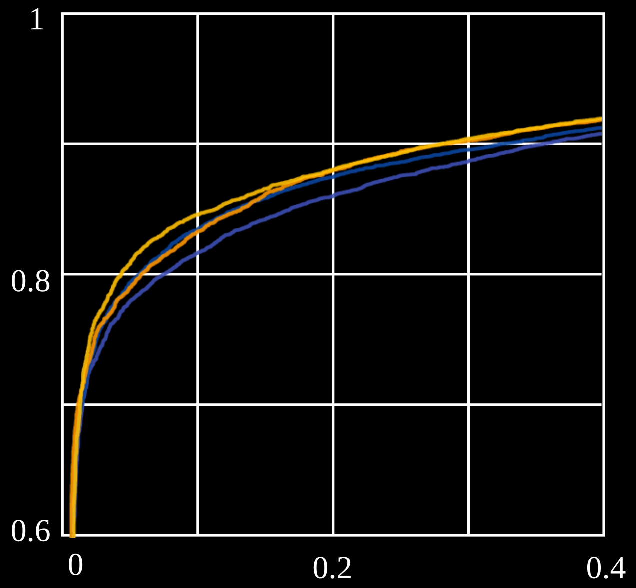
Train Test

Task III: Video Forgery Classification

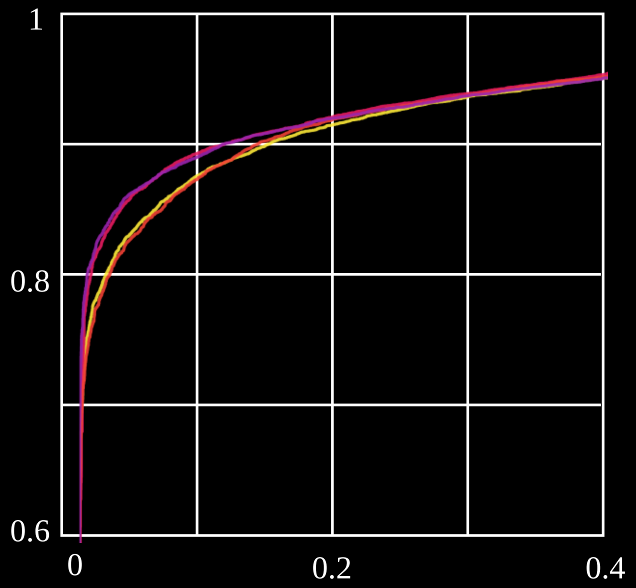
Intra-forgery Evaluation



Train Test

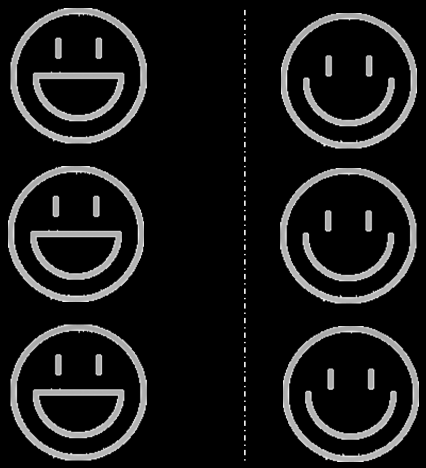


Binary Classification



n-way Classification

Cross-forgery Evaluation

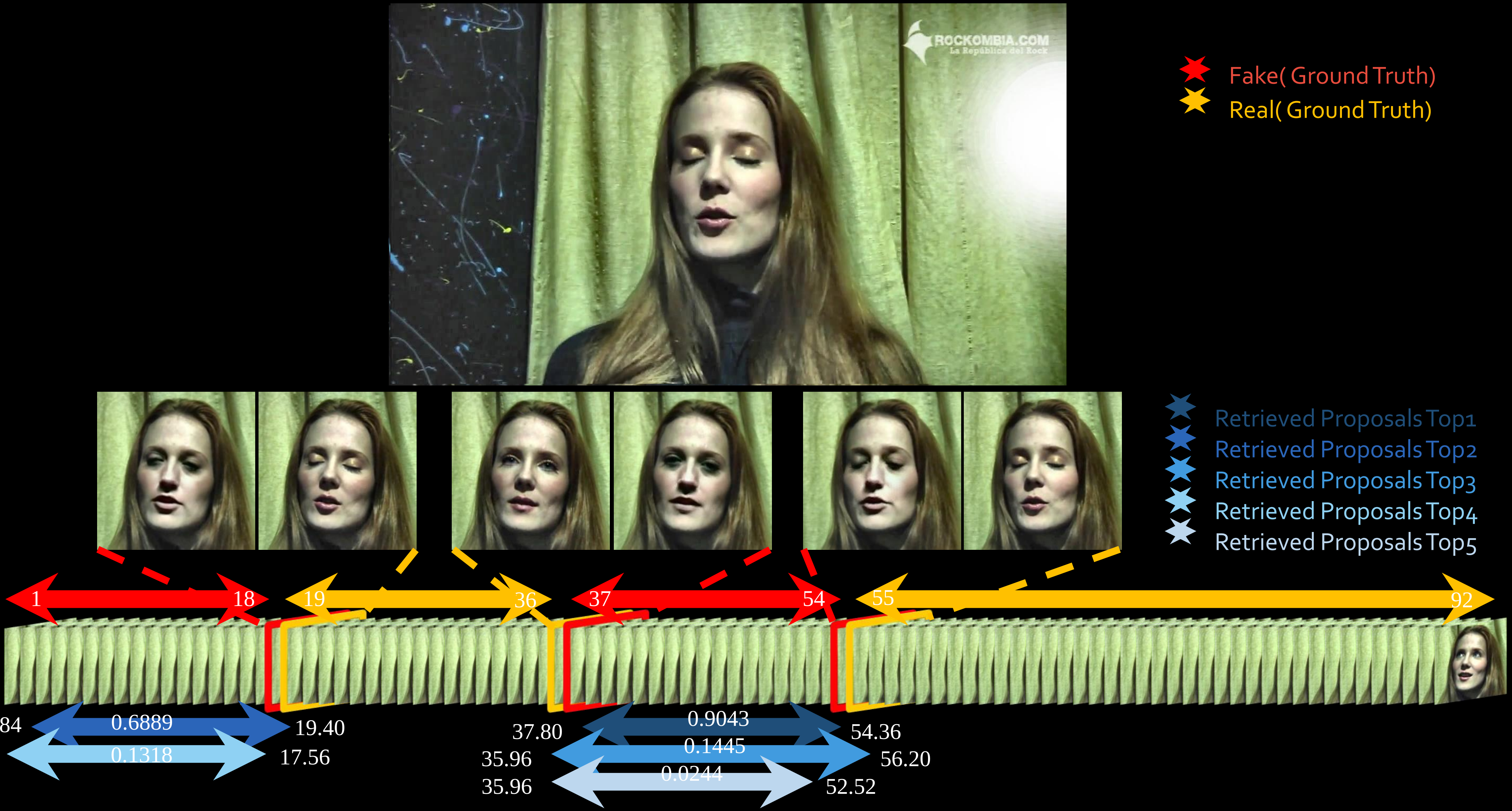


Train Test

		ID-replaced		ID-remained	
		Acc.	AUC	Acc.	AUC
X3D-M	ID-replaced	87.92	92.91	55.25	65.59
	ID-remained	55.93	62.87	88.85	95.40
SlowFast	ID-replaced	88.26	92.88	52.64	64.83
	ID-remained	52.70	61.50	87.96	95.47

Video Forgery Classification (Protocol 2)

Task IV: Temporal Forgery Localization

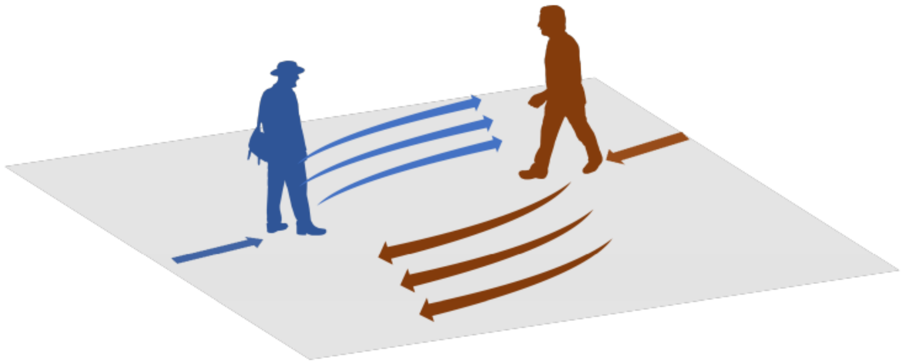


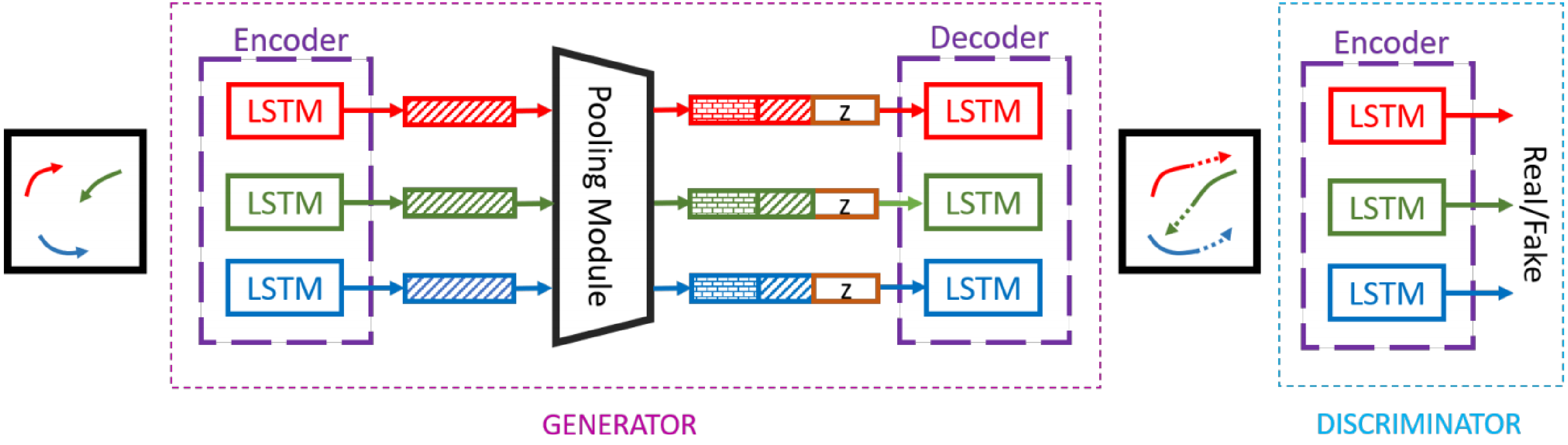
provide temporal boundaries of forgery segments and the corresponding confidence values


104.6 RTL
Berlins Hit-Radio
Es ist gut wieder hier zu sein. Es ist eine Weile her.
FOX NEWS SUNDAY
ABADI: NOW UNDERSTANDING THAT IRAQ IS AN ALLY
IRAQI PRIME MINISTER EXCLUSIVE
zammutv
NEW
OLD
Liz Gannes
Video: The Greatest Pre-Launch Start-Up Pitch Ever?
LU 800
M T D
www.mtdcnc.com
That all peoples may know Him
SALVATION
Provincial Response
Global NEWS

GANs: Not Just for Images

GANs: Trajectory Prediction



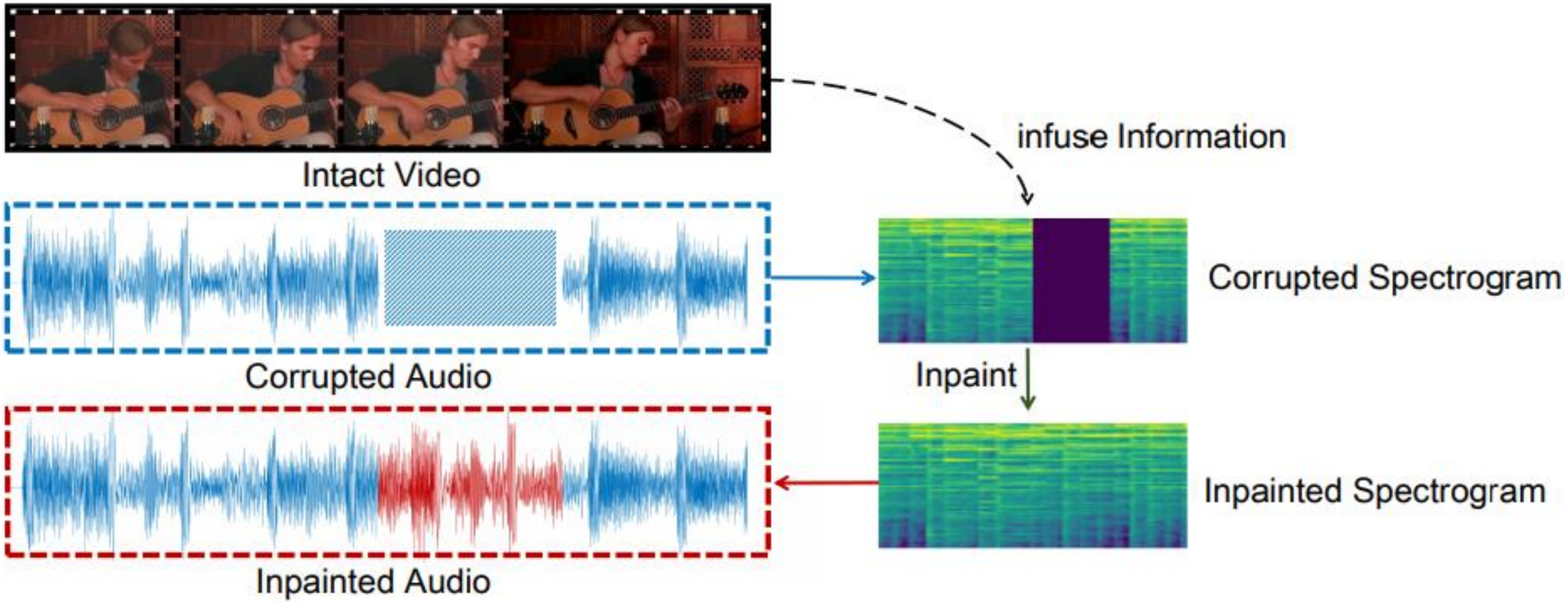


Gupta, **Johnson**, Li, Savarese, Alahi, “Social GAN: Socially Acceptable Trajectories with Generative Adversarial Networks”, CVPR 2018

GANs: Trajectory Prediction

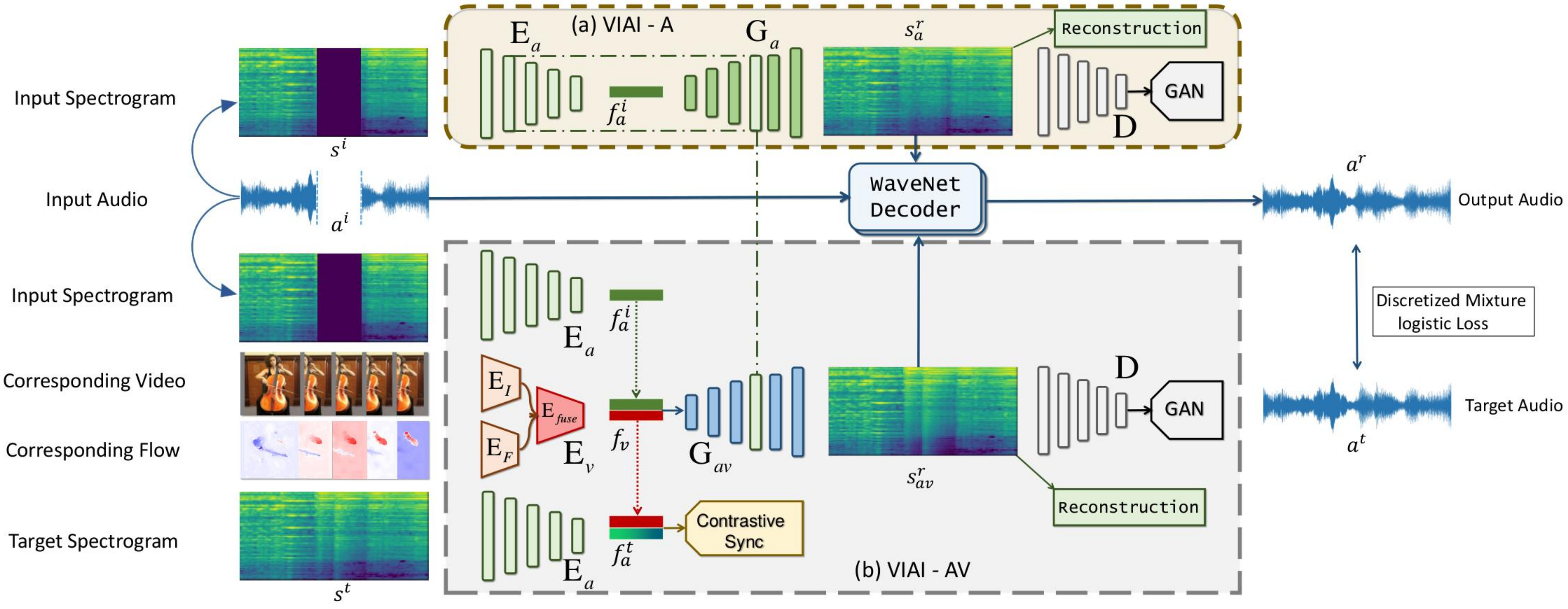
Gupta, **Johnson**, Li, Savarese, Alahi, “Social GAN: Socially Acceptable Trajectories with Generative Adversarial Networks”, CVPR 2018

GANs: Audio Inpainting



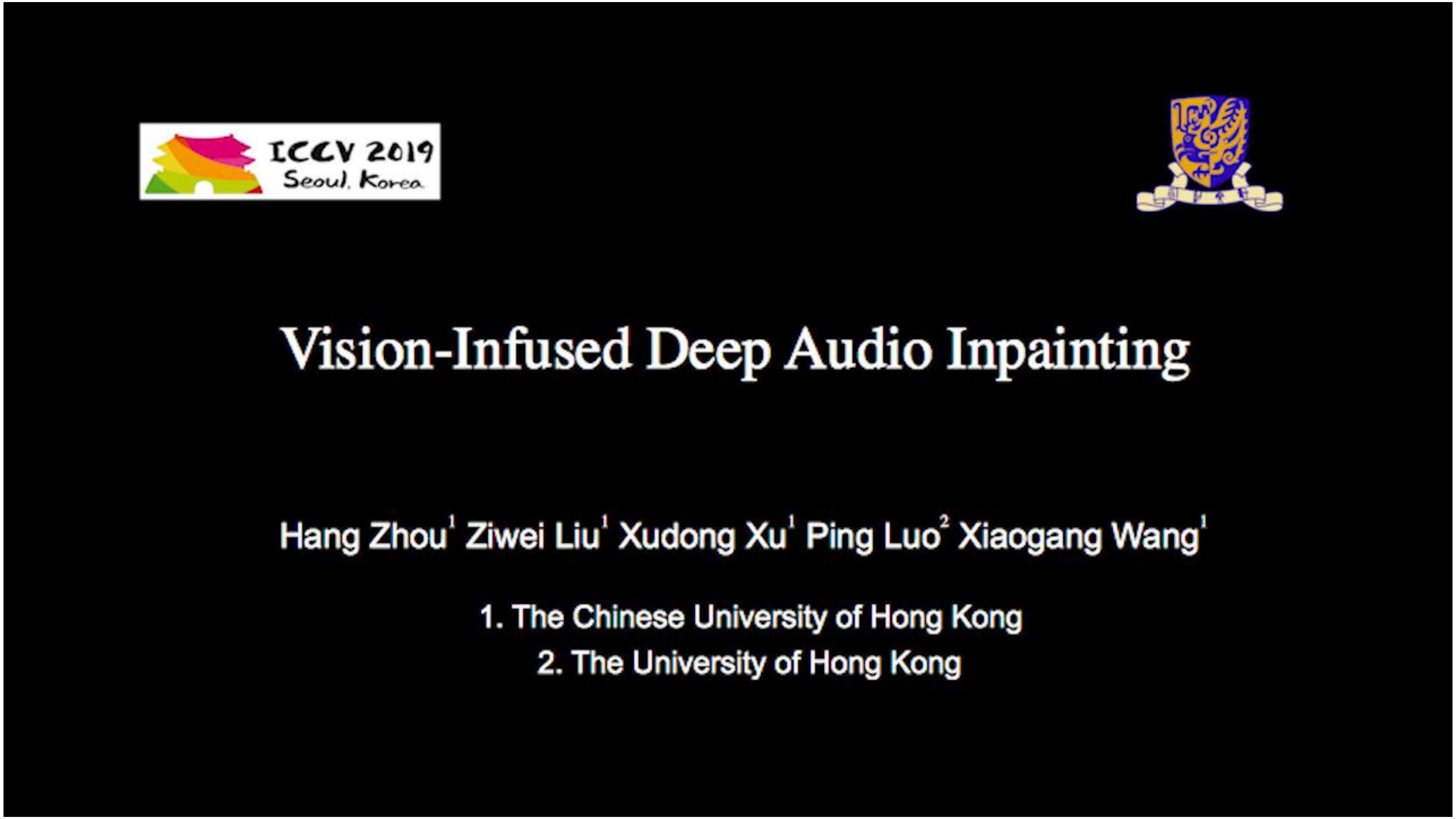
Zhou et al., “Vision-Infused Deep Audio Inpainting”, ICCV 2019

GANs: Audio Inpainting



Zhou et al., “Vision-Infused Deep Audio Inpainting”, ICCV 2019

GANs: Audio Inpainting



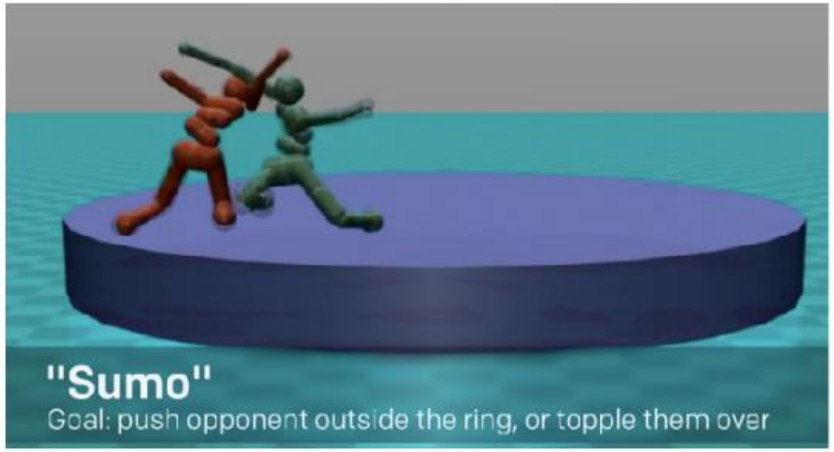
Zhou et al., "Vision-Infused Deep Audio Inpainting", ICCV 2019

GANs: Self Play

1959: Arthur Samuel's checkers agent

(Silver et al, 2017)

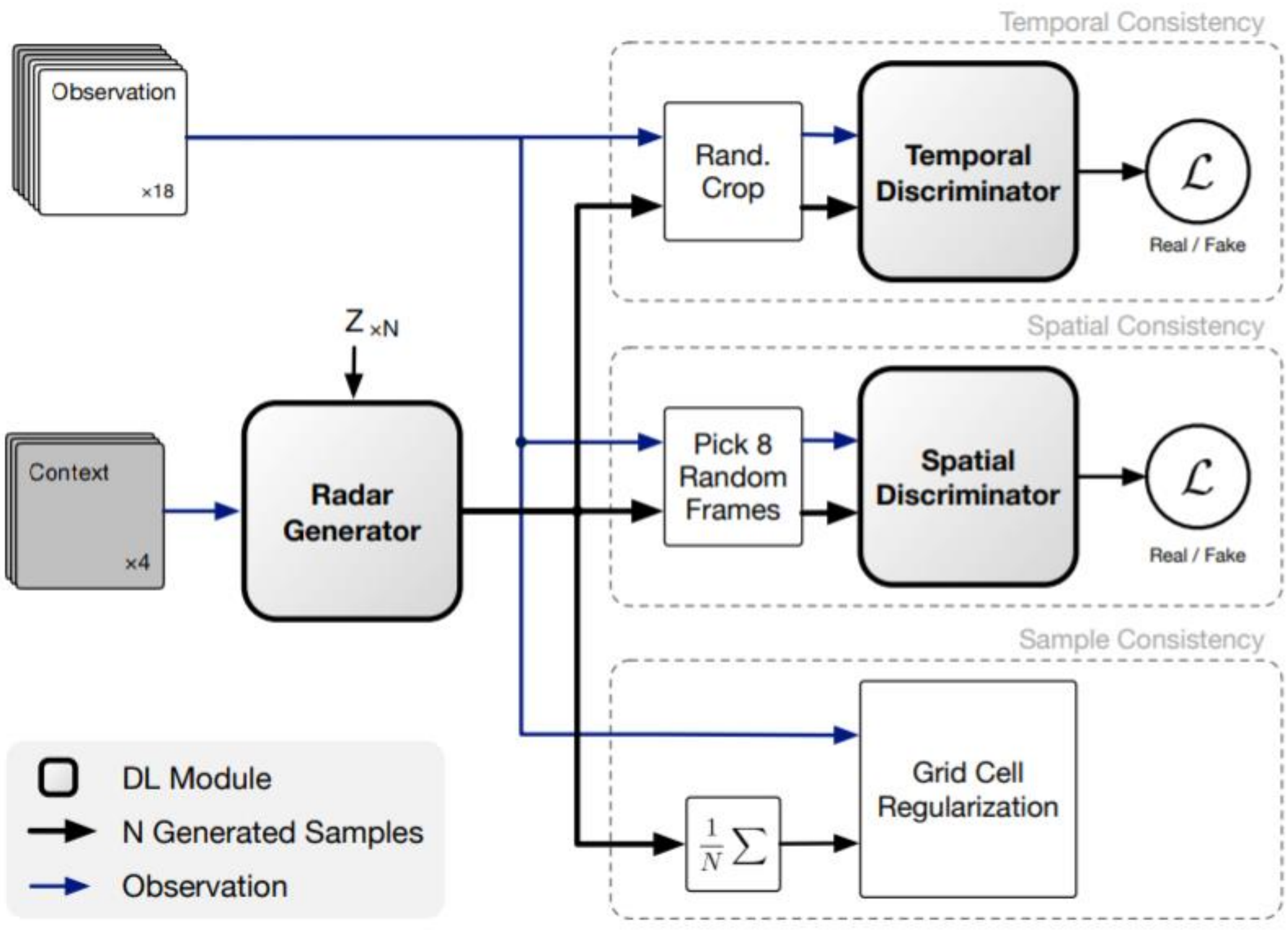
(OpenAI, 2017)



(Bansal et al, 2017)

GANs: Precipitation Nowcasting

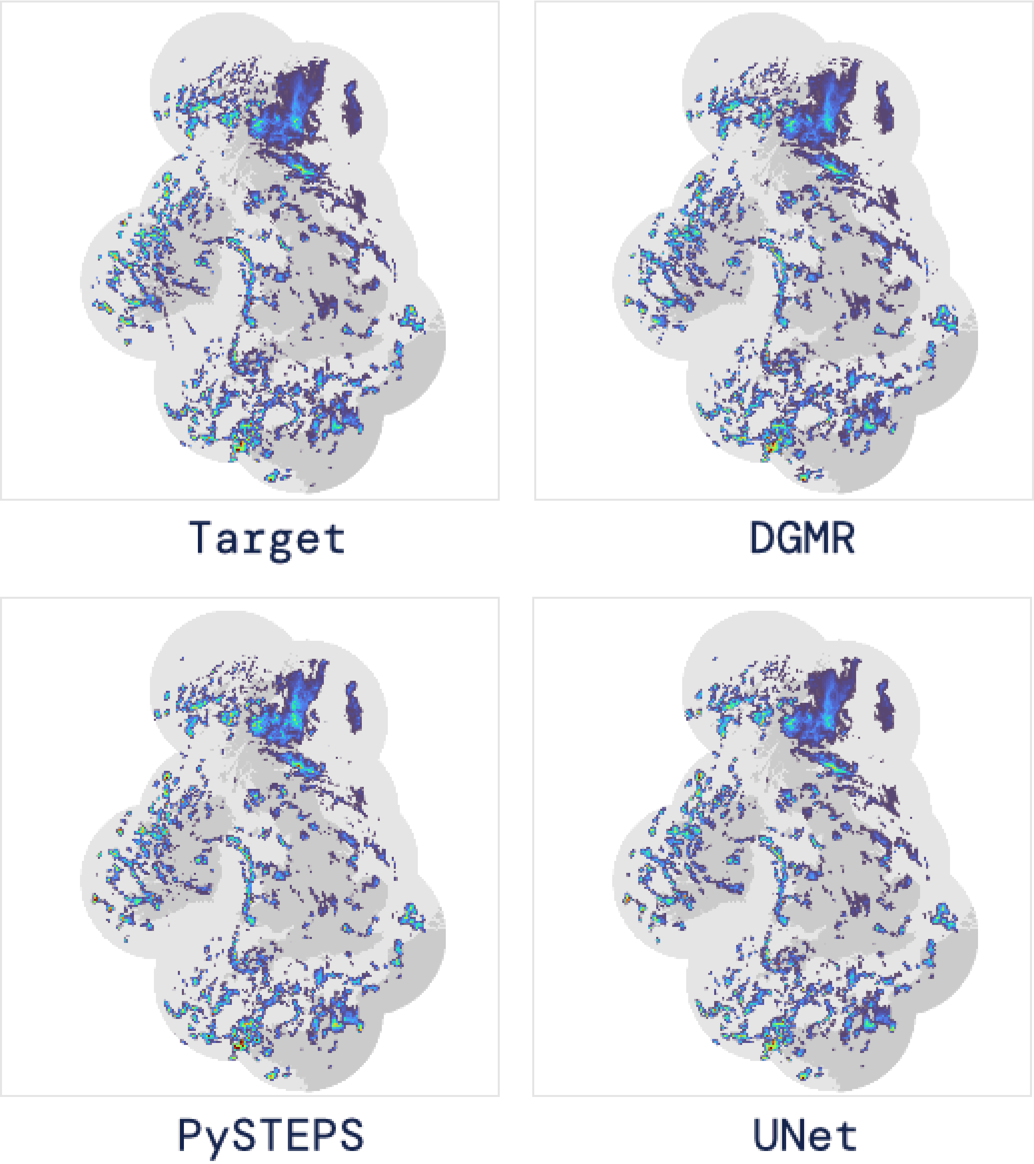
GANs: Precipitation Nowcasting

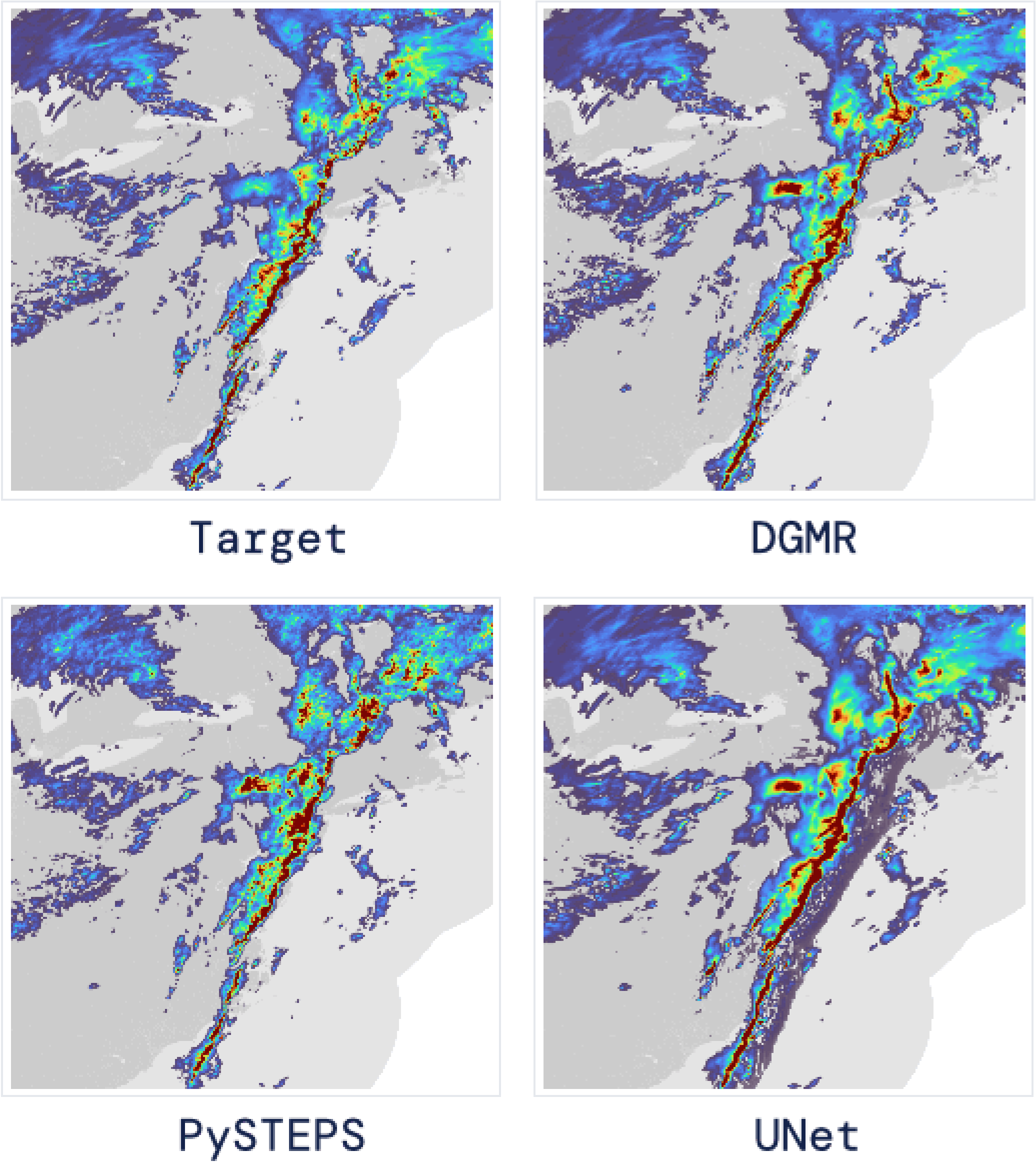


Ravuri et al., "Skillful Precipitation Nowcasting using Deep Generative Models of Radar", Nature 2021

GANs: Precipitation Nowcasting

GANs: Precipitation Nowcasting

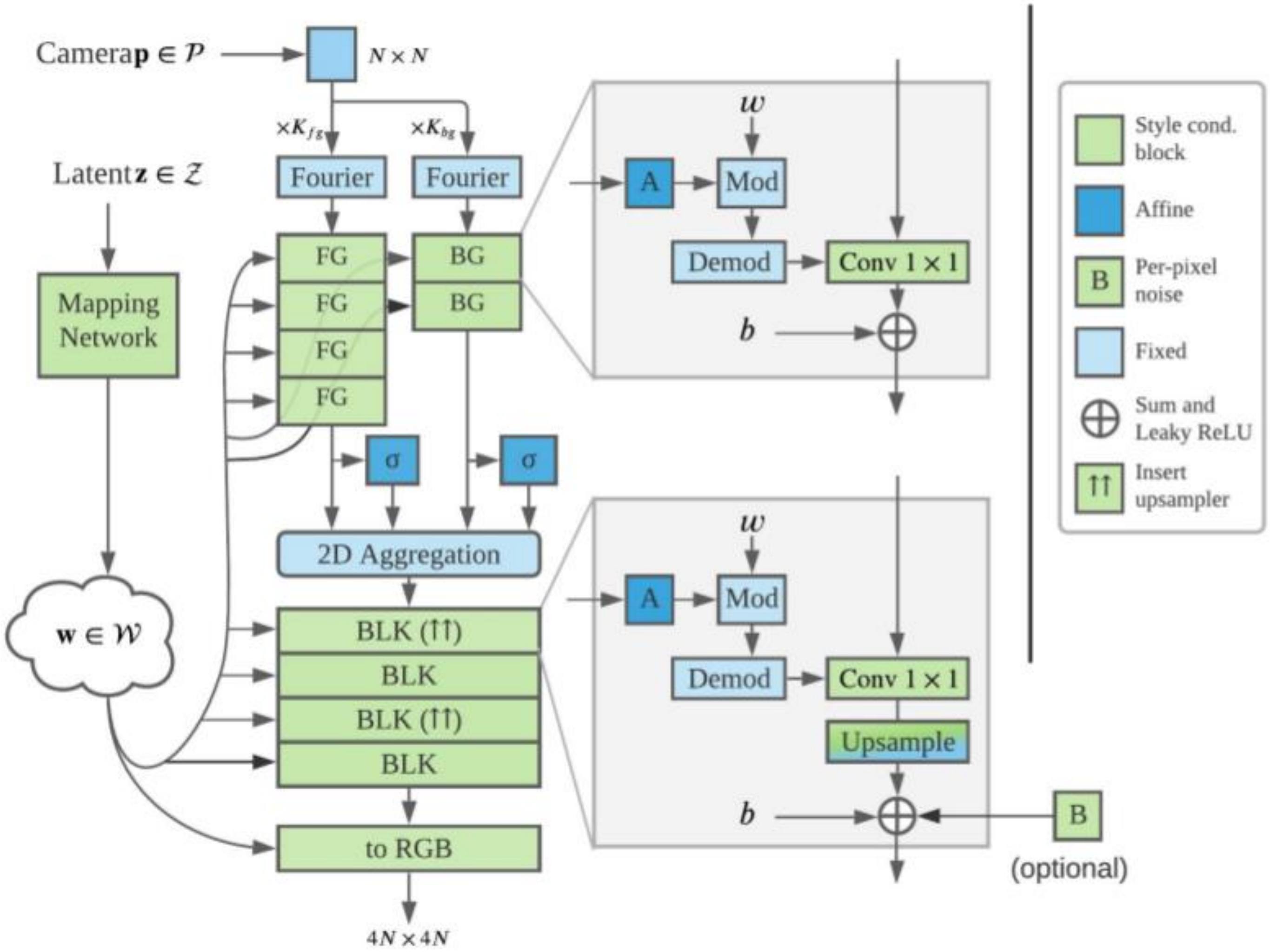




Open Problems

Open Problem I:
3D-Aware Synthesis

StyleNeRF



Gu et al., "StyleNeRF: A Style-based 3D-Aware Generator for High-resolution Image Synthesis", ArXiv 2021

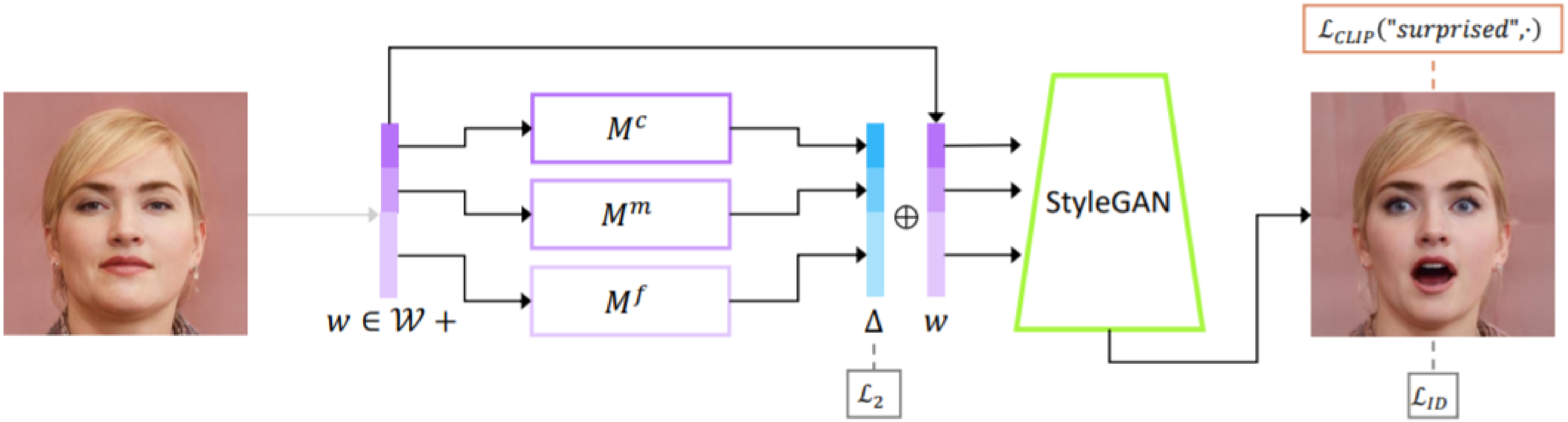
StyleNeRF: A Style-based 3D-aware Generator for High-resolution Image Synthesis

Jiatao Gu[1], Lingjie Liu[2], Peng Wang[3], Christian Theobalt[2]

[1] Facebook AI Research
[2] Max Planck Institute for Informatics
[3] The University of Hong Kong

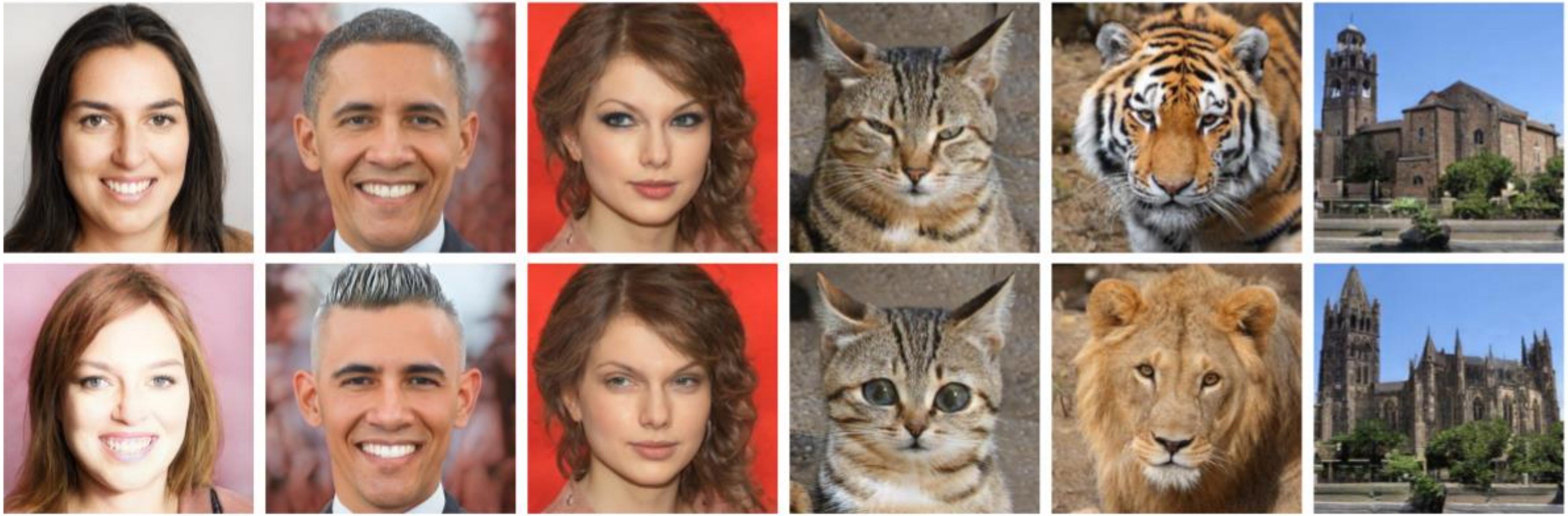
Open Problem II:
Multi-Modality Interaction

StyleCLIP



Patashnik et al., “StyleCLIP: Text-Driven Manipulation of StyleGAN Imagery”, ICCV 2021

StyleCLIP



“Emma Stone” “Mohawk hairstyle” “Without makeup” “Cute cat” “Lion” “Gothic church”

Patashnik et al., “StyleCLIP: Text-Driven Manipulation of StyleGAN Imagery”, ICCV 2021

DALL-E

TEXT PROMPT an armchair in the shape of an avocado. . . .

AI-GENERATED IMAGES

Edit prompt or view more images↓

TEXT PROMPT a store front that has the word 'openai' written on it. . . .

AI-GENERATED IMAGES

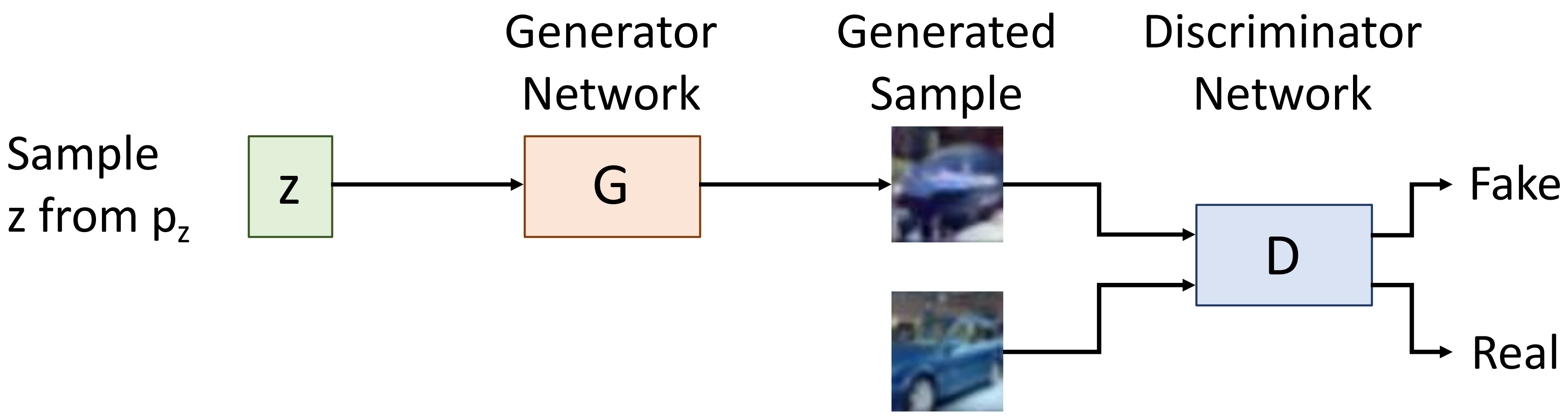
Edit prompt or view more images↓

GAN Summary

Jointly train two networks:

Discriminator: Classify data as real or fake

Generator: Generate data that fools the discriminator



Under some assumptions, generator converges to true data distribution

Many applications! Very active area of research!

Taxonomy of Generative Models

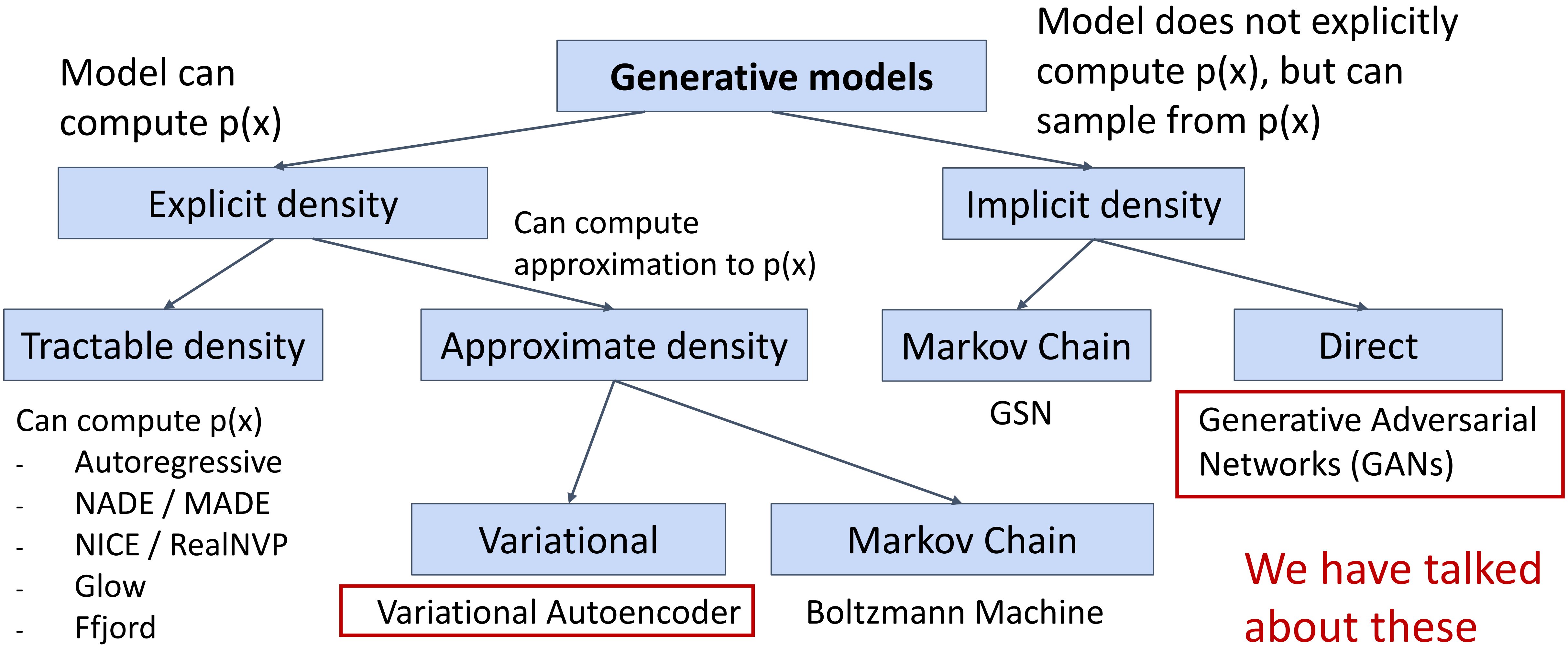


Figure adapted from Ian Goodfellow, Tutorial on Generative Adversarial Networks, 2017.

Generative Models Summary

Variational Autoencoders introduce a latent z, and maximize a lower bound:

$$p_\theta(x) = \int_Z p_\theta(x|z)p(z)dz \geq E_{z\sim q_\phi(z|x)}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x), p(z)\right)$$

Latent z allows for powerful interpolation and editing applications.

Generative Adversarial Networks give up on modeling p(x), but allow us to draw samples from p(x). Difficult to evaluate, but best qualitative results today

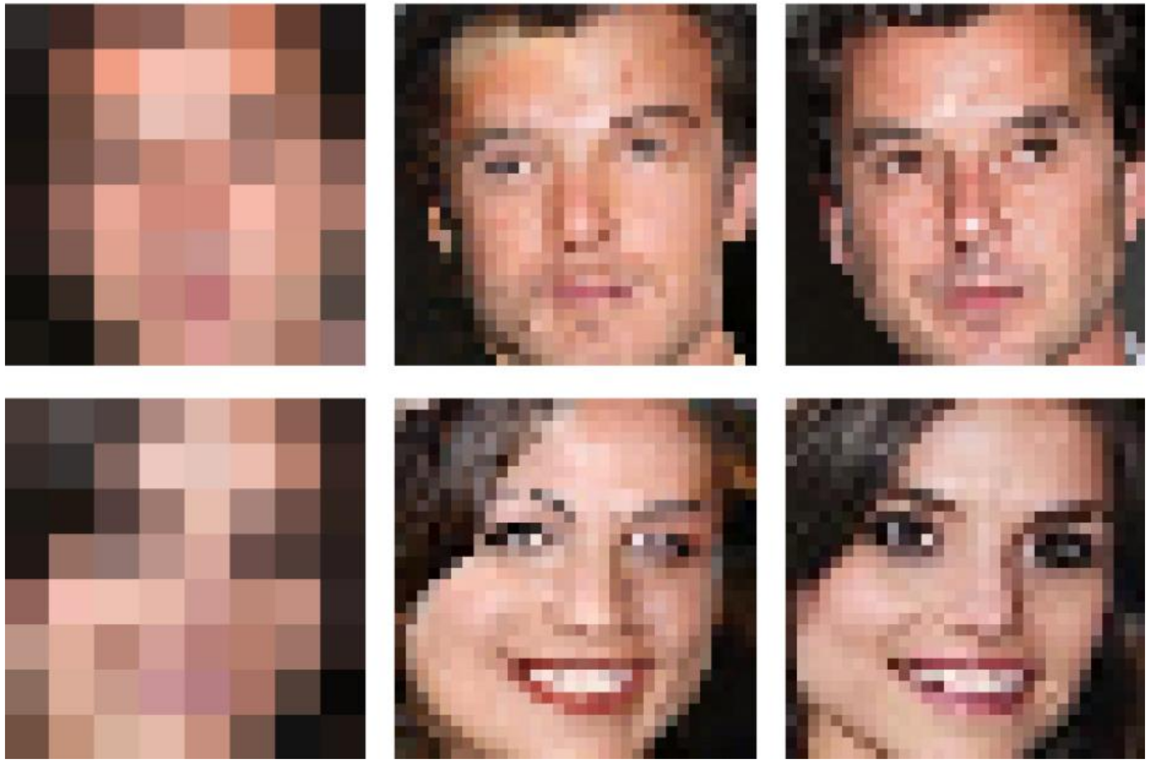
Resources

- MMGeneration: https://github.com/open-mmlab/mmgeneration



- GAN Lab: https://poloclub.github.io/ganlab/

- CVPR 2018 Tutorial on GANs:
  https://sites.google.com/view/cvpr2018tutorialongans/



Next Time:
Image Super-Resolution